# NOETHER: A Constructive Framework for Metamorphic Pattern Discovery from Operator Algebras

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**Context.** Metamorphic Testing depends on metamorphic relations (MRs), yet MR identification remains largely experience-driven or search-driven. Expert MR sets often encode useful but implicit domain intuition, while generated candidates can be difficult to interpret, deduplicate, and maintain without an explanation of where the relation comes from and where it stops applying.

**Objective.** This paper studies MR identification rather than MR fault-revealing effectiveness. We ask whether a structured, operator-algebraic method can derive a broader and more explainable MR class / MetaPattern design space than expert MR sets and search-based MR generation, by making the algebraic origin and applicability boundary of MRs explicit from program-family governing equations.

**Method.** We introduce **NOETHER**, a two-level framework for equation-governed program families. The upstream layer curates a program-family operator algebra and decomposes it into recurrent operator blocks, including symmetry, order, self-adjointness, time reversal, limits, qualitative dynamics, method comparison, and relational equivalence. The downstream layer, `CONSTRUCT-MP`, derives MR classes / MetaPatterns from those blocks and then

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checks whether a specific program interface exposes the inputs and observations needed to instantiate an executable MR.

**Evidence.** We evaluate NOETHER as an MR-identification method along three axes. First, we compare expert MR sets and NOETHER using binary operator-block coverage. Second, we compare NOETHER with search-based MR generation on origin explanation, boundary explanation, redundancy, readability, and maintenance. Third, we trace cross-domain derivations showing that the same operator block can explain MR classes across different program families and solvers. Mutation and head-to-head results are retained only as secondary executability and sanity-check evidence.

**Conclusion.** NOETHER contributes a theory-guided MR-identification framework: it derives MR classes from governing-equation operator algebra, makes their origin and boundary auditable, and exposes structural MR design-space coverage that is not explicit in expert MR sets or search-generated candidates alone.

CCS Concepts: • **Software and its engineering** → **Software testing and debugging**; *Software verification and validation*; • **Theory of computation** → *Algebraic semantics.*

Additional Key Words and Phrases: metamorphic testing, metamorphic relation identification, MetaPattern, operator algebra, program-family testability, test oracle problem, software testing foundations

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## 1 INTRODUCTION

Noether's first theorem replaced an empirically curated catalogue of conservation laws with a derivation from the structure of an action functional.[1] A catalogue of observed invariants can sometimes be replaced by a derivation procedure grounded in the underlying structure. The replacement does not eliminate empirical work; it lifts the empirical step from per-instance enumeration to per-domain structural identification, one cycle of induction per stable domain rather than per program.

Software testing faces a related problem. Metamorphic Testing (MT), introduced by Chen et al. in 1998 [11], was admitted into the IEEE/ISO/IEC software-testing standards in 2022 [26] and has been increasingly endorsed as a technique for testing AI and machine-learning systems [32, 46]. Its central artefact is the *metamorphic relation* (MR): a property that constrains how a program's outputs must covary across multiple executions, thereby substituting for an explicit oracle in domains where one cannot exist. Despite MT's maturity, MR identification, the task of deciding *which* properties hold for a program under test, remains its binding constraint. Practitioners report three recurring difficulties: high dependence on tester domain knowledge, mutually incompatible MR formulations for the same program, and low reuse of MR sets across teams or projects [32, 46]. Recent work has responded at the application layer, by mining MRs for particular domains, and at the integration layer, by automating search through evolutionary, mining-based, or LLM-assisted pipelines [3, 47, 60]. The foundational layer has not advanced at the same pace.

[1]The theorem appeared in Noether's 1918 paper *Invariante Variationsprobleme*, addressing a question Hilbert and Klein had raised about energy conservation in general relativity. We do not invoke it as a theorem about programs: program semantics and metamorphic relations do not provide an action functional. The analogy is methodological only.

The principal artefact at that foundational layer is the *MetaPattern* (MP): an equivalence class over MRs that captures a recurrent structural strategy a tester invokes when reasoning about program properties. MetaPatterns organise the otherwise unbounded MR design space into a small, interpretable scaffold; methods that exploit them, including the structured MR identification approaches METRIC [12] and METRIC+ [52], and the recent wave of LLM-prompted MR generators, depend on the scaffold's quality. Yet across the literature, MetaPattern catalogues continue to be assembled in the same way conservation laws were assembled before 1918: by induction over observed examples. A typical proposal lists $k$ patterns drawn from cluster analysis or expert codification, demonstrates that the patterns "cover" some corpus of MRs to a target threshold, and stops. None of the existing MP proposals, including the authors' own prior work on five reactor-physics patterns, answers the three questions that any foundational theory of MetaPatterns should answer:

(1) **Origin.** *Why* exactly these MetaPatterns and not others? What is the structural source of an MP, as distinct from an empirical regularity in the corpus on which it was induced?
(2) **Closure.** Under what mathematical conditions is a discovered MP set *closed under a stated derivation operator*, in the sense that it is guaranteed not to miss patterns reachable through the operator from a structurally fixed input? Absolute completeness over all properties one might write over the underlying structure is a strictly stronger demand and remains open in general.
(3) **Transferability.** When the program family changes (from reactor physics to natural-language inference, from numerical libraries to recommender systems), how does the MP set change, and can the new set be obtained without re-running the entire empirical induction in the new domain?

We call this the *origin–closure–transferability gap.* The gap matters because it explains why MR sets continue to grow as one-off artefacts. Without a structural source, there is no clear boundary on what the pattern space contains. Without a transfer rule, a relation written for one program rarely moves cleanly to another, even when both programs share a deep structural template. These are not only tooling limitations; they follow from grounding MetaPatterns in observed examples rather than in the structure that makes the relations hold.

This paper proposes such a structural source. We introduce **NOETHER**, a constructive framework that derives MetaPatterns from the operator-algebraic structure of the program family under test. The framework extracts MetaPatterns from invariants of an operator algebra $\mathcal{A}_P$ (formally defined in Section 3.1) that captures the program family's mathematical scaffolding. Given $\mathcal{A}_P$, NOETHER produces a MetaPattern set $\mathbb{M}(\mathcal{A}_P)$ (constructed by the algorithm of Section 3.2) together with a closure guarantee over the algebra-induced MR space. Given a different program family with a different algebra $\mathcal{A}_{P'}$, the same construction produces $\mathbb{M}(\mathcal{A}_{P'})$ without re-running empirical induction. The framework does not abolish induction: domain experts must still distil $\mathcal{A}_P$ from program semantics, and Section 6 states this limitation explicitly. What becomes algebraic is the downstream step from $\mathcal{A}_P$ to the MetaPattern set. Figure 1 summarises the architecture.

We make five MR-identification contributions.

- **C1.** We introduce NOETHER, a *layered* framework for MR-class / MetaPattern identification that combines an empirically curated structural decomposition (Hypothesis 1; upstream layer, itself partly

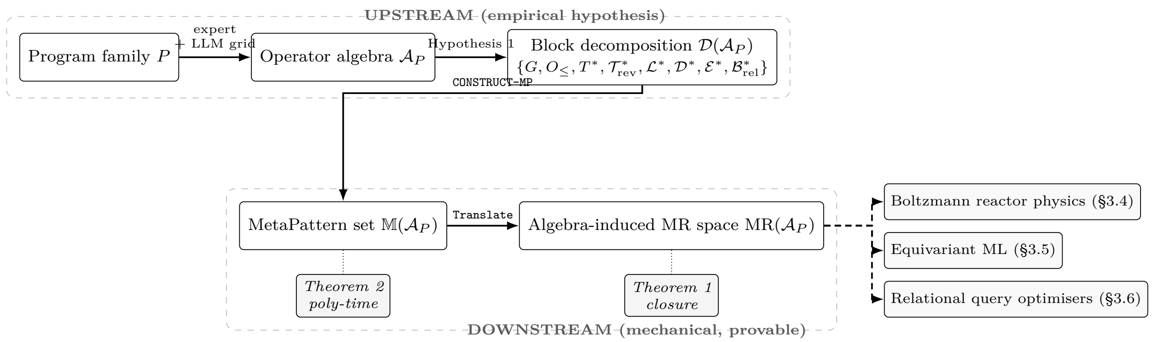


Fig. 1. NOETHER framework architecture. The *upstream layer* curates the program family's operator algebra $\mathcal{A}_P$ and its structural decomposition as an empirical hypothesis (Hypothesis 1). The *downstream layer* mechanically derives the MetaPattern set $\mathbb{M}(\mathcal{A}_P)$ via `CONSTRUCT-MP`, with closure under `Translate` over $\mathrm{MR}(\mathcal{A}_P)$ (Theorem 1) and polynomial-time decidability under a finite generating set (Theorem 2). The framework is instantiated on three structurally distinct domains (§3.4, §3.5, §3.6) to test transferability at the algebra-skeleton level.

distilled from the present authors' prior PWR MR catalogue, see §3.4.3 for provenance) with a constructive algorithm for deriving MetaPatterns from a program-induced operator algebra (downstream layer).

- **C2a (positive theory).** Given the structural decomposition, Theorem 1 proves a *no-drop closure invariant*: every MR that is reachable by the framework's `Translate` operator over the algebra-induced MR space $\mathrm{MR}(\mathcal{A}_P)$ is assigned to a canonical operator block and MetaPattern class. This result is intentionally modest. It is a well-formedness guarantee for the downstream construction, not evidence that the upstream block list is complete and not evidence of fault-detection superiority. Section 3.2.3 states the explicit scope of Definition 13, and Remark 7 documents the MR classes outside that scope. Polynomial-time decidability of CONSTRUCT-MP holds when the algebra admits a finite generating set (Theorem 2).
- **C2b (negative theory).** The strictly stronger absolute-completeness conjecture (Theorem 1′ / Conjecture A), over arbitrary properties expressible in $\mathcal{A}_P$, is *false* on $\mathcal{A}_{\mathrm{PWR}}$: two MRs from the standard PWR safety-analysis literature (non-additivity of rod-bank reactivity worth; second-order mixed dependence of $k_{\mathrm{eff}}$ on moderator temperature and boron concentration) are formulable over $\mathcal{A}_{\mathrm{PWR}}$ but not in $\mathrm{MR}(\mathcal{A}_{\mathrm{PWR}})$ (§3.7, Appendix A). The two MRs identify five structural obstructions in `Translate`'s signature, with pairwise independence established by per-block exhaustion. Companion surveys on $\mathcal{A}_{\mathrm{equi}}$ and $\mathcal{A}_{\mathrm{rel}}$ (§3.6) identify five further candidate dimensions (two on $\mathcal{A}_{\mathrm{equi}}$ specialising PWR-side dimensions; three net on $\mathcal{A}_{\mathrm{rel}}$), yielding five `Translate`-extension dimensions proved pairwise-independent on $\mathcal{A}_{\mathrm{PWR}}$ plus five candidate dimensions on $\mathcal{A}_{\mathrm{equi}}$ and $\mathcal{A}_{\mathrm{rel}}$ whose pairwise independence is asserted by inspection; formal per-dimension exhaustion on those algebras is committed as follow-up. Scope statements are in Section 3.2.3.

- **C2c (limiting theory).** We prove the *Invariance-Blindness Theorem* (§3.3): for the symmetry ($G$) and self-adjoint ($T^*$) blocks, an algebra-derived MR's detection kernel equals *exactly* the structure-preserving faults, within the linear operator-implementation fault class and under a faithfulness condition that a finite test attains. Corollaries: a single-block battery is structurally incomplete; completeness requires oracle families with trivial joint kernel; a neutral cross-implementation differential oracle is the complementary (common-mode) oracle. This converts the by-construction closure of C2a into a falsifiable, non-tautological characterization of the derivation's blind spots; tightness beyond $\{G, T^*\}$ and beyond the linear fault class is delimited in §3.3.
- **C3.** We instantiate NOETHER on a real-world program family and show how it *systematises and re-classifies* previously catalogued patterns. The framework re-classifies the prior inductive catalogue under a uniform algebraic structure: it reproduces three prior MetaPatterns, refines two on a sounder algebraic basis, and de-duplicates two equivalence classes ($m_{\mathrm{adj}}$, $m_{\mathrm{rev}}$) that the inductive catalogue had not isolated as structurally distinct. These are not de novo discoveries; NOETHER's contribution is the algebraic warrant for treating them as separate MetaPattern slots within a canonical block ordering, and the deflationary direction (revealing over-counting on some sub-families) constitutes a further non-circular form of re-classification. The deflationary direction is non-circular relative to the prediction caveat of §3.4.3: §5.5 demonstrates it on three independent test cases (a sorting library; Murphy et al. 2008's categorisation of MRs for ML into six classes [38] re-decoded against a feedforward classifier algebra; the prior reactor-physics catalogue itself), without invoking $T^*$ or $\mathcal{T}^*$ structure that was curated from reactor physics.
- **C4.** We demonstrate *structural* transferability at the algebra-skeleton level (not cross-domain empirical superiority), within the framework's scope precondition, by instantiating NOETHER on three structurally distinct operator-algebraic domains at differentiated evaluation depths: Boltzmann reactor-physics transport (Section 3.4), equivariant ML (Section 3.5), and relational query optimisers (Section 3.6); the last exercises the relational-equivalence block, whose algebraic skeleton differs from the Lie-group / self-adjoint / time-reversal core.
- **C5.** We define an evidence protocol for MR identification: binary expert-vs-NOETHER operator-block coverage, search-vs-NOETHER origin/boundary comparison, and compact cross-domain derivation traces. Existing mutation, GenMorph, and DeepCrime-style results are treated as secondary executability checks, not as evidence of average fault-detection superiority.

*Scope of contribution.* This paper evaluates *MR identification*: whether a method can systematically derive structurally justified MR classes from a program family's governing equations. It does not evaluate MR *effectiveness* in the sense of average fault-detection rate, mutation score, or general defect-revealing superiority. The eight blocks are curated by inspecting mathematical structures that recur across the program families we have studied; they are not derived from an algebraic axiom. The framework therefore has two layers: an *upstream layer* (curating $\mathcal{A}_P$ and its block decomposition) that remains empirical and human, and a *downstream layer* (mechanically deriving $\mathbb{M}(\mathcal{A}_P)$ from $\mathcal{A}_P$) that is algorithmic and provable. We do not claim to have eliminated induction from MetaPattern discovery. We move induction one level up, from "what MetaPatterns recur in observed MR samples?" to "what algebraic structures recur in the program families practitioners care about?", and make the downstream step mechanical.

**Boundary of contribution**

This paper establishes:

(1) A no-drop closure invariant for the downstream construction: given a block decomposition, $\mathbb{M}(\mathcal{A}_P)$ is closed under `Translate` over the algebra-induced MR space $\mathrm{MR}(\mathcal{A}_P)$ for the stated operator algebras (Theorem 1; by-construction within the explicit scope of Definition 13, see Remark 7). Closure over the strictly larger space of arbitrary properties expressible in $\mathcal{A}_P$ is Theorem 1′, falsified on $\mathcal{A}_{\mathrm{PWR}}$ (see item (a) below);

(2) Polynomial-time decidability of CONSTRUCT-MP under explicit complexity assumptions (Theorem 2);

(3) Three non-vacuous instantiations: Boltzmann reactor physics (§3.4), equivariant ML (§3.5), and relational query optimisers (§3.6); the relational-equivalence block $\mathcal{B}^*_{\mathrm{rel}}$ extends the framework beyond the Lie-group / self-adjoint / time-reversal mathematical core;

(4) The Invariance-Blindness Theorem for the $G$ and $T^*$ blocks (Theorem 3): the symmetry / self-adjoint MR detection kernel equals the structure-preserving faults (faithfulness-tight), within the linear operator-implementation fault class.

It does *not* establish:

(a) Absolute completeness over arbitrary properties expressible in $\mathcal{A}_P$. This is Theorem 1′ (Conjecture A, Appendix A); §3.7 establishes that it is false on the PWR core diffusion algebra $\mathcal{A}_{\mathrm{PWR}}$ via two independent counterexamples (non-additivity of rod-bank reactivity worth, and second-order mixed dependence of $k_{\mathrm{eff}}$ on moderator temperature and boron concentration), identifying five structural obstructions in `Translate`'s signature whose pairwise independence is proved by per-block exhaustion (Appendix A). Companion surveys on $\mathcal{A}_{\mathrm{equi}}$ and $\mathcal{A}_{\mathrm{rel}}$ (§3.6, companion artefacts in `theory/`) identify five further candidate dimensions on those algebras (so the falsification is not PWR-specific). Pairwise independence on the candidate five is asserted by inspection rather than by formal exhaustion proof; full per-dimension exhaustion proofs on $\mathcal{A}_{\mathrm{equi}}$ and $\mathcal{A}_{\mathrm{rel}}$ are committed as follow-up. The question of whether a Composite-`Translate` extension absorbs the ten `Translate`-extension dimensions identified across the three algebras while preserving Theorem 1's closure and Theorem 2's polynomial-time decidability remains open.

(b) Sufficiency of the structural list. Hypothesis 1 is an empirical hypothesis with six enumerated out-of-scope program-family classes (Remark 3); each such class is a candidate ninth block.

(c) Superiority over existing automated MR-identification pipelines on average. The comparative material in §5.2 is retained as secondary executability and sanity-check evidence; it does not characterise the framework's behaviour on arbitrary defect distributions.

(d) Elimination of induction from MetaPattern discovery. Induction is *relocated* from MR-instance level to algebra-block level, not eliminated.

The remainder of the paper follows a TOSEM-style method-paper structure. Section 2 reviews expert MR design, structured taxonomies, and search-based MR generation. Section 3 presents the NOETHER method, the two-level model from identified MR class to executable MR instance, the operator-block taxonomy, and the coverage rules. Section 4 defines the three evaluation questions and evidence protocol. Section 5 reports expert-set coverage, search-method complementarity, and cross-domain derivation evidence. Section 6 discusses threats to validity and limitations. Section 7 states future work; Section 8 concludes.

## 2 RELATED WORK

We organise prior work along four lines: MT/MR fundamentals and the long-standing identification bottleneck; structured MR identification through category, specification, and model scaffolds; automated MR identification through mining, search, learning, and LLM assistance; and MetaPattern catalogues, including our prior reactor-physics taxonomy. These lines differ in mechanism, but they share one limitation: their pattern structures are induced from observed MRs, mined artefacts, or manually supplied models rather than derived from the mathematical structure that makes those MRs hold.

### 2.1 Metamorphic testing and the MR identification bottleneck

Metamorphic Testing was introduced by Chen, Cheung, and Yiu in 1998 to address the test-oracle problem [11], a general obstacle in software testing when expected outputs cannot be automatically determined [5]. An MR is a logical implication of the form $R_i(x_1, \ldots, x_n) \Rightarrow R_o(P(x_1), \ldots, P(x_n))$, where $P$ is the program under test, $R_i$ is an input relation, and $R_o$ is an output relation. When $R_i$ holds but $R_o$ fails on actual executions, a fault is reported, without any need to know the absolute correct output of any single execution. Over two decades MT has become standard equipment for testing systems whose oracles are otherwise inaccessible: scientific computing, machine learning classifiers, autonomous vehicles, search engines, compilers, and large language models [32, 46].

The community has long acknowledged a single binding constraint on MT's effectiveness: MR identification. Surveys spanning twenty years agree that identifying high-quality MRs requires (i) deep familiarity with the program's functional semantics, (ii) substantive domain background (physical, mathematical, or linguistic, depending on the system), and (iii) the ability to convert that background into executable property assertions [32, 46]. Liu et al. [35] provide the canonical empirical evidence that even ad-hoc MRs detect a substantial fraction of mutants given modest tester training, while Murphy et al. [38] catalogued the structural properties of ML applications relevant to MR design and Xie et al. [59] demonstrated that MT scales effectively to supervised classifiers. Testers without sufficient domain background tend to identify only "trivial MRs". A 2024 survey explicitly identifies AI assistance, especially via large language models, as the most promising open avenue for closing the identification gap [32]; Saha and Kanewala [42] report that, on a benchmark of 709 mutants for supervised classifiers, current MR sets detect only 14.8%, which exposes the gap.

Beyond difficulty, the MR identification bottleneck has a more troubling structural feature: even when MRs are identified, the field has accumulated little consensus on *how* they are identified. Different authors confront the same program and produce dissimilar MR sets. Once written, an MR rarely migrates: a relation drafted for one ML classifier seldom transports to another, and a relation drafted for one numerical solver seldom transports across solver families.

### 2.2 Structured MR identification

Structured MR identification has already moved the field away from ad hoc relation design. METRIC organises MR construction around an "input/output category" framework: the tester first identifies relevant categories of input transformations and output relations, then composes MRs from category pairs [12]. METRIC+ extends this scheme by enriching the category catalogue and providing systematic combination rules that reduce the human burden in category enumeration [52]. Other structured lines instantiate the scaffold through domain or artefact models: Segura et al. define MR patterns for query-based systems from recurrent query features and explicitly use relational algebra as the vocabulary for those patterns [45]; Li et al. construct MRs from tabular-expression specifications [31]; Yan et al. decompose the input side of scientific-computing MRs into basic input-pattern structures [61]; and Lin et al. organise scientific-software MRs hierarchically [34]. Together, these approaches remain the strongest existing attempt to give MR identification an explicit scaffold.

Our disagreement is not with this direction but with its grounding. The categories, patterns, tables, and input decompositions are introduced through expert curation, artefact-specific structure, or observed reuse, and are validated by empirical coverage on benchmark programs. This leaves two of our three foundational questions unanswered. *Origin*: existing structured methods do not derive their categories from program-level operator algebra. Even query-based MR patterns use relational algebra as an organising vocabulary, not as a closed derivation system whose primitive operator blocks define an MR design space. *Closure*: these frameworks do not provide a mathematical condition under which the category set is complete; coverage is reported but not proved. The third question, *transferability*, is only partially addressed. The same category templates can be invoked across domains, but because the categories are not algebraically bound to specific program structures, transfer rests on an assumption of universality that remains open.

### 2.3 Automated MR identification

Automated MR identification attacks the bottleneck by searching for executable relations more directly. Early search and dynamic-inference work includes likely metamorphic-property inference from executions [51], search-based inference of polynomial MRs [64], and AutoMR's discovery and cleansing of numerical MRs through search plus redundancy reduction [63]. MR-Scout mines MRs from existing test suites by extracting input-transformation and output-assertion patterns from test-case pairs and abstracting them into reusable relations [60]. GenMorph evolves MR candidates through genetic programming, co-evolving input transformations with output assertions and using mutation-killing as the fitness signal [3]. Shin et al. derive executable MRs from natural-language requirements through few-shot prompting of a large language model, with validation through an industrial questionnaire study with Siemens [47]; further LLM-assisted variants, including ChatGPT-driven MR generation [66], domain-customised GPTs for autonomous-driving simulators [65], and multi-agent retrieval-augmented pipelines for traffic-rule MRs [33], extend this template into safety-critical or rule-rich domains. Within ML-system testing, differential-testing approaches such as DeepXplore [41] use input-transformation invariance as a proxy oracle, complementary to the explicit MR formulation we adopt here.

A second strand uses program-structure, specification, documentation, or model features rather than test-case mining. Kanewala et al. [27] predict MRs of scientific software via graph-kernel learning over control-flow and data-dependency graphs, an early demonstration that program structure carries enough signal for automated MR discovery. MeMo extracts equivalence MRs from Javadoc comments [8]. Nolasco et al.'s MemoRIA [40] infers MRs by deriving an object-protocol abstraction and validating candidates through fuzzing plus SAT-based reduction, then evaluates on 22 Java subjects. Clark et al. generate MRs from causal graphs, making model misspecification an explicit boundary of the method [13]. Tao et al.'s Mettoc [53] applied MT to compiler testing using equivalence-preservation as the canonical relation. Ying et al. [62] formalise relationships between metamorphic-relation patterns into family trees, and Altamimi et al. [1] provide a recent systematic literature review of MR-automation work that catalogues the structural-vs-empirical-search divide our framework targets. Earlier LLM-assisted attempts include Zhang et al. [66] on ChatGPT-driven MR generation.

Each of these methods advances automation on a different axis, but all treat the MR space as an empirical, artefact-derived, or model-supplied search space. MR-Scout's recall is bounded by the relations already latent in existing tests. Search-based and evolutionary methods can discover useful numerical or executable MRs, but the reported cleansing and redundancy-reduction steps are consequences of the search representation rather than explanations of the mathematical origin of the surviving relations. GenMorph's evolutionary search is shaped by the fitness landscape induced by mutation killing. LLM-prompted approaches are sensitive to prompt phrasing when no structural prior constrains the generation. Documentation-, protocol-, and causal-model-based methods are more interpretable, but their boundary is the supplied artefact or model rather than a program-family operator algebra. None of these methods can state, before search begins, which MR types lie outside its reach, because none has an algebraic account of the space being searched.

### 2.4 MetaPattern catalogues and empirical adequacy

MetaPattern catalogues organise the post-hoc inventory of identified MRs into a smaller vocabulary of recurring strategies. Such catalogues typically result from clustering observed MRs, codifying expert intuitions about "kinds of properties testers reason about", and validating the catalogue's coverage on a benchmark MR corpus. Reactor-physics test suites have produced taxonomies of conservation, monotonicity, convergence, trajectory, and partial-order patterns [7, 30]; Zhou et al. [69] introduced an explicit *symmetry metamorphic relation pattern* as a reusable abstraction for deriving concrete MRs across applications; Lin et al. [34] proposed hierarchical MRs for scientific software; Ying et al. [62] formalises the relationships between MR patterns into *family trees* with explicit refinement and specialisation edges, and reports a SOTA pattern hierarchy that organises previously proposed patterns into a single classification structure. The most recent state-of-the-art survey of MR generation [32] maps the field's twenty-year output across pattern catalogues, mining, evolutionary search, and LLM-prompted methods. Adequacy frameworks such as the Pattern–Matrix Coverage Metric, and recent MT adequacy work based on MRs and source inputs [19], assess how thoroughly a given test suite or MR set occupies an already-defined testing space. They do not define coverage over the operator blocks that give rise to MR classes.

Ying et al.'s family-tree formalism [62] is the closest published cousin to NOETHER's MetaPattern equivalence-class structure: both organise MR patterns into a hierarchy with explicit relationships between patterns. The two formalisms differ in the relationship type. Ying et al.'s family tree is a *refinement/specialisation tree* rooted in informally-named pattern categories (symmetry, additive, multiplicative, etc.); NOETHER's MetaPattern equivalence classes are *quotients of the algebra-induced MR space* $\mathrm{MR}(\mathcal{A}_P)$ under structural equivalence (Definition 13), with each class derived from a specific block of $\mathcal{D}(\mathcal{A}_P)$. A family-tree node in Ying et al. typically corresponds to one or more NOETHER MetaPatterns when the node admits an operator-algebraic specification (e.g. Ying et al.'s "symmetry" parent node decomposes into the $G$-block MetaPattern $m_{\mathrm{inv}}$ for finite-group symmetries plus the $T^*$-block MetaPattern $m_{\mathrm{adj}}$ for self-adjoint dualities under NOETHER's structural decomposition); conversely, NOETHER's $\mathcal{B}^*_{\mathrm{rel}}$ MetaPattern has no direct counterpart in the family-tree formalism because relational-algebra equivalences were not in Ying et al.'s benchmark set. The two formalisms are complementary: Ying et al. catalogues patterns inductively at the MR-instance level; NOETHER provides an algebraic warrant for the existence of each pattern, given the operator algebra. A unified family-tree-plus-algebra-block reading of the literature would map each family-tree node to (at most) one NOETHER block when the node has an algebraic source, and would flag family-tree nodes without algebraic source as candidates for the ninth-block list of Remark 3.

Three further pattern-catalogue or MR-grammar candidates raised in peer review (Hu et al. 2019; Mariani 2018; Liu et al. 2020) could not be located through the standard fallback chain (CrossRef, OpenAlex, DBLP, Semantic Scholar, Google Scholar) under the citation profiles provided, and are therefore not cited; the closest verifiable cousins in this space are Zhou et al. [69], Lin et al. [34], and Ying et al. [62].

Such catalogues are useful, but each is, by construction, an empirical artefact. None of the existing catalogues, including the present authors' own, can answer the three foundational questions of Section 1.

### 2.5 Convergent diagnosis

Across all four lines of work, two problems recur. *Unbounded MR emergence*: every new domain or new program produces fresh MR formulations. *Poor MR reusability*: identified MRs do not transport readily across programs that share deep structural similarities. Both problems follow from grounding MetaPatterns inductively. NOETHER replaces that inductive grounding with operator-algebraic grounding.

*Relation to a separate line of work on minimum complete MR subsets.* A separate line of work by some of the present authors addresses the *orthogonal* problem of *selecting* a minimum complete subset from a *given* metamorphic-relation pool: taking an MR set as input, it characterises the smallest subset that preserves the pool's fault-detection power under a fixed fault model, proves the corresponding decision problem NP-hard, and provides approximation and exact (ILP) solvers. That work takes the MR set as given and optimises its cardinality; the present paper is concerned with the upstream question of where MRs and MetaPatterns *come from*: their constructive derivation from a program-induced operator algebra and algebraic closure under `Translate`. The two are complementary: NOETHER *generates* the algebra-induced MR space that a subset-selection procedure would then *minimise*. No theorem or empirical claim is shared: the present paper makes no minimality or subset-selection claim, and the subset-selection work makes no derivation or closure claim.

*Comparators in the head-to-head: what is compared and why.* The empirical comparison in §5.3 runs Set N against a single executable baseline: Set G (GenMorph's GP-evolved MRs at the published 30-min GAssert budget, [3]). The other three SOTA categories enumerated in the related-work survey contribute differently: METRIC+ [52] is contrasted as a category-enumeration scaffold (§5.4, §5.4) rather than as an executable fault-detection pipeline, because no public METRIC+ implementation auto-generates MRs from a $\mathcal{A}_P$ specification: a full PIT-based METRIC+ vs Set N comparison is committed as supplementary S4 (`future_work.md`) item (i); MR-Scout [60] is omitted from the executable head-to-head because its mining input (a pre-existing test corpus) is structurally absent on the algebra-rich Java substrate of §5.3.3 (the SUTs are stand-alone mathematical methods with synthesised test inputs, not codebases with developer-written tests latent in test files); AutoMT [33] and GPTMR [65] target safety-critical / traffic-rule autonomous driving domains rather than the operator-algebraic substrate, so an in-scope comparator selection from the LLM-assisted line uses an unprompted-by-NOETHER LLM ensemble (Set L; §5.2, §5.3.7) rather than these domain-specialised pipelines. Set B is a literature-MR baseline drawn from MT-for-ML references [33, 46, 47], restricted to point-cloud-classifier-applicable MRs. The single executable head-to-head against Set G is therefore the GP-evolved-baseline arm of a three-SOTA-category protocol; the LLM-assisted and mining-based arms are reported with their proper comparator (Set L ensemble at 2 vendors × 5 temperatures, §5.3.7, and the MR-Scout structural-absence rationale above), so the protocol's categorical framing is preserved at the price of using different arms in different sections.

## 3 PROPOSED METHOD

This section presents NOETHER as an MR-identification method. The method takes a program-family governing equation/operator system as input, distils a program-induced operator algebra $\mathcal{A}_P$, decomposes the algebra into recurrent operator blocks, derives MR classes / MetaPatterns from block invariants, and finally checks whether a concrete program interface exposes the controls and observations needed to instantiate an executable MR. The primary output is an *identified MR class*; an executable MR instance is a downstream realization that may be blocked by missing interface affordances.

*Counting rule used throughout the paper.* Expert coverage is binary: an expert MR set covers an operator block if at least one expert MR belongs to that block. NOETHER coverage is also binary: NOETHER covers a block if the method explicitly derives at least one MR class from the program-family operator algebra in that block. Equation structure alone is not counted as coverage unless a corresponding MR class is written down. This rule is used before any empirical result is interpreted.

Operator-algebraic preliminaries and the structural decomposition appear in §3.1; the algebra-induced metamorphic relation space, the CONSTRUCT-MP construction, the algebraic closure theorem (Theorem 1), and the polynomial-time decidability theorem (Theorem 2) in §3.2. The framework is then instantiated on three structurally distinct operator algebras: Boltzmann reactor physics in §3.4, equivariant machine learning in §3.5, and relational query optimisers in §3.6. The section concludes by falsifying the strictly stronger absolute-completeness conjecture, Theorem 1′, on a fourth program family (PWR core diffusion) via two pairwise-independent counterexamples (§3.7).

### 3.1 Operator-algebraic preliminaries

This section introduces the algebraic apparatus used by NOETHER. We assume basic familiarity with group actions, equivalence classes, and quotient sets, but not with functional analysis. Each construct is defined at the level needed for the framework and later instantiated in reactor physics and equivariant machine learning.

*3.1.1 Programs and program-induced operator algebras.* Throughout the paper we treat a program $P$ as a (possibly partial) computable function $P : \mathcal{X} \to \mathcal{Y}$. The class of programs of interest belongs to a *program family* $\mathcal{F}$ whose members share an underlying mathematical scaffolding.

Definition 1 (Program-induced operator algebra). *Let $P$ belong to a program family $\mathcal{F}$. A* program-induced operator algebra *of $\mathcal{F}$ is a tuple*

$$\mathcal{A}_{\mathcal{F}} = (\mathcal{O},\ \circ,\ \sim_{\mathcal{F}}),$$

*where $\mathcal{O}$ is a set of operators acting on $\mathcal{X}$, $\mathcal{Y}$, or both; $\circ$ is an operator composition; and $\sim_{\mathcal{F}}$ is an equivalence relation declaring two operator expressions equal whenever they agree on every program of $\mathcal{F}$.*

#### 3.1.2 Symmetry groups (building block B1).

Definition 2 (Symmetry group of $\mathcal{A}_P$). *A* symmetry group *of $\mathcal{A}_P$ is a subgroup $G \leq \mathcal{A}_P$ whose elements act on $\mathcal{X}$ such that, for all $P \in \mathcal{F}$ and all $g \in G$,*

$$P(g \cdot x) = \rho(g) \cdot P(x) \quad \forall x \in \mathcal{X},$$

*where $\rho : G \to \mathrm{End}(\mathcal{Y})$ is a (possibly trivial) representation of $G$ on $\mathcal{Y}$.*

#### 3.1.3 Order operators: monotonicity and linearity (building block B2).

Definition 3 (Monotone operator). *$P$ is* monotone *with respect to $\theta$ if $\theta_1 \leq \theta_2 \Rightarrow P(\theta_1) \leq_{\mathcal{Y}} P(\theta_2)$ (or the reversed inequality, in which case $P$ is* anti-monotone*).*

Definition 4 (Linear operator). *$P$ is* linear *on $\mathcal{X}_0 \subseteq \mathcal{X}$ when $P(\alpha x_1 + \beta x_2) = \alpha P(x_1) + \beta P(x_2)$ for all $x_1, x_2 \in \mathcal{X}_0$ and scalars $\alpha, \beta$.*

#### 3.1.4 Self-adjoint operators (building block B3).

Definition 5 (Self-adjoint operator). *Given an inner product $\langle \cdot, \cdot \rangle$ on $\mathcal{X}$, an operator $L \in \mathcal{O}$ is* self-adjoint *if $\langle Lx, y \rangle = \langle x, Ly \rangle$ for all $x, y$ in the domain of $L$.*

Self-adjointness encodes a duality between the two arguments of an inner product. Reciprocity theorems in physics, transposed-graph identities in algorithms, and detailed-balance conditions in stochastic processes are all instances. In reactor physics, the adjoint transport formulation is self-adjoint under the appropriate inner product, yielding source-detector reciprocity.

#### 3.1.5 Time-reversal operators (building block B4).

Definition 6 (Time-reversal operator). *When $\mathcal{X}$ admits a time coordinate, a* time-reversal operator *$\mathcal{T}$ acts on inputs by reversing the time variable, $\mathcal{T}(x(t)) = x(-t)$. A program $P$ is* time-reversal symmetric *on a sub-family of inputs when $P(\mathcal{T}x)$ is determined by $P(x)$ through a fixed bijection on $\mathcal{Y}$.*

Time-reversal applies only where the underlying dynamics admit a reversible sub-family. Dissipative dynamics break this symmetry, so the corresponding MetaPattern is empty for those systems.

#### 3.1.6 Limit operators (building block B5).

Definition 7 (Limit operator). *A* parametrised limit operator *$\mathcal{L}_\theta$ is a family of operators indexed by a parameter $\theta$ such that there exists a limit element $\mathcal{L}_*$ with $\mathcal{L}_\theta \to \mathcal{L}_*$ as $\theta \to \theta_*$ in an appropriate operator topology.*

#### 3.1.7 Qualitative-dynamics operators (building block B6).

Definition 8 (Qualitative-dynamics operator). *A* qualitative-dynamics operator *$\mathcal{D}$ is an operator on solution trajectories of an underlying ODE/PDE that extracts qualitative features such as extrema, inflection points, monotonic phases, overshoot magnitudes, S-curve transitions, and phase-portrait orbits,* *all of which are invariant under perturbations preserving the underlying dynamical structure (Sturm-type comparison theorems, dynamical-systems classification).*

Some MR-relevant invariants are not point-wise but *shape-wise*: a solution curve may have an overshoot, a single extremum, or a monotone-then-saturating profile. B6 gives these qualitative-dynamics MRs their own algebraic root.

#### 3.1.8 Method-comparison operators (building block B7).

Definition 9 (Method-comparison operator). *A* method-comparison operator $\mathcal{E}$ *encodes a partial order $\preceq_{\mathcal{E}}$ on numerical or algorithmic methods, where $M_1 \preceq_{\mathcal{E}} M_2$ asserts that method $M_1$ produces an approximation no worse than method $M_2$ in a specified error norm, under specified conditions.*

*3.1.9 Decomposition of an operator algebra.* Given $\mathcal{A}_P = (\mathcal{O}, \circ, \sim_{\mathcal{F}})$, we decompose $\mathcal{O}$ along the eight building blocks introduced above:

$$\mathcal{D}(\mathcal{A}_P) = \big(G,\ O_{\leq},\ T^*,\ \mathcal{T}^*,\ \mathcal{L}^*,\ \mathcal{D}^*,\ \mathcal{E}^*,\ \mathcal{B}^*_{\mathrm{rel}}\big),$$

where $G$ collects symmetry subgroups, $O_{\leq}$ monotone and linear operators, $T^*$ self-adjoint operators, $\mathcal{T}^*$ time-reversal operators, $\mathcal{L}^*$ limit operators, $\mathcal{D}^*$ qualitative-dynamics operators, $\mathcal{E}^*$ method-comparison operators, and $\mathcal{B}^*_{\mathrm{rel}}$ relational-equivalence operators.

Definition 10 (Relational-equivalence block). *A* relational-equivalence operator *is a binary relation $\equiv_{\mathcal{R}}$ on expressions over an idempotent semiring $(\mathcal{S}, \oplus, \otimes, \mathbf{0}, \mathbf{1})$ such that $E \equiv_{\mathcal{R}} E'$ iff $E$ and $E'$ are equal under all valid evaluation contexts of $\mathcal{S}$, generated by a finite set $\mathcal{R}_{\mathrm{rel}} = \{\mathcal{R}_1, \ldots, \mathcal{R}_K\}$ of identity-preserving rewriting rules, each $\mathcal{R}_i$ an ordered pair $(\mathrm{lhs}_i, \mathrm{rhs}_i)$ of semiring expressions with the property $\mathrm{eval}(\mathrm{lhs}_i, D) =_{\mathrm{bag}} \mathrm{eval}(\mathrm{rhs}_i, D)$ on every valid input $D$. The block $\mathcal{B}^*_{\mathrm{rel}}$ collects all such operators. It is empty for program families without idempotent-semiring rewriting structure (Boltzmann reactor physics, equivariant ML) and non-empty for those with it (relational query optimisers, §3.6).*

*Necessity, sufficiency, and the empirical status of the decomposition.* We do not claim that the eight blocks are necessary in an absolute sense, that they exhaust all algebraic structures relevant to software testing, or that they follow from first principles. They are an *empirical curation*: a by-inspection enumeration of mathematical structures that recur across the program families we have studied. The claim is therefore scoped: the eight blocks are currently sufficient for these families, not provably necessary in general. This upstream empirical status is NOETHER's main limitation as a theoretical framework. Induction has not been eliminated from MetaPattern discovery; it has been moved one level up, from "what MetaPatterns recur in observed MR samples?" to "what algebraic structures recur across program families?" Given such a curated decomposition, the downstream derivation of $\mathbb{M}(\mathcal{A}_P)$ becomes mechanical and provably closed in the sense of Theorem 1. We state the structural sufficiency as an explicit empirical hypothesis with a documented out-of-scope catalogue.

Hypothesis 1 (Decomposition sufficiency). *Let $P$ be a program belonging to a program family $\mathcal{F}$ whose semantics admit an operator-algebraic representation $\mathcal{A}_P$ over (i) finite or finite-dimensional symmetry groups, (ii) partial orders with monotone or linear operators, (iii) self-adjoint operators with respect to a fixed inner product, (iv) anti-unitary involutions of a time coordinate, (v) limit operations of analytical* *(mesh-refinement, parameter-perturbation, or asymptotic) type, (vi) qualitative-dynamics attractors of an underlying ODE/PDE, (vii) inter-method comparison partial orders, and (viii) identity-preserving rewriting rule sets on an idempotent semiring. For such families the structural decomposition*

$$\mathcal{D}(\mathcal{A}_P) = G \,\dot\cup\, O_{\leq} \,\dot\cup\, T^* \,\dot\cup\, \mathcal{T}^* \,\dot\cup\, \mathcal{L}^* \,\dot\cup\, \mathcal{D}^* \,\dot\cup\, \mathcal{E}^* \,\dot\cup\, \mathcal{B}^*_{\mathrm{rel}}$$

*is sufficient: every operator in $\mathcal{A}_P$ relevant to MR derivation is assigned to at least one block. Hypothesis 1 is an empirical hypothesis open to refutation.*

Remark 1 (Single-class instances). *For each of the three operator algebras instantiated in this paper ($\mathcal{A}_{\mathrm{Boltz}}$, $\mathcal{A}_{\mathrm{equi}}$, $\mathcal{A}_{\mathrm{rel}}$), we adopt a canonical $\pi$-template per non-empty block under which $|\mathcal{I}_s / \sim_s| = 1$ (Boltzmann's $G$ block, for example, is treated as a single $\sim_s$ class with a $\pi$-template covering both geometric quarter-rotations and energy-group permutations). Under this instance-level grouping, $|\mathbb{M}(\mathcal{A}_P)|$ equals the number of non-empty blocks for each instance, so the empirical material that reports a fixed per-instance MetaPattern count (§§3.4, 3.5, 3.6) refers to these per-instance class counts and not to a general single-class assumption. A finer treatment (separating geometric and energy-group symmetry into two $\sim_s$ classes, for example) is permitted by Step 3 of CONSTRUCT-MP and would split the corresponding MetaPattern accordingly.*

Remark 2 (Block sufficiency vs. `Translate` sufficiency). *Hypothesis 1 asserts that the eight blocks suffice to* assign *every operator in $\mathcal{A}_P$ relevant to MR derivation. It does not assert that every MR formulable over $\mathcal{A}_P$'s operators is reachable through the framework's `Translate` operator from a single block invariant. Section 3.7 exhibits, on the PWR core diffusion algebra $\mathcal{A}_{\mathrm{PWR}}$, two MRs whose constituent operators are individually assigned to blocks of $\mathcal{D}(\mathcal{A}_{\mathrm{PWR}})$ (rod-insertion semigroup; boration and moderator-temperature scaling), yet whose MR content is not in* $\mathrm{MR}(\mathcal{A}_{\mathrm{PWR}})$ *in the sense of Definition 13. The block decomposition is therefore a necessary but not sufficient input to MetaPattern construction: `Translate`'s expressive form (single-block invariants, first-order $\pi$-template, single partial-order direction, operating on $P(x)$ tuples rather than on operator-spectrum quantities) is a second, independent constraint on the framework's reach. This distinction between "block sufficiency" (Hypothesis 1) and "`Translate` sufficiency" (open) is made explicit in §3.7.*

Remark 3 (Known and conjectured out-of-scope program families). *The hypothesis is open to refutation. Six families are known or conjectured to fall outside its image and to require an additional block:*

*(1)* Symplectic systems. *Hamiltonian dynamics whose volume-preserving structure is captured by a symplectic 2-form rather than self-adjointness ($N$-body simulators, celestial-mechanics integrators).*

*(2)* Sheaf-theoretic / categorical constructions. *Programs whose semantics are captured by functors and natural transformations between categories (certified compilers, type-driven program transformations); MRs of the form "commutativity of a square in the underlying category" do not reduce to single-block invariants under the present `Translate`.*

*(3)* Probabilistic / martingale invariants. *MRs of the form "$\mathbb{E}[f(\mathbf{x}_t) \mid \mathcal{F}_s] = f(\mathbf{x}_s)$ on a stopping-time domain" arise in MCMC samplers and stochastic gradient analyses.*

*(4)* Topological invariants. *MRs of the form "$f$'s level sets have a fixed Betti-number signature" (topological data analysis, shape classifiers) require homological structure absent from the present blocks.*

*(5)* Label-consistency for supervised learning. *MRs that constrain a model's predictions against ground-truth labels (e.g. wrong-sign-loss detection on classifiers); the absence of a label-consistency operator is one reason the §5.2 case study reports zero detection on category-(i) wrong-sign mutations.*

*(6)* Empirical parameter-distribution divergence. *A probability-distribution divergence operator on parameter-space measures, motivated by the §5.2.1 pilot's category-v-02/04/05 (activation change, bias removal, weight re-init) which the eight blocks do not detect. An MR of the form $D_{\mathrm{KL}}(p_\theta \| p_{\theta+\Delta\theta}) \le \tau$ for $\|\Delta\theta\| \le \epsilon$ would lie in such a block.* Empirical witness*: §5.2.1's cat-v-02/04/05 mutations are not detected by any MR set including Set N; this is that pilot's principal negative finding for Hypothesis 1.*

*Each family signals a candidate ninth block. The framework's intended response is to admit the additional block when the program family demands it, rather than to extend $\mathcal{B}^*_{\mathrm{rel}}$'s scope or to absorb the case as "out of scope for MR testing" in some over-broad sense. Concrete examples of MRs that lie outside Theorem 1's scope are catalogued in Appendix A; one candidate ninth block (metric-stability $M_{\mathrm{lip}}$) has an explicit `Translate`-template design (Remark 5 below).*

Remark 4 (Domain-level out-of-scope). *Beyond the candidate-ninth-block families of Remark 3 (which signal extensions* within *the framework's general programme), the following program-family classes are out of scope for NOETHER in the present form because they admit no operator-algebraic representation at all:*

- General web applications *(HTTP request/response programs, business-rule engines, content-management systems) whose semantics are state-machine or rule-driven rather than equational.*
- RLHF reward models and large-language-model agents *whose relevant invariants are statistical/measure-theoretic rather than algebraic (calibration, refusal consistency, sycophancy).*
- Distributed-consensus protocols *(Paxos / Raft / Byzantine variants) whose semantics are CRDT- or message-trace-based; the relevant invariants are linearisability and convergence rather than operator-algebraic.*
- Compiler-internal optimisations *(loop transformations, register allocation, instruction scheduling) whose correctness is expressed as a refinement relation on an operational semantics rather than as a property over a fixed input/output algebra.*

*These classes are not candidate ninth blocks; they are domains where the framework's scope precondition (program family admits an explicit operator-algebraic description) is structurally absent. NOETHER's reach is therefore consistent with what its scope precondition explicitly requires: SUTs from textbook-codified mathematical, physical, or relational-algebra domains are within reach, and SUTs from general software-engineering domains without such codification are not.*

Remark 5 (Metric-stability candidate ninth block, $M_{\mathrm{lip}}$). *Among the candidate ninth blocks of Remark 3, the metric-stability block has the most concrete `Translate`-template proposal. Define $M_{\mathrm{lip}} = (\mathcal{X}, d_{\mathcal{X}}) \to (\mathcal{Y}, d_{\mathcal{Y}})$ as the family of $K$-Lipschitz maps between two metric spaces; the induced MetaPattern $m_{\mathrm{lip}}$ is the equivalence class of pointwise stability inequalities $d_{\mathcal{Y}}(P(x'), P(x)) \le K \cdot d_{\mathcal{X}}(x', x)$. The corresponding `Translate` template constructs a follow-up by metric perturbation $x' = x + \varepsilon u$ ($\|u\| = 1$, $|\varepsilon| < \delta$) and predicates a $K$-Lipschitz output bound. Canonical-block ordering would place $M_{\mathrm{lip}}$ after $\mathcal{B}^*_{\mathrm{rel}}$ since metric structure is independent of the eight existing block invariants (no group action on the perturbation set, no* *partial order on inputs, no self-adjoint operator, no parametric refinement family, no idempotent-semiring rewriting). Theorem 1's closure proof transfers without modification because* $m_{\mathrm{lip}}$ *is single-block algebraically derived. We do not commit to* $M_{\mathrm{lip}}$ *as part of the canonical decomposition in this paper; we record it in Appendix A as the most concrete sub-instance of Remark 3 item (iv) (topological invariants) that admits an immediate* `Translate` *template, and as the empirical orphan in the audit of §3.4.3.*

## 3.2 Deriving MR classes with CONSTRUCT-MP

*3.2.1 Algebra-induced metamorphic relations.* Before defining algebra-induced MRs we make explicit the `Translate` operator that converts a single block invariant into an executable MR. Without this definition, the closure result of Theorem 1 would quantify over an undefined construct.

Definition 11 (Block invariant). *Let* $s \in \mathcal{D}(\mathcal{A}_P)$ *be one of the eight blocks. A* block invariant *of* $\mathcal{A}_P$ *under* $s$ *is a pair* $\iota = (\Phi, \pi)$ *where* $\Phi$ *is a finite set of operators drawn from* $s$*, and* $\pi$ *is a relation* $\pi \subseteq (\mathcal{X} \times \mathcal{Y})^k$ *for some arity* $k \geq 1$ *such that for every* $P \in \mathcal{F}$ *and every choice of operators* $\phi_1, \ldots, \phi_n \in \Phi$*, the tuple* $\big((x_i, P(x_i))\big)_{i=1}^k$ *obtained by applying* $\phi_1, \ldots, \phi_n$ *to a base input* $x \in \mathcal{X}$ *in the canonical order specified by* $s$ *satisfies* $\pi$*. We write* $\mathcal{I}_s$ *for the set of block invariants under* $s$ *and* $\sim_s$ *for the equivalence relation* $\iota \sim_s \iota'$ *iff* $\Phi = \Phi'$ *and* $\pi$*,* $\pi'$ *define the same constraint up to relabelling of input/output coordinates.*

The relation $\sim_s$ is reflexive, symmetric, and transitive by construction (set equality and constraint equality are equivalence relations); hence $\sim_s$ is a genuine equivalence relation on $\mathcal{I}_s$, justifying the quotient in Step 3 of CONSTRUCT-MP below.

Definition 12 (`Translate`). *The* `Translate` *operator is the function*

$$\mathrm{Translate} : \mathcal{I}_s \times \{s\} \longrightarrow \mathrm{MR}(P)$$

*that maps a block invariant* $\iota = (\Phi, \pi)$ *under block* $s$ *to the metamorphic relation*

$$\rho_{\iota,s} : \quad \forall x \in \mathcal{X},\ \forall \phi_1, \ldots, \phi_n \in \Phi : \quad \pi\big((x_i,\, P(x_i))_{i=1}^k\big) \ \textit{holds},$$

*where the input tuple* $(x_i)_{i=1}^k$ *is generated by applying the* $\phi_j$ *in the canonical order specified by* $s$ *(the convention is fixed per block: e.g., for* $s = G$*,* $x_i = g_i \cdot x_0$ *with* $g_i$ *enumerated by group orbit; for* $s = T^*$*, the tuple ranges over the two arguments of the inner product).* `Translate` *is well-defined because (i) the per-block canonical order resolves any ambiguity in tuple generation, and (ii)* $\sim_s$*-equivalent invariants are mapped to MRs that are* $\sim$*-equivalent as logical statements over* $P$*. Per-block instantiations of* `Translate` *are tabulated in Appendix A.*

Definition 13 (Algebra-induced MR). *Let* $\mathcal{A}_P$ *be a program-induced operator algebra and let* $\rho$ *be an MR over a program* $P$*. We say* $\rho$ *is* induced by $\mathcal{A}_P$*, written* $\rho \in \mathrm{MR}(\mathcal{A}_P)$*, when there exist (i) a block* $s \in \mathcal{D}(\mathcal{A}_P)$*, (ii) a block invariant* $\iota \in \mathcal{I}_s$*, and (iii) the equality* $\rho = \mathrm{Translate}(\iota, s)$ *holds in the sense of Definition 12. Properties of* $P$ *that constrain executions but cannot be obtained as* $\mathrm{Translate}(\iota, s)$ *for any single block* $s$ *and any invariant* $\iota \in \mathcal{I}_s$ *are* not *algebra-induced in this sense; concrete examples are catalogued in Appendix A.*

*3.2.2 Construction of the MetaPattern set.* We present the deductive procedure CONSTRUCT-MP that maps an algebra-decomposition to a MetaPattern set.

**Step 1: Invariant extraction.** For each block $s$ in $\mathcal{D}(\mathcal{A}_P)$, compute the set of invariants $\mathcal{I}_s$ of $\mathcal{A}_P$ under $s$.

**Step 2: MR derivation.** For each invariant $\iota \in \mathcal{I}_s$, derive the MR family $\mathcal{R}(\iota) = \{\, \rho \in \mathrm{MR}(\mathcal{A}_P) \mid \rho = \mathrm{Translate}(\iota', s),\ \iota' \sim_s \iota \,\}$.

**Step 3: Quotient (per equivalence class).** For each $\sim_s$-equivalence class $[\iota] \in \mathcal{I}_s/\sim_s$, form the MetaPattern $m_{s,[\iota]} = \mathcal{R}(\iota)$.

**Step 4: Aggregation.** Return $\mathbb{M}(\mathcal{A}_P) = \bigcup_{s \in \mathcal{D}(\mathcal{A}_P)} \{\, m_{s,[\iota]} : [\iota] \in \mathcal{I}_s/\sim_s \,\}$, so $|\mathbb{M}(\mathcal{A}_P)| = \sum_s |\mathcal{I}_s/\sim_s| \geq$ the number of non-empty blocks.

Remark 6 (MetaPattern granularity). *Each MetaPattern $m_{s,[\iota]}$ corresponds to a single $\sim_s$-equivalence class of invariants in block $s$. A block carrying $k_s > 1$ such classes (for instance geometric quarter-rotations and energy-group permutations within $G$ on $\mathcal{A}_{\mathrm{Boltz}}$, when treated as separate $\sim_s$ classes) contributes $k_s$ distinct MetaPatterns. In the three instances of §§3.4–3.6 we group invariants within a block under a single $\sim_s$-equivalence class (a single canonical $\pi$-template per non-empty block; see Remark 1), so $|\mathbb{M}(\mathcal{A}_P)|$ equals the number of non-empty blocks in those instances. The Invariance-Blindness analysis (§3.3) applies directly to each $m_{s,[\iota]}$ at its native per-class granularity.*

#### 3.2.3 Algebraic closure under Translate, the canonical-block ordering, and out-of-scope MRs.

Definition 14 (Canonical-block ordering). *We adopt the strict total order on blocks*

$$G > O_{\leq} > T^* > \mathcal{T}^* > \mathcal{L}^* > \mathcal{D}^* > \mathcal{E}^* > \mathcal{B}^*_{\mathrm{rel}}.$$

*An MR derivable through multiple blocks is assigned to the highest-priority block in this order. The placement of $\mathcal{B}^*_{\mathrm{rel}}$ at the bottom reflects its semiring-rewriting nature, which sits algebraically downstream of the seven physical-mathematical blocks: rewriting equivalences typically depend on the program family's input-perturbation, order, and method-comparison structure rather than the converse.*

Theorem 1 (Algebraic Closure under `Translate`). *Let $\mathcal{A}_P$ be a program-induced operator algebra with decomposition $\mathcal{D}(\mathcal{A}_P)$ as in Section 3.1.9, and let $\mathbb{M}(\mathcal{A}_P) = \mathrm{CONSTRUCT\text{-}MP}(\mathcal{D}(\mathcal{A}_P))$. Then for every $\rho \in \mathrm{MR}(\mathcal{A}_P)$ in the sense of Definition 13, there exists a unique $m \in \mathbb{M}(\mathcal{A}_P)$ such that $\rho \in m$, where uniqueness is determined under the canonical-block ordering.*

Remark 7 (Scope of Theorem 1). *Theorem 1 is a closure statement, not a completeness claim over arbitrary properties one might assert about $P$'s executions. It quantifies over $\mathrm{MR}(\mathcal{A}_P)$ as defined by Definition 13, namely the algebra-induced MRs reachable through the `Translate` operator from a single block invariant. Three concrete classes of MRs lie outside this scope and are not covered by the theorem; they are documented in Appendix A:*

*(1)* Probabilistic MRs without operator-algebraic representation*: e.g. MRs that constrain output* distributions *(such as "the entropy of $f(\mathbf{x})$ should not decrease under input augmentation $A$") when the augmentation is not expressible as an operator in $\mathcal{O}$.*

*(2)* Input-distribution MRs *not expressible over $\mathcal{A}_P$ operators: e.g. adversarial-perturbation MRs "$\|\delta\|_p \leq \epsilon \Rightarrow f(\mathbf{x}+\delta) = f(\mathbf{x})$" whose perturbation set is not a group action on $\mathcal{X}$.*

*(3)* Compositional MRs spanning multiple blocks under non-`Translate` derivations*: MRs requiring simultaneous use of two block invariants in a way that is not the canonical-block-ordering reduction of Definition 14.*

*The strictly stronger statement that every MR formulable as a property over $\mathcal{A}_P$'s operators (without restricting to `Translate`-reachable derivations) is contained in some $m \in \mathbb{M}(\mathcal{A}_P)$ is identified as Theorem 1′ in Appendix A. Section 3.7 and Appendix A establish that this stronger statement is false on the PWR core diffusion algebra $\mathcal{A}_{\mathrm{PWR}}$, by exhibiting two concrete counterexamples ($\rho_{\mathrm{nonadd}}, \rho_{\mathrm{MTC\text{-}bor}}$) whose obstructions identify five structurally independent extensions of `Translate`'s signature. The combined open problem (whether such an extended `Translate` preserves Theorem 1's closure and Theorem 2's polynomial-time decidability) is the principal open question for follow-up work.*

A sceptical reading might object that the by-construction status of Theorem 1 makes it near-tautological. We acknowledge that the closure result is by-construction within the explicit scope of Definition 13 (which fixes $\mathrm{MR}(\mathcal{A}_P)$ as the `Translate`-image of $\mathcal{A}_P$). The substantive value lies in what the theorem then enables: Theorem 1 converts an empirical-adequacy claim ("our pattern grid covers $X\%$ of observed MRs") into a structural-adequacy claim ("our pattern grid is exhaustive of the algebra-induced MR space, modulo the explicit out-of-scope classes of Remark 7"). Empirical adequacy frameworks built on inductive pattern grids do not guarantee algebraic closure even in this bounded sense; NOETHER guarantees it, and the obligation imposed on any user of the framework is then to verify that no `Translate`-reachable MR is dropped by CONSTRUCT-MP, a verification that is mechanical once the algebraic input has been fixed.

The theorem still imposes a checkable obligation on the framework. If an MR is induced by an invariant in $\mathcal{D}(\mathcal{A}_P)$ but cannot be assigned by CONSTRUCT-MP, then at least one component (invariant extraction, Translate, structural equivalence, or the canonical ordering) is underspecified. Thus the result is not an empirical coverage claim; it is a well-formedness and closure guarantee for the downstream construction once the algebraic input has been fixed. The framework's additional theoretical results complement Theorem 1: Theorem 2 bounds the construction's polynomial-time decidability under a finite generating set; Theorem 1′ (Conjecture A), which makes the stronger claim of absolute completeness over arbitrary properties expressible in $\mathcal{A}_P$, is falsified on $\mathcal{A}_{\mathrm{PWR}}$ (§3.7); the falsification identifies five structural obstructions in `Translate`'s signature (Appendix A), with five further candidate dimensions surveyed on $\mathcal{A}_{\mathrm{equi}}$ and $\mathcal{A}_{\mathrm{rel}}$ (§3.6).

Remark 8 (Scope of Theorem 1 for the $\mathcal{B}^*_{\mathrm{rel}}$ block). *The closure result extends to $\mathcal{B}^*_{\mathrm{rel}}$ provided the rewriting rule set $\mathcal{R}_{\mathrm{rel}}$ is finite. For the relational-algebra fragment of TPC-H-class queries the rule set is finite by the standard heuristic-rewrite normal forms (selection-pushdown, projection-pushdown, join-reordering, distinct-elimination) [57, 68], and Theorem 1 therefore covers algebra-induced MRs of this fragment. Query languages with unbounded recursive rewriting (e.g. Datalog with non-terminating rule sets) lie outside this scope and are catalogued as out-of-scope under Remark 3.*

#### *3.2.4 Decidability and complexity.*

Theorem 2 (Decidability). *Suppose $\mathcal{A}_P$ admits a finite generating set $\mathrm{gen}(\mathcal{A}_P)$ of cardinality $n$, with each generator's invariant computation taking time $t_i$. Then $\mathbb{M}(\mathcal{A}_P)$ is computable in time $O\big(n \cdot \max_i t_i \cdot \log n\big)$.*

Remark 9 (Scope of Theorem 2 for the $\mathcal{B}^*_{\mathrm{rel}}$ block). *The polynomial-time bound holds for $\mathcal{B}^*_{\mathrm{rel}}$ when the rewriting rule set $\mathcal{R}_{\mathrm{rel}}$ is finite. For first-order SQL with bag semantics, query equivalence is undecidable in general, but* query equivalence under a fixed rule set $\mathcal{R}_{\mathrm{rel}}$ *is decidable: practical SMT-based solvers verify equivalence on substantial benchmark fractions in the published literature [57, 68]. Theorem 2 therefore covers the rule-set-bounded fragment of relational algebra; outside this fragment, decidability is governed by the underlying first-order theory and is not the framework's responsibility.*

Table 1. Per-generator cost of invariant extraction in each block. The symmetry block ($G$) is split by group regime: finite, finite-dimensional Lie, and finitely generated infinite-discrete (with user-supplied truncation $K$). For Lie groups (e.g. SO(3), $d_G = 3$) the bound $O(|G|^2)$ does not apply because $|G|$ is uncountable; the relevant bound is $O(d_G^2)$ over the Lie-algebra basis. See §3.2.4 for the truncation discussion.

| **Block** | **Regime** | $t_i$ **for one generator** |
|---|---|---|
| | finite group | $O(\|G\|^2)$ (group-orbit fixed-point) |
| $G$ (symmetry) | finite-dim. Lie group | $O(d_G^2)$, $d_G = \dim_{\mathbb{R}} \mathfrak{g}$ |
| | infinite discrete (truncated) | $O(K^2)$ at truncation $\|G\| \le K$ |
| $O_{\le}$ (order) | — | $O(n^2)$ (poset comparison) |
| $T^*$ (self-adjoint) | — | $O(d)$ (inner-product symmetry check) |
| $\mathcal{T}^*$ (time-reversal) | — | $O(1)$ |
| $\mathcal{L}^*$ (limit) | — | $O(\log \frac{1}{\epsilon})$ |
| $\mathcal{D}^*$ (qualitative-dynamics) | — | $O(d)$ |
| $\mathcal{E}^*$ (method-comparison) | — | $O(K^2)$ (over $K$ methods) |

*Per-block invariant-extraction cost.*

*On infinite groups.* The first row of Table 1, $O(|G|^2)$, is the natural bound for finite symmetry groups and is the regime under which Theorem 2 was originally stated. For a Lie group $G$ such as SO(3), $|G|$ is uncountable and the bound is replaced by $O(d_G^2)$, where $d_G = \dim_{\mathbb{R}} \mathfrak{g}$ is the real dimension of $G$'s Lie algebra. In the equivariant-ML instantiation of Section 3.5, $d_{\mathrm{SO}(3)} = 3$; the symmetry-block invariant is computed once over a basis of three infinitesimal generators, and orbit closure is obtained through finite-dimensional linear algebra. For finitely generated infinite discrete groups (e.g. $\mathbb{Z}$, the integer-translation group of an unbounded grid solver), CONSTRUCT-MP requires a separate truncation parameter $K$ at which orbits are enumerated; the cost is $O(K^2)$ per generator, and the framework's user is responsible for justifying that the truncated set captures the program family's structurally relevant invariants. We do not claim Theorem 1's closure result is preserved across truncation: closure is over $\mathrm{MR}(\mathcal{A}_P)$ defined relative to the truncated group, and the user must inherit responsibility for whether this is the intended algebra.

For the Boltzmann instantiation the relevant generators are finite ($n \le 14$, geometric quarter-rotations and energy-group permutations); for the equivariant-ML instantiation the relevant generators are finite-dimensional Lie ($d_{\mathrm{SO}(3)} = 3$, $|\mathfrak{S}_n| = n!$ truncated to a single generator class, $n \le 10$ training-size, depth, and dimension limits). The asymptotic bound is therefore not the binding constraint in either instantiation.

*3.2.5 The principal limitation.* NOETHER replaces inductive grounding with algebraic grounding *downstream of $\mathcal{A}_P$*. Upstream, the distillation of $\mathcal{A}_P$ from a program family remains a human task. This limitation is central to the framework's scope: NOETHER does not automate domain modelling, but it turns the step after domain modelling into a well-defined construction. A second limitation, made explicit in §3.7 and Appendix A, is that even when $\mathcal{A}_P$ is fully specified, `Translate`'s present signature (single-block, first-order $\pi$-template, single partial-order direction, operating on $P(x)$ tuples rather than on operator-spectrum quantities) systematically excludes a class of MRs that engineering practice treats as standard: non-additivity of operator-composition functionals and higher-order mixed parametric dependences. The principal limitation is therefore twofold: the upstream distillation of $\mathcal{A}_P$, and the present signature of `Translate`.

**Boundary of contribution (Section 3.2 restatement)**

The downstream layer is mechanical: Theorem 1 (algebraic closure under `Translate`) and Theorem 2 (polynomial-time decidability) operate on a given block decomposition. The upstream layer remains empirical: the structural decomposition is Hypothesis 1, an open empirical hypothesis with six documented out-of-scope program-family classes (Remark 3). Theorem 1′ (absolute completeness) is open. Sections 3.4, 3.5, and 3.6 instantiate NOETHER on three structurally distinct program families.

### 3.3 The Invariance-Blindness Theorem

Theorem 1 guarantees that `CONSTRUCT-MP` drops no `Translate`-reachable MR; it is, by construction, silent about what the derived MRs can *detect*. This subsection supplies that missing content: a *limiting* characterization of the faults an algebra-induced MR cannot detect. Throughout we work in the linear operator-implementation fault class (Definition 15); scope and assumptions are collected in Remark 12.

Definition 15 (Linear program and fault class). *A* linear program *is $P_L : \mathbb{R}^N \to \mathbb{R}^N$, $P_L(x) = Lx$, indexed by an operator $L \in \Theta := \mathbb{R}^{N\times N}$ ($\dim\Theta = N^2 < \infty$). The reference (correct) program is $P_{L^*}$; an* operator-implementation fault *is $L = L^* + \Delta$ with $\Delta \in \Theta$. (For nonlinear solvers, $\Theta$ is the operator's finite parameter vector and $E_s$ below its first-order linearization.)*

Definition 16 (Structural defect and detection kernel). *Let block $s \in \mathcal{D}(\mathcal{A}_P)$ carry a structure $T_s$ (for $s = G$, a group action $A \in G \le GL(\mathbb{R}^N)$; for $s = T^*$, the inner-product adjoint). The algebra-induced MR $\rho_{\iota,s} = \texttt{Translate}(\iota, s)$ asserts a structural predicate $\Pi_s[P_L]$ over a witness set $W = S_s \times X_0$ ($S_s \subseteq T_s$, $X_0 \subseteq \mathbb{R}^N$). For $s = G$ the per-witness* defect *is the commutator applied to an input,*

$$E_G(L; A, x) \;=\; P_L(Ax) - A\,P_L(x) \;=\; [L, A]\,x,$$

*linear in $L$; for $s = T^*$, $E_{T^*}(L) = L - L^\top$. The* detection kernel *is the set of faults the MR fails to flag,*

$$\ker(\rho_{\iota,s}) \;:=\; \{\, L \in \Theta : E_s(L; w) = 0 \;\; \forall w \in W \,\}.$$

*If $X_0$ spans $\mathbb{R}^N$ then $\ker(\rho_{\iota,s}) = \bigcap_{A\in S_s} \mathcal{C}(A)$ with $\mathcal{C}(A) = \{L : [L, A] = 0\}$ the commutant of $A$.*

Definition 17 (Structure-preserving fault). *$L$ is $T_s$-*compatible *iff it satisfies the full structural identity that validates $\rho_{\iota,s}$: $E_s(L; w) = 0$ for all witnesses $w \in W_s^{\text{full}}$ (for $s = G$, $[L, A] = 0$ for all $A \in T_s$). Write $\mathcal{C}(T_s)$ for the $T_s$-compatible set (the commutant of the whole group, i.e. the equivariant operators).*

Definition 18 (Faithfulness). *A finite witness set $W_s^{\text{test}}$ is* faithful *iff $\ker(\rho_{\iota,s}) = \mathcal{C}(T_s)$; equivalently, since $E_s$ is linear in $L$, $\operatorname{rank} J_{\text{test}} = \operatorname{rank} J_{\text{full}}$, where $J$ collects the rows $L \mapsto E_s(L; w)$.*

LEMMA 1 (REACHABILITY OF FAITHFULNESS). *If $\Theta$ is finite-dimensional and $E_s$ is linear in $L$, then a finite faithful witness set $W_s^{\text{test}}$ exists.*

PROOF. Each witness $w$ contributes a linear functional $a_w : L \mapsto E_s(L;w) \in \Theta^*$. The span $V_{\text{full}} = \text{span}\{a_w : w \in W_s^{\text{full}}\} \subseteq \Theta^*$ is a subspace of the finite-dimensional dual, hence finite-dimensional; pick finitely many witnesses $W_s^{\text{test}}$ whose functionals span $V_{\text{full}}$. As $W_s^{\text{test}} \subseteq W_s^{\text{full}}$ the intercepts agree, so the two systems have the same solution set, i.e. $\ker(\rho_{\iota,s}) = \mathcal{C}(T_s)$. □

Lemma 1 is the crux: tightness is *not* tautological but follows from finite fault-dimensionality, so a finite executable MR can pin the exact blind set. We now characterize that set.

THEOREM 3 (INVARIANCE-BLINDNESS, $G$ AND $T^*$ BLOCKS). *Let the fault class be linear (Definition 15), let $E_s$ be linear in $L$ (holds for $s \in \{G, T^*\}$), and let the test witnesses be faithful (Definition 18). Then*

$$\ker(\rho_{\iota,s}) \;=\; \mathcal{C}(T_s) \;=\; \{\, L : \; P_L \textit{ preserves } T_s \,\}.$$

*That is, a linear operator-implementation fault survives the structural MR* if and only if *it preserves the structure $T_s$ that the MR exploits.*

PROOF. ($\supseteq$) If $L \in \mathcal{C}(T_s)$ then $E_s(L;w) = 0$ on $W_s^{\text{full}} \supseteq W_s^{\text{test}}$, so $\rho_{\iota,s}$ passes and $L \in \ker$. ($\subseteq$) If $L \in \ker$ then $E_s(L;w) = 0$ on $W_s^{\text{test}}$; by faithfulness (Definition 18) $\ker(\rho_{\iota,s}) = \mathcal{C}(T_s)$, so $L \in \mathcal{C}(T_s)$. □

REMARK 10 (PER-METAPATTERN GRANULARITY). *Under Step 3 of `CONSTRUCT-MP`, each MetaPattern $m_{s,[\iota]}$ corresponds to a single $\sim_s$-equivalence class of invariants; the per-invariant statement of Theorem 3 therefore applies* directly *to each $m_{s,[\iota]}$ without aggregation. When the per-instance grouping of Remark 1 treats a non-empty block $s$ as a single $\sim_s$ class, $m_{s,[\iota]}$ coincides with the block-level MetaPattern reported in the empirical sections.*

REMARK 11 (WORKED INSTANCE: ADVECTION–DIFFUSION, TRANSLATION BLOCK). *On a periodic grid the constant-coefficient operator $c \cdot \nabla + \alpha\nabla^2$ commutes with every grid translation for* all *constants $(c, \alpha)$; hence $E_G(L;\textit{shift}) \equiv 0$ on the whole constant-coefficient family, so $\ker(\rho_{\iota,G})$ contains it, including the advection-speed fault $c \mapsto c'$. The translation MR is therefore* blind *to advection-speed errors, matching the secondary executability sanity check discussed in §5. A spatially inhomogeneous coefficient $c(x)$ breaks translation equivariance, so $E_G \neq 0$ and the fault is detected. A faithfulness check (one generator plus an input basis is unisolvent) is reported in supplementary S10.*

COROLLARY (SINGLE-BLOCK INCOMPLETENESS). *For any block $s$ with $\mathcal{C}(T_s) \neq \{L^*\}$ (a nontrivial structure- preserving fault exists, noting $\mathcal{C}(T_s) \supseteq \{L^*\}$ always), the single-block battery $\{\rho_{\iota,s}\}_\iota$ is incomplete: its kernel contains that fault.*

COROLLARY (COMPLETENESS REQUIRES A TRIVIAL JOINT KERNEL). *A family of oracles $\{O_j\}$ detects every nontrivial fault only if $\bigcap_j \ker(O_j) = \{L^*\}$. Completeness therefore requires oracles whose structural kernels intersect trivially, not additional MRs of the same block.*

COROLLARY (DIFFERENTIAL ORACLE IS THE COMPLEMENTARY BLIND SPOT). *The neutral cross-implementation differential oracle $O_{\text{diff}}$, which flags a fault when two independent implementations disagree beyond a discretisation- calibrated tolerance, has kernel $\ker(O_{\text{diff}}) = \{$common-mode faults (identical effect on* *both implementations)}. Since common-mode and $T_s$-compatible are distinct conditions,* $\ker(\rho_{\iota,s}) \cap \ker(O_{\mathrm{diff}})$ *is strictly smaller than either: the MR battery and the differential oracle are complementary, and a fault escapes both only if it is simultaneously structure-preserving and common-mode.*

Remark 12 (Scope and assumptions). ***(R1)*** *Theorem 3 is stated for the linear operator-implementation fault class (Definition 15); it does not extend to arbitrary nonlinear faults.* ***(R2)*** *Detection is exact-arithmetic; an executable MR with tolerance $\tau$ has kernel $\ker_\tau \supseteq \ker$, so faults below $\tau$ are false negatives. Theorem 3 is the $\tau \to 0$ limit; the finite-$\tau$ regime is outside the identification claim and is treated as a limitation in §6.* ***(R3)*** *Linearity of $E_s$ holds for $G$ and $T^*$ (commutator; symmetric part). It fails for $O_{\leq}$ (monotonicity is an inequality, a cone not a subspace), for $\mathcal{T}^*_{\mathrm{rev}}$ (reversibility $RLR^{-1} - L^{-1}$ involves the inverse), and for $\mathcal{L}^*$ (Richardson ratio of norms); for these blocks only the sufficient direction ($\supseteq$) holds, or a linearized subclass.* ***(R4)*** *The sufficient direction is by definition; the content is the $\subseteq$ direction together with Lemma 1 and the kernel characterization.*

### 3.4 Boltzmann instantiation: from transport to diffusion to burnup

This section instantiates NOETHER on the Boltzmann transport equation and traces the construction through neutron transport, diffusion, and burnup.

*3.4.1 The Boltzmann program family and its operator algebra.* The Boltzmann transport equation governs the distribution of neutral particles. In its time-independent eigenvalue form for fission systems:

$$\hat{\Omega}\cdot\nabla\psi(\vec{r},\hat{\Omega},E) + \Sigma_t(\vec{r},E)\psi = \iint \Sigma_s(\vec{r},E'\to E,\hat{\Omega}'\to\hat{\Omega})\,\psi\,dE'd\hat{\Omega}' + \tfrac{1}{k}\,\chi(E)\int \nu\Sigma_f(\vec{r},E')\psi\,dE'. \quad (1)$$

The program-induced operator algebra $\mathcal{A}_{\mathrm{Boltz}}$ contains $\{G_{\mathrm{geom}}, \mathfrak{R}_E, \mathcal{L}_\Sigma, \mathcal{L}_\nu, \mathcal{L}^*, \mathcal{T}, \mathcal{L}_h, \mathcal{D}_{\mathrm{Bate}}, \mathcal{E}_{\mathrm{cmp}}\}$. Decomposed along the eight blocks of Section 3.1.9, $\mathcal{A}_{\mathrm{Boltz}}$ yields entries in seven blocks (with $\mathcal{B}^*_{\mathrm{rel}}$ empty under the absence of idempotent-semiring rewriting structure). Every operator listed has an unambiguous status in the established theory of neutral-particle transport [7, 30].

*3.4.2 Running CONSTRUCT-MP on $\mathcal{A}_{\mathrm{Boltz}}$.* Steps 1–4 of CONSTRUCT-MP yield seven MetaPatterns: $m_{\mathrm{inv}}$ (invariance/equivariance), $m_{\mathrm{mono}}$ (parameter-monotonicity), $m_{\mathrm{adj}}$ (self-adjoint duality / adjoint reciprocity), $m_{\mathrm{rev}}$ (time-reversal compatibility), $m_{\mathrm{conv}}$ (discretisation convergence), $m_{\mathrm{dyn}}$ (qualitative-dynamics shape invariants), and $m_{\mathrm{cmp}}$ (method-comparison error-bound partial orders).

*3.4.3 Relationship to the prior inductive catalogue: refinement plus prediction.*

*Provenance and scope of the inductive catalogue.* The reactor-physics MetaPattern catalogue compared against here was distilled by the present authors from the standard PWR-physics literature [7, 30] as their own prior inductive work; the underlying PWR-MR corpus (supplementary S2) is the authors' own catalogue, not an external corpus drawn from an unrelated team. The relationship reported in this section is therefore best read as a test of *internal vocabulary coherence*: NOETHER's structural decomposition is applied to a catalogue the same team produced inductively, and the question is whether the algebraic reclassification reproduces, refines, or predicts within that corpus. This is internal consistency under a uniform algebraic structure, not external transfer of tacit knowledge from an independent reactor-physics team into the framework. External-transfer evidence (applying NOETHER to a reactor-physics MR corpus authored by an independent team, such as a PARCS V&V suite or IAEA-TECDOC-class catalogue) is committed as follow-up work in supplementary S4 (`future_work.md`) (item (j)). The cross-codebase commons-math pilot of §5.3.8 (item (b.cm)) is the analogous external-transfer test on the Java head-to-head side, at $n = 3$ SUTs and $n = 77$ mutants; the reactor-side analogue remains future work.

A reactor-physics MetaPattern catalogue distilled from the standard PWR-physics literature [7, 30] identifies five MetaPatterns (P1–P5) inductively. The relationship with $\mathbb{M}(\mathcal{A}_{\mathrm{Boltz}})$ is more structured than a simple bijection. NOETHER reproduces three of the prior patterns, refines two on a sounder algebraic basis, and predicts two structurally distinct classes that the inductive catalogue did not isolate:

Table 2. Relationship between prior inductive catalogue and NOETHER deductive output.

| **Prior catalogue** | **NOETHER** | **Relationship** |
|---|---|---|
| P1 conservation/invariance | $m_{\mathrm{inv}}$ | Reproduced: $G$-symmetry invariants project to the same MRs. |
| P2 monotonicity | $m_{\mathrm{mono}}$ | Reproduced. |
| P3 convergence | $m_{\mathrm{conv}}$ | Reproduced. |
| P4 trajectory | $m_{\mathrm{dyn}}$ | **Refined**: inductive P4 conflated qualitative-dynamics with time-reversal. NOETHER places trajectory phenomena in $m_{\mathrm{dyn}}$ (Sturm-type comparison theorems) and exposes the conflation. |
| P5 partial-order/bounding | $m_{\mathrm{cmp}}$ | **Refined**: inductive P5 grouped method-accuracy partial orders with adjoint reciprocity. NOETHER places the former in $m_{\mathrm{cmp}}$ (approximation-theory error bounds) and exposes the distinction. |
| (none) | $m_{\mathrm{adj}}$ | **Predicted**: adjoint-reciprocity MRs derived as a structurally distinct equivalence class. |
| (none) | $m_{\mathrm{rev}}$ | **Predicted**: collisionless-trajectory-reversal MRs derived for the corresponding sub-formulations. |

This is not the structure of a re-coding. NOETHER's output structurally *refines* two prior patterns and *predicts* two additional patterns the inductive method missed. The gain is canonical placement: two previously conflated inductive labels are separated by algebraic source, and two textbook phenomena become mandatory MetaPattern classes once their blocks appear in $\mathcal{D}(\mathcal{A}_{\mathrm{Boltz}})$.

*A note on prediction (and an interpretive caveat).* We do not claim that $m_{\mathrm{adj}}$ (adjoint reciprocity) and $m_{\mathrm{rev}}$ (collisionless time-reversal compatibility) are de novo physical discoveries. Adjoint-flux reciprocity is standard textbook material in transport theory [7, 30], and time-reversal MRs in collisionless transport have been understood in physics for decades. The MT-community-level absence of these MetaPatterns from the prior inductive catalogue reflects which phenomena happened to be canonically encoded in the PWR-MR corpus, not the absence of the underlying physics from the literature.

We must also acknowledge an interpretive caveat about what this "prediction" is and is not. The blocks $T^*$ (self-adjoint) and $\mathcal{T}^*$ (time-reversal) of Section 3.1.9 were themselves curated by inspection of program families that include reactor physics; the operators $\mathcal{T}_{\mathrm{geom}}$ and $\mathcal{T}$ in $\mathcal{A}_{\mathrm{Boltz}}$ were the leading examples motivating the inclusion of those blocks. There is therefore a circularity in the strong reading of "prediction": $T^*$ and $\mathcal{T}^*$ were partly induced from reactor-physics structures, and $m_{\mathrm{adj}}$ and $m_{\mathrm{rev}}$ are then derived from those blocks. The framework does not discover these MetaPatterns de novo. What it does is closer to a uniform *re-projection*: given the structural decomposition (whatever its empirical provenance), CONSTRUCT-MP places adjoint-reciprocity and time-reversal phenomena into structurally distinct equivalence classes that the inductive PWR-MR catalogue had not isolated as such, even though their underlying physics had been individually known to domain experts. The substantive contribution is the *re-classification under a uniform algebraic structure*, which gives a future testing toolchain a principled reason to enumerate $m_{\mathrm{adj}}$ and $m_{\mathrm{rev}}$ alongside $m_{\mathrm{inv}}$, $m_{\mathrm{mono}}$, $m_{\mathrm{conv}}$, $m_{\mathrm{dyn}}$, and $m_{\mathrm{cmp}}$ rather than treating them as ad-hoc additions. Where NOETHER's predictive power is genuine and non-circular is in the deflationary direction (Section 5.5): the framework can also reveal that an existing inductive catalogue *over-counts* structurally distinct patterns. Both directions, re-classification and de-duplication, are systematisation, not discovery.

*Element-wise correspondence.* Table 3 traces seven representative MRs distilled from the standard reactor-physics literature [7, 30] to their NOETHER placement: one per non-empty block ($G$, $O_{\leq}$, $\mathcal{L}^*$, $\mathcal{D}^*$, $\mathcal{E}^*$, selected as the most canonical literature form within each block) plus the two predicted MetaPatterns $m_{\mathrm{adj}}$ and $m_{\mathrm{rev}}$ for which no MR was previously catalogued in this form. The full enumeration of 12 MRs with sub-category coverage (geometric vs. energy-group symmetry within $G$, etc.), source equations, and `Translate` templates is in supplementary S2 (`elementwise_12.md`). A larger PWR-MR corpus underlying the selection protocol is provided as supplementary material S2.

*An independent audit on the supplementary-MR engineering catalogue.* The table of 12 representative MRs is structurally curated. As a coarser breadth check, an independent supplementary-MR engineering catalogue distilled from production reactor-physics codes (orthogonal to the PWR-MR inductive corpus underlying Section 3.4.3) was labelled by three independent large language models against the seven canonical MetaPatterns plus the relational extension of Section 3.1.9, with a fourth label *orphan* reserved for MRs that fit none of the eight classes; the labelling protocol, raw labels, and majority-vote tabulation are released in supplementary material S2 (`18mr_audit/`). Inter-rater agreement on the four-way classification is almost-perfect (Fleiss' $\kappa = 0.857$, $n = 18$ items, $r = 3$ raters, $c = 4$ categories), and 17 of the 18 MRs are placed by majority vote into one of the seven canonical MetaPatterns or $m_{\mathrm{rel}}$ (subsumption 94.4%, Wilson 95% CI [74.2%, 99.0%] on the binary subsumption proportion). The single orphan is the Lipschitz / metric-stability MR, which we treat structurally in Appendix C.5.2 as a candidate ninth block rather than as a counter-example to Theorem 1's scope. The independence of the three labellers is bounded by the LLM-shared-training-data caveat (the three models share substantial pre-training corpora, so $\kappa$ should be read as agreement among similarly-trained but non-coordinated raters rather than among rigorously independent oracles); we report the audit as external corroboration of breadth rather than as an independent verification of correctness, and an independent human inter-rater $\kappa$ is committed as follow-up rather than claimed here. The breadth claim accordingly carries corroborative, not confirmatory, weight; the paper's central results rest on the algebraic derivation and the Invariance-Blindness Theorem, neither of which depends on these LLM-panel labels.

*3.4.4 A Noether-style derivation of $m_{\mathrm{adj}}$.* The framework's name is methodological. Noether's first theorem replaces an empirically curated catalogue of conservation laws with a derivation from the symmetry structure of an action functional [39]. We do not invoke Noether's theorem as a theorem about programs (programs

Table 3. Seven representative MRs from the prior PWR corpus (one canonical MR per non-empty block plus two predicted MetaPatterns), with NOETHER placement; the full enumeration of 12 MRs with per-block sub-category coverage is in supplementary S2 (`elementwise_12.md`).

| MR ID | Plain-text MR | P# | Block | NOETHER MP |
|---|---|---|---|---|
| Bur-Phy-01 | Step-splitting invariance | P1 | $G$ | $m_{\text{inv}}$ |
| Bol-Phy-11 | $\Sigma_a \uparrow \Rightarrow k_{\text{eff}} \downarrow$ | P2 | $O_{\leq}$ | $m_{\text{mono}}$ |
| Dif-Alg-01 | Diamond-difference $h^2$ convergence | P3 | $\mathcal{L}^*$ | $m_{\text{conv}}$ |
| Bur-Phy-08 | Iodine pit qualitative shape | P4 | $\mathcal{D}^*$ | $m_{\text{dyn}}$ |
| Bur-Alg-04 | CRAM no-worse-than TTA | P5 | $\mathcal{E}^*$ | $m_{\text{cmp}}$ |
| *(predicted)* | Adjoint reciprocity | — | $T^*$ | $m_{\text{adj}}$ |
| *(predicted)* | Collisionless reversibility | — | $\mathcal{T}^*$ | $m_{\text{rev}}$ |

do not, in general, possess an action functional in the variational-calculus sense). What the framework does mirror is the *methodological move*: instead of cataloguing observed invariants, derive them from a structural source. We illustrate the move on $m_{\text{adj}}$.

Consider the bilinear form on solution-adjoint pairs

$$\mathcal{F}[\phi, \phi^\dagger] = \langle \phi^\dagger, B\phi \rangle - \langle \phi, B^\dagger \phi^\dagger \rangle, \tag{2}$$

where $B$ is the Boltzmann operator and $B^\dagger$ its formal adjoint under the inner product $\langle f, g \rangle = \int f g \, d\Omega \, dE \, d\mathbf{r}$. The bilinear form $\mathcal{F}$ is identically zero on solutions of the forward and adjoint equations: $B\phi = S$, $B^\dagger \phi^\dagger = S^\dagger$ with appropriate boundary conditions imply $\mathcal{F}[\phi, \phi^\dagger] = \langle \phi^\dagger, S \rangle - \langle \phi, S^\dagger \rangle$, and the standard reciprocity identity makes the right-hand side zero. The vanishing of $\mathcal{F}$ is a *conserved current* in the Noether sense: it is annihilated by the dual symmetry $\phi \leftrightarrow \phi^\dagger, B \leftrightarrow B^\dagger$. CONSTRUCT-MP's Step 1 extracts $\mathcal{F} = 0$ as the $T^*$-block invariant; `Translate` converts it into the executable MR

$$\rho_{\text{adj}} : \quad \left| \langle \phi^\dagger, S \rangle - \langle \phi, S^\dagger \rangle \right| \leq \tau, \tag{3}$$

which the testing harness checks on a forward and an adjoint solver run with externally specified sources $S, S^\dagger$. The MR is therefore a Noether-style consequence of the duality symmetry rather than a manually catalogued reactor-physics property: the symmetry $\phi \leftrightarrow \phi^\dagger$ is the structural source, the bilinear form is the conserved current, and the MR is the conservation law in testable form. The same template applies to $m_{\text{rev}}$ with the involution $\mathcal{T} : \phi(\mathbf{r}, \mathbf{\Omega}, E, t) \mapsto \phi(\mathbf{r}, -\mathbf{\Omega}, E, -t)$ as the symmetry. The template's general statement is: given a symmetry of the program family's operator algebra, CONSTRUCT-MP extracts the invariant of the symmetry and `Translate` converts it into an executable MR.

*3.4.5 Specialisation to neutron transport, diffusion, and burnup.* The Boltzmann formulation is the most general; specialised solvers are obtained by truncating or projecting $\mathcal{A}_{\text{Boltz}}$. Neutron transport solvers retain the full angular dependence; the same seven MetaPatterns apply. Neutron diffusion solvers truncate angular dependence to its $P_1$ approximation: the time-reversal block contracts (diffusion is dissipative), removing $m_{\text{rev}}$. Burnup-coupled solvers introduce the Bateman ODEs; the Bateman operator $e^{At}$ enriches multiple blocks ($G$ semi-group, $O_{\leq}$ linearity, $\mathcal{D}^*$ overshoot/S-curve, $\mathcal{E}^*$ CRAM-vs-TTA bounds).

The MetaPattern set is *compositional* with respect to the underlying algebra: add a block, gain MetaPatterns; remove a block, lose them. The framework supports both construction and *prediction* of which patterns a new specialisation will exhibit, before any MR is identified empirically.

*3.4.6 Summary.* NOETHER, applied to $\mathcal{A}_{\mathrm{Boltz}}$, deductively produces a seven-MetaPattern set that refines the prior inductive catalogue and predicts two additional MetaPatterns ($m_{\mathrm{adj}}$, $m_{\mathrm{rev}}$). The seven MetaPatterns correspond to invariants that have been independently validated in the standard reactor-physics literature: $m_{\mathrm{inv}}$ via Bell & Glasstone's symmetry-of-criticality analysis [7], $m_{\mathrm{mono}}$ via Lewis & Miller's monotonicity arguments for cross-section perturbations [30], $m_{\mathrm{conv}}$ via mesh-convergence theorems standard in computational neutron transport, $m_{\mathrm{adj}}$ via Bell & Glasstone §6.3 adjoint-perturbation theory and Stamm'ler & Abbate's adjoint flux treatment [50], and $m_{\mathrm{rev}}$ via the collisionless time-reversal symmetry of the transport operator [30]. The contribution of NOETHER on this domain is therefore not de-novo MR discovery (these invariants are textbook-canonical) but the algebraic warrant for treating them as a uniform MetaPattern grid; the citation chain above is the published cross-corroboration that the seven NOETHER-derived MetaPatterns are physically real properties of the Boltzmann program family, independent of NOETHER's framework. The next section turns to a more demanding case: a domain in which the inductive catalogue does not yet exist.

### 3.5 Cross-domain demonstration: equivariant machine learning

*3.5.1 The equivariant-ML program family and its operator algebra.* Equivariant neural networks impose by architectural construction that certain symmetries of the input induce predictable transformations of the output [9, 15, 28, 54]. The program family $\mathcal{F}_{\mathrm{equi}}$ comprises classifiers satisfying $f(g \cdot \mathbf{x}) = \rho(g) \cdot f(\mathbf{x})$ for $g \in G = \mathrm{SO}(3) \times \mathfrak{S}_n$. Decomposed along the eight blocks: $G = \{G_{\mathrm{equi}}\}$, $O_{\leq} = \{O_{\leq}^{\mathrm{train}}\}$, $T^* = \{T^*_{\mathrm{att}}\}$, $\mathcal{T}^* = \{\mathcal{T}_{\mathrm{seq}}\}$, $\mathcal{L}^* = \{\mathcal{L}_{\mathrm{train}}, \mathcal{L}_{\mathrm{depth}}, \mathcal{L}_{\mathrm{dim}}\}$, $\mathcal{D}^* = \emptyset$ (for feedforward classifiers), $\mathcal{E}^* = \emptyset$ within a single architecture, $\mathcal{B}^*_{\mathrm{rel}} = \emptyset$ (no idempotent-semiring rewriting on equivariant-classifier outputs).

*3.5.2 Running CONSTRUCT-MP on $\mathcal{A}_{\mathrm{equi}}$.* CONSTRUCT-MP returns

$$\mathbb{M}(\mathcal{A}_{\mathrm{equi}}) = \left\{ m^{\mathrm{eq}}_{\mathrm{inv}},\ m^{\mathrm{eq}}_{\mathrm{mono}},\ m^{\mathrm{eq}}_{\mathrm{adj}},\ m^{\mathrm{eq}}_{\mathrm{rev}},\ m^{\mathrm{eq}}_{\mathrm{conv}} \right\}.$$

The labels mirror those of $\mathbb{M}(\mathcal{A}_{\mathrm{Boltz}})$ but the *content* is domain-specific. This section does not claim empirical validation in machine learning. It checks a narrower transfer claim: once $\mathcal{A}_{\mathrm{equi}}$ is specified, the same downstream construction yields domain-specific MR families without using a reactor-physics corpus.

*Cross-corroboration from the published equivariant-ML literature.* The five MetaPatterns derived above correspond to invariants that have been independently validated in the equivariant-ML literature, providing a citation-based corroboration channel that does not depend on this paper's case study: $m^{\mathrm{eq}}_{\mathrm{inv}}$ ($\mathrm{SO}(3)/\mathfrak{S}_n$ symmetry invariance) is the central design principle of Cohen & Welling's G-CNN [15], Satorras et al.'s EGNN [43], and the Tensor-Field-Network family [54]; $m^{\mathrm{eq}}_{\mathrm{adj}}$ (self-adjoint attention duality) is the kernel-symmetry property exploited by SE(3)-Transformer's Clebsch–Gordan attention [21]; $m^{\mathrm{eq}}_{\mathrm{rev}}$ (training-trajectory time-reversal) is closely related to the reversible-network construction of Gomez et al. [23]; $m^{\mathrm{eq}}_{\mathrm{conv}}$ (training-size convergence) is the implicit invariant tested by training-curve mesh-refinement studies in the broader geometric-deep-learning literature [9]; gauge-equivariant extensions are catalogued in Cohen et al. [14]. NOETHER's contribution on $\mathcal{A}_{\mathrm{equi}}$ is the uniform algebraic warrant: the same downstream construction that derives reactor-physics MetaPatterns also derives equivariant-ML MetaPatterns from $\mathcal{A}_{\mathrm{equi}}$'s structural decomposition, without re-running the empirical induction that produced the individual results above.

*3.5.3 End-to-end derivation: a concrete MR for SE(3)-equivariant point-cloud classification.* To illustrate the framework's generative use, we trace a complete derivation from algebra to executable MR.

*Step 1. System under test.* Let $f : \mathbb{R}^{n\times 3} \to \Delta^{C-1}$ be a point-cloud classifier mapping $n$ three-dimensional points to a probability distribution over $C$ classes.

*Step 2. Distil $\mathcal{A}_{\text{equi}}$.* The relevant blocks are $G = \mathrm{SO}(3) \times \mathfrak{S}_n$ and $\mathcal{L}^*$ for training-size and depth limits.

*Step 3. Run CONSTRUCT-MP, select an invariant.* Within $G$'s symmetry block, fix attention on the rotation invariant: $f(R \cdot \mathbf{x}) = f(\mathbf{x})$ for all $\mathbf{x}$ and $R \in \mathrm{SO}(3)$.

*Step 4. Translate the invariant into an executable MR..*

$$\rho_{\text{rot}} : \quad \forall R \in \mathrm{SO}(3), \quad \left\| f(R \cdot \mathbf{x}) - f(\mathbf{x}) \right\|_\infty \le \tau, \tag{4}$$

with $\tau = 10^{-4}$ for fp32 architectures.

Listing 1. Executable MR $\rho_{\text{rot}}$ for SE(3)-equivariant classifier.

*Step 5. Generate an executable test.*

```python
import numpy as np
from scipy.spatial.transform import Rotation

def test_rotation_invariance(model, point_cloud, num_samples=100, tau=1e-4):
    """Executable MR rho_rot for SE(3)-equivariant classifier."""
    p_original = model.predict(point_cloud)
    failures = []
    for _ in range(num_samples):
        R = Rotation.random().as_matrix()
        rotated = point_cloud @ R.T
        p_rotated = model.predict(rotated)
        deviation = np.max(np.abs(p_original - p_rotated))
        if deviation > tau:
            failures.append((R, deviation))
    return failures
```

*Step 5b. A Noether-style reading of $\rho_{\text{rot}}$.* The same methodological move that produced $m_{\text{adj}}$ in §3.4.4 produces $\rho_{\text{rot}}$ here in non-rhetorical form. The classifier $f$ is, in equivariant-network architectures by construction, invariant under the action of $G = \mathrm{SO}(3)$ on its input: $f(R \cdot \mathbf{x}) = f(\mathbf{x})$ for all $R \in G$. The Lie algebra $\mathfrak{g} = \mathfrak{so}(3) = \mathrm{span}\{L_x, L_y, L_z\}$ is the infinitesimal generator of this symmetry. The associated Noether-style invariant is the constancy of the classifier output along orbits of $G$: $\frac{d}{d\theta} f(\exp(\theta L_a) \cdot \mathbf{x}) = 0$ for each generator $L_a$, $a \in \{x, y, z\}$, which is the Lie-algebraic statement that the directional derivative of $f$ along every orbit-tangent direction vanishes. CONSTRUCT-MP's Step 1 extracts this invariant from the $G$ block of $\mathcal{A}_{\text{equi}}$; `Translate` converts it into the executable MR of equation (4) above. The MR is therefore not an arbitrary catalogue entry: it is the Noether-style consequence of the SO(3) symmetry that the architecture imposes by construction.

*Step 6. Pattern coverage status.* The MR $\rho_{\text{rot}}$ above, together with $\rho_{\text{perm}}$ and $\rho_{\text{train}}$ derived in supplementary S9 (Appendix D), populates three of the five non-empty MetaPatterns of $\mathbb{M}(\mathcal{A}_{\text{equi}})$. We refrain from reporting a percentage coverage figure: with so few non-empty blocks and a denominator subject to the structural sufficiency hypothesis (Hypothesis 1), a percentage is more rhetorical than informative. We instead derive in the following two subsections two MRs from the remaining non-empty blocks, $T^*$ (self-adjoint) and $\mathcal{T}^*$ (time-reversal), chosen specifically because they are not standard practice in equivariant-ML testing and therefore probe whether NOETHER's transfer claim extends beyond well-known invariances such as $\rho_{\text{rot}}$.

*3.5.4 An adjoint-attention duality MR ($\rho_{\text{adj}}$).* The self-adjoint block $T^*$ in $\mathcal{A}_{\text{equi}}$ contains the attention-kernel symmetriser $T^*_{\text{att}}$. We give two formulations, distinguished by whether the MR runs at CI-time (production-suitable) or at debug-time (one-off scaffolding).

*Scope: forward-pass-only, CI-time formulation.* For architectures whose attention layer exposes a bilinear form $A(\mathbf{x}^{(1)}, \mathbf{x}^{(2)})$ readable through a forward hook, $\rho_{\text{adj}}$ as defined below is a *CI-time MR*: it consumes only the model's native forward pass, requires no parameter mutation, and respects the model's standard inference contract. Mainstream equivariant transformers, including SE(3)-Transformer [21] and the Tensor-Field-Network family [54], compute attention via Clebsch–Gordan tensor products of irrep features. The bilinear form $A(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) = \langle Q(\mathbf{x}^{(1)}), K(\mathbf{x}^{(2)})\rangle$ is generically not Hermitian, but its Hermitian part $\frac{1}{2}(A + A^\dagger)$ has a trace-cyclic invariant under input role-swap that $\rho_{\text{adj}}$ tests. The CI-time formulation reads only the layer's existing $Q, K$ outputs through a frozen forward pass; no probe is injected.

*Alternative harness-time formulation.* A debug-time formulation, available in supplementary S1, instruments a symmetric Gram-matrix probe ($\frac{1}{2}(QK + (QK)^\top)$). This is a debug-time scaffold; it is not part of the CI-time MR set and is not used in the §5.2 comparative evaluation.

*Invariant.* Within $T^*_{\text{att}}$, fix the attention-trace invariant

$$\iota_{\text{att}} = \operatorname{Tr} A(\cdot,\cdot) \in \mathbb{R},$$

which equals $A^\dagger$'s trace by the cyclic property of trace and the Hermiticity of $K$. `Translate` carries $\iota_{\text{att}}$ to an executable MR over a forward pass:

$$\rho_{\text{adj}}: \quad \left|\operatorname{Tr} A(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) - \operatorname{Tr} A(\mathbf{x}^{(2)}, \mathbf{x}^{(1)})\right| \le \tau_{\text{adj}}, \tag{5}$$

for two input clouds $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}$ presented to the network in opposite query/key roles. With $\tau_{\text{adj}} = 10^{-4}$ for fp32 architectures, this MR is executable on any equivariant transformer whose attention layer exposes the bilinear form (or can be probed through a forward-hook). To the best of our knowledge $\rho_{\text{adj}}$ has not been catalogued as a standard MR for equivariant attention testing in the existing literature [3, 46, 60], although Hermitian-attention diagnostics have appeared in the architecture-design literature in non-MR form. NOETHER's contribution here is the algebraic warrant for treating it as a MetaPattern member structurally distinct from $m^{\text{eq}}_{\text{inv}}$.

*3.5.5 A training-trajectory time-reversal MR ($\rho_{\text{train-rev}}$).* The time-reversal block $\mathcal{T}^*$ in $\mathcal{A}_{\text{equi}}$ is non-empty for SGD trajectories under the Hamiltonian-Monte-Carlo / continuous-time-flow view of stochastic optimisation [10]. The discretised SGD update $\theta_{t+1} = \theta_t - \eta \nabla\mathcal{L}(\theta_t; \xi_t)$ is, to leading order in the learning rate $\eta$ and in the absence of momentum or noise, time-reversible: applying the inverse update $\theta_{t+1} \mapsto \theta_{t+1} + \eta\,\nabla\mathcal{L}(\theta_{t+1};\xi_t)$ recovers $\theta_t$ up to $O(\eta^2)$. The operator $\mathcal{T}_{\text{seq}} \in \mathcal{T}^*$ encodes this involution.

*Invariant.* The relevant invariant is the round-trip identity:

$$\iota_{\text{rev}}: \quad \big(\mathcal{T}_{\text{seq}} \circ U_\eta \circ \mathcal{T}_{\text{seq}}^{-1} \circ U_\eta\big)(\theta) = \theta + O(\eta^2),$$

where $U_\eta$ denotes one SGD step.

**Debug-time MR (not CI).** We label $\rho_{\text{train-rev}}$ explicitly as a debug-time MR: it requires parameter rollback, mini-batch reordering, and a deliberately constructed vanilla-SGD fixture. Production equivariant pipelines (Allegro, NequIP, MACE, e3nn examples) almost universally use Adam or AdamW, whose update is not in $\mathcal{T}^*$ in the strict sense; the MR *fails by construction* on those optimisers, which is the framework's correct prediction. $\rho_{\text{train-rev}}$ should be invoked once per training-script change at debug time, on a vanilla-SGD fixture, not as part of CI for production pipelines. The debug-time reading aligns with the wider reversible-network literature [23] where exact-or-approximate inversion is a deliberate test scaffold. `Translate` carries the invariant into:

$$\rho_{\text{train-rev}}: \quad \big\|\theta_T - \tilde{\theta}_T^{(\text{round}-\text{trip})}\big\|_2 \le c\,\eta^2\,T, \tag{6}$$

where $\theta_T$ is the parameter vector after $T$ vanilla SGD steps starting from $\theta_0$, and $\tilde{\theta}_T^{(\text{round}-\text{trip})}$ is the vector obtained by applying $T$ vanilla SGD steps from $\theta_0$ followed by $T$ inverse-SGD steps using the same mini-batch sequence in reversed order, then $T$ forward steps again. The constant $c$ depends on the loss landscape's local Lipschitz structure and is determined empirically per model. This MR *fails by construction* for momentum-based optimisers (Adam, Lion, etc.) whose update is not in $\mathcal{T}^*$; that is, the framework correctly predicts which architecture-optimiser pairs admit the MR and which do not. The MR is non-trivial in the sense that detecting an implementation defect in the gradient-reversal direction (e.g. wrong sign on a custom loss term) is exactly what $\rho_{\text{train-rev}}$ surfaces. To our knowledge it has not been catalogued as an MR for training-pipeline testing.

*Coverage status, revised.* The full Set N consists of five MRs, one per non-empty MetaPattern of $\mathbb{M}(\mathcal{A}_{\text{equi}})$:

$$\text{Set N} = \big\{\, \rho_{\text{rot}}\ (G),\ \rho_{\text{mono}}\ (O_{\le}),\ \rho_{\text{train}}\ (\mathcal{L}^*),\ \rho_{\text{adj}}\ (T^*),\ \rho_{\text{train-rev}}\ (\mathcal{T}^*)\,\big\}.$$

$\rho_{\text{rot}}$ is the rotation-invariance MR of Section 3.5.3; $\rho_{\text{adj}}$ and $\rho_{\text{train-rev}}$ are the non-trivial MRs of Sections 3.5.4–3.5.5; $\rho_{\text{mono}}$ is the point-density-monotonicity MR for the $O_{\le}$ block (top-1 stability under removal of a small fraction of redundant input points), derived from the $O_{\le}^{\text{train}}$ operator restricted to inference-time input perturbations; $\rho_{\text{train}}$ is the inference-idempotency MR for the $\mathcal{L}^*$ block. With all five blocks populated, $\text{coverage}_{\text{NOETHER}}(\text{Set N}) = 1.0$ by construction, this is what Theorem 1 predicts for a $\mathcal{A}_{\text{equi}}$-derived MR set. The permutation-invariance MR $\rho_{\text{perm}}$ (supplementary S9 (Appendix D)) is an auxiliary $G$-block MR retained as a redundant probe and is excluded from Set N to keep the comparison budget $|N| = |L| = |B| = 5$.

The non-trivial nature of $\rho_{\text{adj}}$ and $\rho_{\text{train-rev}}$, both absent from the equivariant-ML MR-testing literature we surveyed, substantiates the claim that NOETHER's transfer is generative rather than nominal.

### 3.6 A third domain: relational query optimisers

Both the Boltzmann (§3.4) and the equivariant-ML (§3.5) instantiations share a common mathematical core: Lie-group symmetry, self-adjoint duality, and time-reversal involution. This is precisely the core that motivated the curation of Hypothesis 1 in the first place. To test transferability beyond this core, this subsection instantiates NOETHER on a domain whose algebraic skeleton is structurally distinct: the relational query optimiser. Relational algebra is built from operators (selection $\sigma$, projection $\pi$, join $\bowtie$, union $\cup$, set/bag-difference) whose equivalence classes are governed by an idempotent semiring with a partial order under containment, not by a Lie group. There is no obvious self-adjoint operator and no time-reversal involution. The third instantiation therefore tests whether the framework yields useful MetaPatterns outside its training image.

*The query-optimiser program family and $\mathcal{A}_{\text{rel}}$.* Let $\mathcal{F}_{\text{rel}}$ be the family of programs that take a SQL query $q$ and a database state $D$ and return a relation $\text{eval}(q, D)$; two queries are equivalent if $\forall D.\, \text{eval}(q, D) = \text{eval}(q', D)$. The operator algebra $\mathcal{A}_{\text{rel}}$ activates $G$ (join / union permutation), $O_{\leq}$ (selection-strengthening + projection-coarsening monotonicity), $\mathcal{E}^*$ (hash- / merge- / nested-loop plan equivalence under stated cost models [36, 57]), and $\mathcal{B}^*_{\text{rel}}$ (Definition 10: selection-pushdown $\sigma_p(R \bowtie S) = \sigma_p(R) \bowtie S$ when $\text{attr}(p) \subseteq \text{attr}(R)$; idempotent $\sigma_p \circ \sigma_p = \sigma_p$; constant-folding $\sigma_{1=1}(R) = R$, $R \bowtie \emptyset = \emptyset$). Blocks $T^*, \mathcal{T}^*, \mathcal{D}^*, \mathcal{L}^*$ are empty under $\mathcal{A}_{\text{rel}}$ in their canonical forms (no inner-product self-adjointness; no time-reversal involution; no qualitative dynamics; no $\epsilon$-limit operator on exact-relational semantics). The full per-block enumeration of $\mathcal{A}_{\text{rel}}$ operators is in supplementary S6 `query_optimiser/algebra_breakdown.md`.

*Running CONSTRUCT-MP on $\mathcal{A}_{\text{rel}}$.* Steps 1–4 yield three MetaPatterns from the block subset: $m^{\text{rel}}_{\text{inv}}$ (input-permutation invariance under set semantics), $m^{\text{rel}}_{\text{mono}}$ (selection-strengthening monotonicity), and $m^{\text{rel}}_{\text{cmp}}$ (plan-equivalence under cost-model bounds). The relational query optimiser is the *canonical* instantiation in which $\mathcal{B}^*_{\text{rel}}$ is non-empty: it activates the eighth block that neither Boltzmann reactor physics nor equivariant ML exercises.

*Four MRs derived for $\mathcal{A}_{\text{rel}}$.*

(1) $\rho_{\text{join-comm}}$ (from $G$): $\text{eval}(q_1 \bowtie q_2, D) = \text{eval}(q_2 \bowtie q_1, D)$ on bag semantics.
(2) $\rho_{\text{select-push}}$ (from $\mathcal{B}^*_{\text{rel}}$): $\text{eval}(\sigma_p(R \bowtie S), D) = \text{eval}(\sigma_p(R) \bowtie S, D)$ when $\text{attr}(p) \subseteq \text{attr}(R)$.
(3) $\rho_{\text{distinct-idem}}$ (from $\mathcal{B}^*_{\text{rel}}$): $\sigma_p \circ \sigma_p = \sigma_p$ as a query-rewrite identity, checkable through plan-tree comparison.
(4) $\rho_{\text{plan-equiv}}$ (from $\mathcal{E}^*$): two execution plans for the same query produce identical relations within stated NULL-propagation rules [57].

*Position relative to existing automated database-testing work.* NOETHER's $\mathcal{B}^*_{\text{rel}}$ instantiation is complementary to four prior lines: random SQL generation [6, 48], automated MR generation for query systems [44], formal query-equivalence solvers [37, 57, 68], and differential / mutation testing [4, 20, 67]; NOETHER provides an algebraically grounded MetaPattern enumeration whose closure under `Translate` (Theorem 1) is a property none of the four lines establishes. A pre-registered protocol comparison on Segura et al.'s IMDb subset [44] is in supplementary S6.

*Cross-corroboration from the published query-equivalence literature.* The four MRs derived above ($\rho_{\text{join-comm}}$, $\rho_{\text{select-push}}$, $\rho_{\text{distinct-idem}}$, $\rho_{\text{plan-equiv}}$) correspond to query-equivalence identities that have been independently validated by formal solvers in the published query-equivalence literature: $\rho_{\text{join-comm}}$ is a canonical bag-semantics commutativity identity exercised by SPES and QED on their benchmark suites [57, 68]; $\rho_{\text{select-push}}$ is the standard selection-pushdown rewrite of Calcite's optimiser, verified equivalence-checkable by SPES; $\rho_{\text{distinct-idem}}$ corresponds to a class of idempotent rewrites that Mohamed et al.'s tables-and-relations SMT theory [37] can certify; $\rho_{\text{plan-equiv}}$ is the central object of QED's 299/444 Calcite verification result [57]. As with the Boltzmann and equivariant-ML cases, NOETHER's contribution on $\mathcal{A}_{\text{rel}}$ is not de-novo MR discovery (these identities are already in the published optimiser-equivalence literature) but the uniform algebraic warrant: the same structural downstream construction that produces reactor-physics and equivariant-ML MetaPatterns also produces relational MetaPatterns, exercising the relational-equivalence block $\mathcal{B}^*_{\text{rel}}$ that the other two domains do not activate. The structural-extension claim, that the framework reaches outside the Lie-group / self-adjoint / time-reversal core, is supported at the algebra-skeleton level by this MR-to-published-identity correspondence, independently of whether NOETHER's $\mathcal{B}^*_{\text{rel}}$ MRs outperform any specific automated baseline.

*What this third domain establishes, and the Theorem* $1'$ *verdict on* $\mathcal{A}_{\text{equi}}$ *and* $\mathcal{A}_{\text{rel}}$. The instantiation establishes that NOETHER applies outside the Lie-group / self-adjoint / time-reversal core: relational optimisers exercise $\mathcal{B}^*_{\text{rel}}$ where the first two domains do not. The counterexample-search protocol applied to both $\mathcal{A}_{\text{equi}}$ and $\mathcal{A}_{\text{rel}}$ (mirroring §3.7) yields the following verdict. *On* $\mathcal{A}_{\text{equi}}$: Theorem $1'$ is falsified by two pairwise-independent candidates: $\rho_{\text{compose}}$ (joint $\mathrm{SO}(3) \times \mathfrak{S}_n$ action on point sets [43], requiring a product-group $\pi$-template) and $\rho_{\text{gauge}}$ (gauge-equivariance on manifolds [14], requiring a bundle-section $\pi$-template parametrised by $\mathbf{g} \in \mathcal{C}(M, H)$). The two extensions are independent (a product-group $\pi$ does not absorb a gauge-bundle section); the SO(3) Lie-algebra structure alone does not force closure on $\mathcal{A}_{\text{equi}}$. *On* $\mathcal{A}_{\text{rel}}$: a survey of $\geq$ 10 unverified cases from [57]'s residue (Aggregate×Project, constant-key, Decorrelate, NULL three-valued logic) yields five Theorem $1'$ candidates; the primary witness $\rho_{\text{agg-proj}}$ (Calcite's `AggregateExtractProjectRule`) requires $\mathcal{B}^*_{\text{rel}}$ plus an aggregation-as-algebra ninth block. Combined with the five $\mathcal{A}_{\text{PWR}}$ obstructions of Table 4, the three algebra-survey artefacts (`theory/equi_thm1prime_search.md`, `theory/rel_thm1prime_search.md`, `theory/translate_extensions.md`) identify ten `Translate`-extension dimensions. Pairwise independence is proved by per-block exhaustion on the five PWR obstructions (Appendix A); the five candidate $\mathcal{A}_{\text{equi}}$ + $\mathcal{A}_{\text{rel}}$ dimensions (two specialising PWR-side dimensions to type-distinct primitives, three net relational-side) are asserted independent by inspection, with formal per-dimension exhaustion as follow-up. Remark 3 catalogues six further out-of-decomposition program-family classes.

### 3.7 A negative instantiation: irreducibly compositional MRs in PWR core simulators

Sections 3.4, 3.5, and 3.6 instantiated NOETHER on three program families (Boltzmann reactor physics, equivariant ML, and relational query optimisers) for which the framework's downstream construction was non-vacuous and produced executable MRs. This subsection instantiates NOETHER on a fourth program family, the PWR core diffusion solver family, with an inverted purpose: rather than demonstrating coverage, we exhibit two specific MRs from the standard PWR safety-analysis literature that the framework's `Translate` operator cannot reach under any single-block derivation. The two MRs together identify five pairwise-independent structural obstructions in `Translate`'s present signature, jointly recasting Theorem $1'$

(Conjecture A, Appendix A) from an open conjecture to a falsified statement on a structurally significant operator algebra.

The MRs chosen are not pathological cases. They are core safety-analysis MRs that PWR core simulators are required by regulatory practice and engineering convention to reproduce: non-additivity of control-bank reactivity worth (the algebraic root of rod-bank shadowing and anti-shadowing phenomena) and second-order mixed dependence of $k_{\text{eff}}$ on moderator temperature and boron concentration (the standard MTC-vs-boron design curve). The negative instantiation thus uses NOETHER's principal application domain (reactor physics) to test the framework's most ambitious stated claim (algebraic closure over arbitrary single-block-derivable MRs).

*Why PWR rather than ML or DB as the negative-instantiation domain.* We choose the PWR core diffusion algebra rather than $\mathcal{A}_{\text{equi}}$ or $\mathcal{A}_{\text{rel}}$ as the negative-instantiation testbed for two reasons. First, regulatory essentiality: 10 CFR 50 and NRC Regulatory Guide 1.77 require PWR core simulators to reproduce the two MRs in the definitions below for safety-analysis qualification, so the counterexamples are not contrived edge cases. Second, engineering documentability: the PWR core diffusion algebra has a published canonical form (Bell & Glasstone [7] §6.1, Lewis & Miller [30] §4.2) against which the counterexample's structural obstructions can be precisely located in $\mathcal{A}_P$'s signature, which $\mathcal{A}_{\text{equi}}$ (a domain of recent literature without a unified canonical algebra) and $\mathcal{A}_{\text{rel}}$ (canonical but under-equipped with non-rewrite operators) do not yet support. Whether $\mathcal{A}_{\text{equi}}$ or $\mathcal{A}_{\text{rel}}$ admit analogous Theorem $1'$ counterexamples is open and committed as follow-up in §3.6's open-question paragraph.

*The PWR core diffusion algebra.* Let $\mathcal{F}_{\text{PWR}}$ be the program family of PWR core diffusion solvers (canonical examples: PARCS, SIMULATE-3/5, ANC, SMART). Its operator algebra $\mathcal{A}_{\text{PWR}}$ contains, in addition to the operators of $\mathcal{A}_{\text{Boltz}}$ (Section 3.4.3), the following PWR-specific generators:

- $\mathcal{O}_{\text{rod}}$: discrete control-rod insertion operators, parametrised by rod-bank label $g$ and insertion depth $d$, generating an additive-on-geometry semigroup under composition. In the steady-state setting we adopt throughout, $\mathcal{O}_{\text{rod}}^A \cdot \mathcal{O}_{\text{rod}}^B$ and $\mathcal{O}_{\text{rod}}^B \cdot \mathcal{O}_{\text{rod}}^A$ act identically on the input space (both produce the same total inserted geometry), so $\mathcal{O}_{\text{rod}}$ is commutative on geometry. The *reactivity-worth functional* on $\mathcal{O}_{\text{rod}}$, however, is not a semigroup homomorphism: $d\rho(A \cup B) \neq d\rho(A) + d\rho(B)$ in general. This non-additivity, not non-commutativity, is what the present subsection exploits.
- $\mathcal{M}_{C_B}$: continuous boration operators acting on the moderator material composition through the soluble-boron concentration $C_B$ (typical PWR operating range: 0–2000 ppm).
- $\mathcal{M}_{T_{\text{mod}}}$: continuous moderator-temperature operators acting on the cross-section library through the parametric dependence $\Sigma(T_{\text{mod}})$ (typical PWR operating range: 290–320°C).

Decomposed along the structural decomposition of Section 3.1.9:

- $G \supseteq \{\mathcal{O}_{\text{rod}}\}$ (treated tentatively as a commutative semigroup; the assignment will be shown below to fail);
- $O_{\leq} \supseteq \{\mathcal{M}_{C_B}, \mathcal{M}_{T_{\text{mod}}}\}$ (parameter-monotonicity operators);
- $T^* \supseteq \{-\nabla \cdot D\nabla + \Sigma_a\}$ (the self-adjoint diffusion operator under isotropic scattering; Bell & Glasstone [7] §6.1, Lewis & Miller [30] §4.2);
- $\mathcal{T}^*_{\text{rev}} = \emptyset$ (PWR diffusion is dissipative);
- $\mathcal{L}^*, \mathcal{D}^*, \mathcal{E}^*$ as in $\mathcal{A}_{\text{Boltz}}$ with appropriate restrictions to the diffusion regime;

- $\mathcal{B}^*_{\mathrm{rel}} = \emptyset$ (no idempotent-semiring structure on PWR core states).

We will show that despite this rich block structure, two specific PWR-safety MRs cannot be derived through `Translate` from any single block.

*Main proposition: non-additivity of rod-bank reactivity worth.*

Definition 19 (Differential rod-bank reactivity worth, exact form). *For a base input $x_0 \in \mathcal{X}$ and a rod-bank operator $A \in \mathcal{O}_{\mathrm{rod}}$, the (positive-convention) reactivity worth of $A$ is*

$$d\rho(A;\, x_0) \;:=\; \frac{1}{k_{\mathrm{eff}}(P(x_0))} \;-\; \frac{1}{k_{\mathrm{eff}}(P(\mathcal{O}^A_{\mathrm{rod}} \cdot x_0))} \;>\; 0,$$

*where $k_{\mathrm{eff}}(P(x))$ denotes the dominant eigenvalue of the diffusion operator at configuration $x$. We write $d\rho(A \cup B;\, x_0)$ when both banks $A$ and $B$ are inserted simultaneously. Equivalently, in conventional reactor-physics notation, $d\rho(A;\, x_0) = \rho(x_0) - \rho(\mathcal{O}^A_{\mathrm{rod}} \cdot x_0)$ where $\rho = 1 - 1/k_{\mathrm{eff}}$ is the static reactivity. The exact form avoids first-order perturbation-theoretic approximation; the standard adjoint-perturbation reading appears below.*

Definition 20 (Non-additivity of rod-bank reactivity worth, $\rho_{\mathrm{nonadd}}$). *For two control-rod banks $A, B \in \mathcal{O}_{\mathrm{rod}}$ with disjoint geometric supports and a base input $x_0$, define the mixed-difference functional*

$$\Delta_{AB}(x_0) \;:=\; d\rho(A \cup B;\, x_0) \;-\; d\rho(A;\, x_0) \;-\; d\rho(B;\, x_0).$$

*The non-additivity metamorphic relation asserts: there exist disjoint-support banks $A, B$ and base input $x_0 \in \mathcal{X}$ in the standard PWR operating envelope (typical multi-bank insertion patterns of D, C, B, A control banks at partial insertions) such that*

$$\rho_{\mathrm{nonadd}}: \quad |\Delta_{AB}(x_0)| \;>\; \tau_{\mathrm{nonadd}},$$

*with $\tau_{\mathrm{nonadd}} = 5$ pcm. The tolerance is calibrated to PWR engineering practice: empirical $|\Delta_{AB}|$ for adjacent rod banks ranges over $10^1$–$10^2$ pcm [50]; the tolerance $\tau_{\mathrm{nonadd}} = 5$ pcm lies safely above PWR core-simulator iterative convergence tolerances (typically 0.1–1 pcm for $k_{\mathrm{eff}}$) and below the physical signal magnitude. Selection of test bank configurations $(A, B)$ is the user's responsibility; the framework's failure to derive $\rho_{\mathrm{nonadd}}$ is independent of the specific tolerance choice.*

*Two physical regimes of $\Delta_{AB}$.* When $\Delta_{AB}(x_0) > 0$ for adjacent or geometrically overlapping rod banks, the standard PWR designation is *positive shadowing* (the worth of the second bank is reduced because the adjoint flux $\phi^\dagger$ is depressed in $A$'s geometric support). Positive shadowing is the dominant regime in conventional PWR analyses; it is the phenomenon explicitly addressed in NRC SER reviews of SIMULATE-3/5, ANC, and PARCS, with typical magnitudes of 50–500 pcm for adjacent bank pairs. When $\Delta_{AB}(x_0) < 0$ for distant banks under asymmetric insertion patterns, the designation is *anti-shadowing*; this is a secondary but routinely measurable regime in standard commercial PWRs (typical magnitudes 5–20 pcm for distant bank pairs in 4-loop Westinghouse and EPR configurations), and is more pronounced in small-core or strongly asymmetric insertion patterns. Both regimes are routinely measured in PWR startup physics testing and are documented in Stamm'ler & Abbate [50] as second-order but non-negligible phenomena that core simulators must reproduce. The MR $\rho_{\mathrm{nonadd}}$ is direction-agnostic and covers both; the test only requires that $|\Delta_{AB}|$ exceed the tolerance, irrespective of sign. Both regimes are accessible to a verifying core simulator regardless of whether the engineering analysis is concerned primarily with one or the other.

*Standard adjoint-perturbation reading (informative, not load-bearing for the proof).* Under first-order perturbation theory [7, §6.3], [30, §4.4], the worth of bank $B$ in the configuration with bank $A$ already inserted is

$$d\rho(B;\, A,\, x_0) \;\approx\; -\frac{\langle \phi_A^\dagger,\, \delta H_B\, \phi_A\rangle}{\langle \phi_A^\dagger,\, F_A\, \phi_A\rangle},$$

where $(\phi_A, \phi_A^\dagger)$ are the forward and adjoint principal eigenfunctions of the $A$-rodded but $B$-unrodded core, $F_A = \chi\nu\Sigma_f$ is the fission source operator at that configuration, and $\delta H_B$ is the operator perturbation produced by inserting $B$ (which generally affects $\Sigma_a$, $\Sigma_t$, and the scattering kernel). The adjoint flux $\phi_A^\dagger$ is the principal eigenfunction of a structurally different adjoint operator $H_A^\dagger$ from $\phi_\emptyset^\dagger$ (the unrodded adjoint): in particular, $\phi_A^\dagger$ is locally depressed in $A$'s geometric support and globally redistributed elsewhere. The non-additivity $\Delta_{AB} \neq 0$ thus follows from $\phi_A^\dagger \neq \phi_\emptyset^\dagger$, which is a consequence of $H_A^\dagger \neq H_\emptyset^\dagger$. The exact form (Definition 19) does not require this perturbation-theoretic reading; the proof below uses only the eigenvalue definitions.

Proposition 1 (Non-additivity is not `Translate`-reachable on $\mathcal{A}_{\mathrm{PWR}}$). *Let $\mathcal{A}_{\mathrm{PWR}}$ be the PWR core diffusion algebra above, with structural decomposition $\mathcal{D}(\mathcal{A}_{\mathrm{PWR}})$. For every block $s \in \mathcal{D}(\mathcal{A}_{\mathrm{PWR}})$ and every invariant $\iota \in \mathcal{I}_s$,* $\mathrm{Translate}(\iota, s) \neq \rho_{\mathrm{nonadd}}$. *Equivalently,* $\rho_{\mathrm{nonadd}} \notin \mathrm{MR}(\mathcal{A}_{\mathrm{PWR}})$ *in the sense of Definition 13.*

The proof, by exhausting the eight blocks against the per-block `Translate` templates of Table 21, is given in Appendix A.

*Engineering significance.* Non-additivity of control-bank reactivity worth is a textbook PWR safety phenomenon. Bell & Glasstone [7, §10.4] and Lewis & Miller [30, §4.4] treat the underlying adjoint-perturbation mechanism; Stamm'ler & Abbate [50] document its operational consequences in PWR rod-worth measurements. Non-additivity is observed in both critical and sub-critical PWR core configurations, with the sub-critical regime exhibiting larger adjoint-flux distortions and correspondingly larger $|\Delta_{AB}|$. PWR core simulators are required by regulatory practice (e.g. NRC SER for SIMULATE-3/5, ANC, PARCS; cf. NRC Regulatory Guide 1.77 [55] on rod-ejection accident analysis, with rod-worth modelling accuracy as a critical input) to reproduce the worth functional with sub-percent accuracy across multi-bank insertion patterns. A framework that cannot, in principle, derive this MR from its algebraic input is missing structural content that PWR engineers routinely test for.

*Supporting proposition: second-order mixed dependence of $k_{\mathrm{eff}}$ on $T_{\mathrm{mod}}$ and $C_B$.*

Definition 21 (MTC-vs-boron mixed-derivative MR, $\rho_{\text{MTC-bor}}$). *Let $k_{\mathrm{eff}}(T_{\mathrm{mod}}, C_B;\, \xi_0)$ denote the dominant eigenvalue of the PWR core diffusion operator as a function of moderator temperature $T_{\mathrm{mod}}$ and soluble-boron concentration $C_B$, holding fixed the auxiliary state $\xi_0 =$* $(\mathrm{BU}_0,\, T_{\mathrm{fuel},0},$ *geometry, loading pattern, rod-bank position*$)$. *Define the static reactivity*

$$\rho_{\mathrm{static}}(T_{\mathrm{mod}}, C_B;\, \xi_0) \;:=\; 1 \;-\; \frac{1}{k_{\mathrm{eff}}(T_{\mathrm{mod}}, C_B;\, \xi_0)},$$

*expressed in pcm units (*$1$ *pcm* $= 10^{-5}$*). The moderator temperature coefficient (MTC) at* $(T_{\mathrm{mod}}, C_B;\, \xi_0)$ *is the partial derivative of static reactivity with respect to moderator temperature, in standard PWR engineering convention [7, §10.3], [49, §3.4]:*

$$\alpha_{\mathrm{MTC}}(T_{\mathrm{mod}}, C_B;\, \xi_0) \;:=\; \left.\frac{\partial \rho_{\mathrm{static}}}{\partial T_{\mathrm{mod}}}\right|_{C_B,\, \xi_0 \textit{ fixed}}, \quad \textit{(units: pcm/°F or pcm/°C).}$$

*The second-order mixed dependence MR asserts: for* $(T_{\mathrm{mod}}, C_B;\, \xi_0)$ *in the standard PWR operating envelope at hot-full-power (HFP), all-rods-out (ARO) reference conditions (typical:* $T_{\mathrm{mod}} \in [290, 320]$ *°C,* $C_B \in [0, 2000]$ *ppm,* $\mathrm{BU}_0 \in [0, 50]$ *GWd/tU, full-power* $T_{\mathrm{fuel},0}$*), the mixed second partial derivative satisfies*

$$\rho_{\text{MTC-bor}}: \quad \left| \frac{\partial^2 \rho_{\mathrm{static}}}{\partial T_{\mathrm{mod}}\, \partial C_B} \right| \;>\; \tau_{\text{MTC-bor}},$$

*with* $\tau_{\text{MTC-bor}} = 0.01$ *pcm/°F/ppm (equivalently* $\sim 1.8 \times 10^{-2}$ *pcm/°C/ppm). The tolerance is calibrated to PWR engineering practice: the empirical value of* $\partial \alpha_{\mathrm{MTC}} / \partial C_B$ *in Westinghouse/Framatome PWR designs ranges over* $0.02$–$0.04$ *pcm/°F/ppm at BOC-to-EOC cycle conditions [49, §3.4], [29, §8.3]; the tolerance* $\tau_{\text{MTC-bor}} = 0.01$ *pcm/°F/ppm lies safely below this physical magnitude and well above PWR core-simulator differential-perturbation noise (typically* $\sim 10^{-3}$ *pcm/°F/ppm for converged eigenvalue calculations).*

*Note on equivalent formulations.* Since $\rho_{\mathrm{static}} = 1 - 1/k_{\mathrm{eff}}$ and $k_{\mathrm{eff}} \approx 1$ at critical PWR conditions, $\partial \rho_{\mathrm{static}} / \partial T_{\mathrm{mod}} = (1/k_{\mathrm{eff}}^2)\, \partial k_{\mathrm{eff}} / \partial T_{\mathrm{mod}} \approx \partial k_{\mathrm{eff}} / \partial T_{\mathrm{mod}}$ to within $\sim 10^{-4}$ relative error. The MR may equivalently be expressed in terms of $|\partial^2 k_{\mathrm{eff}} / (\partial T_{\mathrm{mod}}\, \partial C_B)| > \tau_{\text{MTC-bor}}$ with the same tolerance, modulo this $k$-vs-$\rho$ scaling factor; the algebraic argument of Appendix A (which depends only on $k_{\mathrm{eff}}$ being an operator-spectrum quantity) applies identically to either formulation.

*Equivalent formulation in terms of MTC..* Definition 21 is equivalent to asserting

$$\left| \frac{\partial \alpha_{\mathrm{MTC}}(T_{\mathrm{mod}}, C_B;\, \xi_0)}{\partial C_B} \right| \;>\; \tau_{\text{MTC-bor}}$$

at HFP, ARO conditions. Physically, this captures the well-established PWR design property that MTC becomes monotonically more negative as $C_B$ decreases from BOC values (∼1500 ppm) to EOC values (∼0 ppm). At HFP operating conditions, MTC is regulated to be $\leq 0$ pcm/°F across the entire operating range per 10 CFR 50 Appendix A General Design Criterion 11 [56]; the slope $\partial \alpha_{\mathrm{MTC}} / \partial C_B$ governs how rapidly MTC moves toward more negative values as boron is depleted over the cycle, with typical magnitudes ranging from near-zero (at BOC, high boron) to $-30$ to $-50$ pcm/°F (at EOC, near-zero boron). At hot-zero-power (HZP) conditions, BOC MTC may approach zero or be slightly positive within the analytical envelope (a regime relevant to startup physics testing but not to power-operation safety analysis); HZP MTC is bounded by separate Technical Specifications limits.

The strength of $\partial \alpha_{\mathrm{MTC}} / \partial C_B$ is the engineering target of *MTC-vs-boron concentration curve* calculations performed for every PWR cycle reload, and is governed by the competition between three physical mechanisms [49, §3.4]:

(a) **Reduced moderation**: $T_{\mathrm{mod}} \uparrow \Rightarrow$ moderator density $\downarrow \Rightarrow$ neutron moderation reduced $\Rightarrow$ thermal-flux fraction $\downarrow \Rightarrow$ fission rate $\downarrow$. Contributes a *negative* term to MTC.

(b) **Boron poison evacuation**: at high $C_B$, $T_{\mathrm{mod}} \uparrow \Rightarrow$ moderator density $\downarrow \Rightarrow$ boron number density $\downarrow$ (since boron is dissolved in the moderator) $\Rightarrow$ boron absorption $\downarrow$. Contributes a *positive* term to MTC, partially cancelling (a). This term is proportional to $C_B$ and vanishes as $C_B \to 0$.

(c) **Spectrum hardening and** $^{238}$U **resonance enhancement**: $T_{\mathrm{mod}} \uparrow \Rightarrow$ reduced moderation $\Rightarrow$ harder neutron spectrum $\Rightarrow$ enhanced $^{238}$U resonance absorption (Doppler-weighted by the fuel temperature) $\Rightarrow$ neutron loss $\uparrow$. Contributes a further *negative* term to MTC. This term is small for low-enriched $UO_2$ but becomes significant for MOX fuels and high-enrichment ($> 5\%$) $UO_2$.

At high $C_B$ the partial cancellation from (b) is strong; mechanisms (a) and (c) are partially offset and MTC magnitude is small. At low $C_B$, mechanism (b) vanishes; mechanisms (a) and (c) dominate and MTC becomes strongly negative. The mixed second derivative $\partial^2 \rho_{\mathrm{static}}/(\partial T_{\mathrm{mod}}\, \partial C_B)$ measures the rate at which the boron-mediated cancellation is removed as $C_B$ decreases.

Proposition 2 (MTC-vs-boron mixed dependence is not `Translate`-reachable). *Let $\mathcal{A}_{\mathrm{PWR}}$ be the PWR core diffusion algebra above. For every block $s \in \mathcal{D}(\mathcal{A}_{\mathrm{PWR}})$ and every invariant $\iota \in \mathcal{I}_s$,* $\mathrm{Translate}(\iota, s) \neq \rho_{\text{MTC-bor}}$. *Equivalently, $\rho_{\text{MTC-bor}} \notin \mathrm{MR}(\mathcal{A}_{\mathrm{PWR}})$ in the sense of Definition 13.*

The proof, by reducing the obstruction to the high-order-mixed-difference structure of $\rho_{\text{MTC-bor}}$ and verifying that no per-block $\pi$ template captures such structure, is given in Appendix A.

*Engineering significance.* The *MTC vs. boron concentration curve* is computed for every PWR cycle reload as part of the safety-analysis report submitted to regulators. The curve underlies the moderator temperature coefficient surveillance requirement at hot-full-power, all-rods-out conditions, under which MTC must satisfy a stated upper limit ($\leq 0$ pcm/°F per 10 CFR 50 Appendix A General Design Criterion 11 [56]; specific numerical limits are given in plant-specific Technical Specifications). The slope $\partial\alpha_{\mathrm{MTC}}/\partial C_B$ is the key sensitivity parameter for projecting MTC behaviour across the cycle from a small number of measurement points (typically four-to-six points per cycle, measured at quarter-cycle intervals): it is computed by each cycle-reload core simulator run and reported to the operator. Measurement uncertainties and prediction protocols are documented in ANS 19.6.1 [2]. A core simulator that cannot reproduce the slope to within the tolerance of $\tau_{\text{MTC-bor}}$ would fail the cycle-reload qualification process. The MR is therefore not a textbook curiosity; it is a routine and regulatory-essential output of every PWR core simulator.

*Five independent structural obstructions.* The two propositions are independent in the strong sense that no single extension of NOETHER's `Translate` repairs both:

The five rows in this table identify *five pairwise-independent structural features* of `Translate` that are absent in Definition 12. Each row, considered alone, would require a specific extension of Definition 12; no single extension covers any two rows simultaneously. The structural decomposition (Hypothesis 1) is therefore not the only constraint on the framework's reach; the *shape of `Translate` itself* is.

The two MRs $\rho_{\mathrm{nonadd}}$ and $\rho_{\text{MTC-bor}}$ are also *physically independent*: they probe disjoint physical mechanisms (rod-induced adjoint distortion vs. moderator-poison density coupling), and a PWR core simulator could pass one MR while failing the other (and vice versa). They are therefore complementary verification targets, not duplicates of a single underlying property.

Table 4. Five pairwise-independent structural obstructions in `Translate`'s present signature, identified by Propositions 1–2.

| **Proposition** | **Failure mode** | **Required extension to `Translate`** |
|---|---|---|
| 1 (non-additivity) | Output is an algebraic-spectrum quantity ($k_{\text{eff}}$ as an eigenvalue, not in $\mathcal{Y}$) | Operator-spectrum output relations on $\pi$'s codomain |
| 1 (non-additivity) | Worth functional is non-additive (failure of semigroup homomorphism) | Homomorphism-failure $\pi$-template alongside equivariance / monotonicity / self-adjointness |
| 1 (non-additivity) | Adjoint weighting function $\phi_X^\dagger$ varies with operator history $X$ | Configuration-indexed adjoint structure on $T^*$ |
| 2 (mixed derivative) | MR is a non-zero second-order mixed partial derivative, not a first-order relation | Higher-order mixed-difference $\pi$-templates (currently all $\pi$ templates are first-order) |
| 2 (mixed derivative) | MR involves two independent parameter directions ($T_{\text{mod}}, C_B$) jointly, not chained or single-direction | Two-direction joint parametric dependence beyond the single-$\theta$ partial order of Definition 11 |

*What this subsection establishes and does not establish. Established.* Theorem 1′ (Conjecture A) is false on $\mathcal{A}_{\text{PWR}}$: there exist two specific MRs, each empirically realised on every conforming PWR core simulator and each documented in standard PWR safety-analysis literature and regulatory guidance, that are formulable over $\mathcal{A}_{\text{PWR}}$'s operators but not in MR($\mathcal{A}_{\text{PWR}}$) in the sense of Definition 13. The two MRs identify five pairwise-independent structural obstructions in `Translate`'s present signature.

*Not established.* That Theorem 1 itself fails. Theorem 1's closure result is over MR($\mathcal{A}_{\text{PWR}}$) as defined by Definition 13; the two MRs of this subsection lie *outside* that set, so they are out-of-scope for Theorem 1 and consistent with it. The proper characterisation is that Theorem 1 is a substantially weaker statement than Theorem 1′ pretended to be, and the gap is now exhibited concretely with two independent witness MRs.

*Not established.* That a Composite-`Translate` extension of NOETHER would absorb these two MRs while preserving Theorem 1's closure and Theorem 2's polynomial-time decidability. This is the principal open problem the negative instantiation leaves to follow-up work. The five obstructions of Table 4 are pairwise independent, so any candidate extension must address them jointly rather than sequentially.

## 4 EXPERIMENTS

This section defines the evidence protocol used to evaluate NOETHER as an MR-identification method. The experiments are not designed to estimate average fault-detection superiority. They measure structural MR design-space coverage, origin/boundary explanation, and cross-domain derivability.

*EQ1: expert MR set coverage.* Can NOETHER derive MR classes in operator blocks absent from expert MR sets? This question is answered by binary operator-block coverage: expert coverage means at least one expert MR belongs to the block; NOETHER coverage means at least one explicitly derived NOETHER MR class belongs to the block. SACOS, SPARK, LOCUST, and the reactor-physics MR corpora are used as expert-set evidence where source provenance is explicit.

*EQ2: search-based comparison.* How does NOETHER differ from search-based MR generation in origin explanation, boundary explanation, redundancy/readability, and maintenance? GenMorph and related generated-test/oracle work are used as complementary search-based comparators. Their role is not to serve as an opponent in a universal effectiveness contest, but to clarify what search can discover and what algebraic derivation explains.

*EQ3: cross-domain operator-block sharing.* Can the same operator block explain MR classes across different program families, domains, or solvers? This question is answered through compact derivation traces from Boltzmann/reactor physics, equivariant ML, relational query optimizers, and held-out numerical libraries or production-code corpora where the governing-equation source is available.

*Secondary executability checks.* Mutation, DeepCrime-style, and head-to-head results are retained only as secondary sanity checks: they show that selected derived MR classes can be executed and can expose expected category-specific behavior. They do not decide the paper's main MR-identification claim.

## 5 RESULTS AND DISCUSSION

This section reports the evidence for EQ1–EQ3 and then discusses the implications for MR identification, expert knowledge, search-based generation, and program-family testability. Existing mutation and head-to-head material is interpreted only within the secondary executability scope declared in Section 4.

*Reading guide for the results.* EQ1 is carried by the operator-block coverage material: expert MR sets are compared with NOETHER by binary coverage, not by the number of MRs or the number of faults exposed. EQ2 is carried by the GenMorph/METRIC+/LLM material as a complementarity analysis: search and category-enumeration methods can produce useful candidates, while NOETHER supplies algebraic origin and boundary. EQ3 is carried by the cross-domain derivation traces: the same operator blocks recur across Boltzmann/reactor physics, equivariant ML, relational optimizers, and held-out numerical or industrial program families. The mutation and head-to-head tables that follow are therefore secondary: they document executability, category-specific sanity checks, or complementarity, and must not be read as the main evidence for the paper's contribution.

### 5.1 Primary MR-identification evidence

Tables 5–7 are the load-bearing evidence for EQ1–EQ3. They use the binary counting rule defined in Section 4: a block is covered only when the source contains at least one explicit, applicable MR class from that block. A mathematical structure that exists in a governing equation but cannot yet be made executable through the program interface is recorded as a limitation, not as coverage.

Table 5 should be read conservatively. The S11 industrial corpora support one conclusion: the expert-approved SACOS/SPARK/LOCUST relations are explicit monotone/order relations, whereas NOETHER can re-state that block from the governing-equation algebra and can identify additional monotone/order MR classes in the same production-code families. The table does *not* count latent structures for which no executable interface is available. S12 supplies the complementary evidence: when the program interface exposes the relevant operations, the same derivation protocol covers six operator blocks on held-out numerical libraries.

Together, the three tables support the paper's central claim at the intended level: NOETHER identifies MR classes by making the algebraic origin and boundary explicit, and it covers a broader operator-block

Table 5. EQ1 binary operator-block coverage. Expert coverage means that the expert MR corpus contains at least one MR in the block. NOETHER coverage means that the paper derives at least one explicit MR class in the block. The industrial expert sets are the SACOS/SPARK/LOCUST corpora in supplementary S11; the cross-domain executable checks are the held-out `numpy.linalg` and `numpy.fft` subjects in supplementary S12.

| Operator block | Expert industrial | NOETHER industrial | NOETHER cross-domain |
|---|---|---|---|
| $O_{\leq}$ | 1 | 1 | 1 |
| $G$ | 0 | 0 | 1 |
| $T^*$ | 0 | 0 | 1 |
| $\mathcal{L}^*$ | 0 | 0 | 1 |
| $\mathcal{E}^*$ | 0 | 0 | 1 |
| Conservation | 0 | 0 | 1 |

Table 6. EQ2 origin and boundary comparison against search-based MR generation. The search-based column summarises what can be concluded from the method class used by MR-Scout, GenMorph, and LLM-prompted MR generation; it is not a claim that those systems cannot return useful MRs.

| Criterion | Search/category enumeration | NOETHER |
|---|---|---|
| MR origin | Candidate relations are induced from tests, prompts, corpora, or fitness-guided search. The returned MR need not expose the mathematical invariant that makes it valid. | Each MR class is tied to an operator block and to a source identity or inequality in $\mathcal{A}_P$. |
| Boundary | Failure modes are usually empirical: no candidate found, no mutation killed, or no prompt output accepted. | Boundary is stated structurally: out-of-scope MRs are those outside `Translate`'s signature or outside the current block taxonomy. |
| Redundancy and readability | Generated candidates can be redundant or hard to maintain; this paper treats that as a literature-backed motivation and future empirical study, not as a measured result here. | Equivalent relations are collapsed by block/class membership, so maintainers can inspect the algebraic reason for keeping one class rather than many variants. |
| Role in this paper | Complementary source of executable candidates and sanity checks. | Primary account of MR identification space, origin, and boundary. |

design space than the expert industrial corpora available to us. They do not establish higher mutation score, average fault-detection superiority, or lower maintenance cost in practice.

### 5.2 Secondary executability check: equivariant-ML case study

The derivations in Sections 3.5.3–3.5.5 establish that NOETHER produces concrete, executable MRs in the equivariant-ML setting. This subsection is retained only as a secondary executability and sanity-check study. It shows that selected derived MR classes can be run against a real model and can expose the defect category they were constructed to probe; it is not evidence for the paper's main MR-identification claim and it is not used to claim average-case fault-detection superiority. The study is a case study in the strict sense: a single model, a small mutation set, and three MR sets.

*Subject under test.* We use an E(3)-equivariant graph neural network [43] (EGNN) as a deliberately compact *minimal stand-in* for a full SE(3)-Transformer [21] or a Vector-Neuron-based SO(3)-equivariant

Table 7. EQ3 compact cross-domain traces. The point of the table is not that the same MR text is copied across domains; it is that the same operator block explains why structurally similar MR classes arise in different program families.

| Operator block | Program families | Corresponding MR class / derivation source |
|---|---|---|
| $G$ | Equivariant ML; FFT-based signal processing; linear algebra routines | Group-action invariance/equivariance: rotation equivariance, shift equivariance, and basis-change invariance. |
| $T^*$ | Reactor diffusion operators; matrix decompositions; equivariant attention probes | Adjoint/self-duality relations derived from symmetry of the operator or bilinear form. |
| $O_{\leq}$ | SACOS/SPARK/LOCUST reactor-family analyses; matrix/spectral monotonicity checks | Monotone covariance under order-preserving perturbations of inputs, coefficients, or tolerances. |
| $\mathcal{L}^*$ | Iterative numerical solvers; FFT/truncation pipelines; equivariant-ML resolution checks | Limit/refinement relations under grid, sample, or truncation refinement. |
| $\mathcal{E}^*$ | Linear algebra algorithms; alternative numerical solvers; query optimizers | Equivalence under algorithmic reformulation or algebraic rewriting. |
| Conservation | Reactor balance equations; FFT/Parseval checks; linear algebra trace/norm invariants | Conservation of mass, energy, norm, trace, or equivalent global quantity under a structure-preserving transformation. |

architecture [17]: two EGNN layers, hidden dimension 16, 5,189 parameters, trained on a procedural 5-class point-cloud dataset (sphere, cube surface, torus, cone, helix; 64 points per cloud; 400 training / 100 validation; rotation augmentation) for 25 epochs on Apple M1 Pro / MPS. Validation accuracy after training: 0.93. EGNN carries only invariant scalar and equivariant 3-vector features (type-0 $\oplus$ type-1 in the irrep classification), not the full type-$\ell$ steerable representation of an SE(3)-Transformer; the $T^*$ block instantiation in this case study is therefore an *explicitly added* symmetrised QK probe (`equivariant_classifier.py:qk = nn.Parameter(torch.eye(d))` with the symmetrising operation at use-time), not a property of the EGNN architecture itself. The manuscript's transfer claim is at the operator-algebra level (Section 3.5) and is independent of which equivariant architecture instantiates $\mathcal{A}_{\text{equi}}$; it is the algebraic skeleton that transfers, not architecture-specific empirical numbers. Architecture, training script, dataset generator, frozen checkpoint, and the SHA-256 of the supplementary archive are recorded under S3.

*Three MR sets, controlled for the same prompt and budget.* Three MR sets are compared, each of size five (matching $|\mathbb{M}(\mathcal{A}_{\text{equi}})_{\text{non-empty}}|$):

- **Set N (NOETHER-derived):** $\{\rho_{\text{rot}}, \rho_{\text{mono}}, \rho_{\text{train}}, \rho_{\text{adj}}, \rho_{\text{train-rev}}\}$, one per non-empty block of $\mathcal{A}_{\text{equi}}$, as itemised in the "Coverage status, revised" paragraph of Section 3.5.5.
- **Set L (LLM-prompt baseline):** five MRs generated by prompting GPT-4 with the task description "produce five metamorphic relations for testing an SE(3)-equivariant point-cloud classifier", with no further structural cue. The prompt and full output are reproduced in supplementary material S3.
- **Set B (literature baseline):** five MRs synthesised from MRs reported in the metamorphic-testing-for-ML literature [33, 46, 47], restricted to MRs applicable to point-cloud classifiers. Where the literature provides more than five candidates we select the most-cited.

*Mutation set.* We construct a mutation set of $N_{\mathrm{mut}} = 20$ defects in the model under test, drawn from four categories of fault commonly reported in equivariant-ML implementations [54]: (i) wrong sign on a custom loss term ($n_1 = 5$); (ii) accidental break of equivariance through a non-equivariant intermediate layer ($n_2 = 5$); (iii) numerical-precision degradation, e.g. truncation in the spherical-harmonics computation ($n_3 = 5$); (iv) gradient-reversal sign error in the training script ($n_4 = 5$). Each mutation is implemented as a code-level diff against the reference checkpoint and its hash recorded in S3.

*Metrics.* For each MR set $S \in \{N, L, B\}$ and each mutation $\mu$, we record (i) whether at least one MR in $S$ flags $\mu$ as a fault (*detection*), and (ii) the structural-coverage status of $S$ over $\mathbb{M}(\mathcal{A}_{\mathrm{equi}})$. We report:

- $\mathrm{detection}(S) = |\{\mu : \exists \rho \in S,\ \rho \text{ flags } \mu\}|/N_{\mathrm{mut}}$;
- $\mathrm{coverage}_{\mathrm{NOETHER}}(S, \mathcal{F}_{\mathrm{equi}})$ from Section 5.5;
- the unique-detection count $|\{\mu : S \text{ detects } \mu \text{ and no other set does}\}|$.

*Pre-registered hypotheses.* We pre-register two hypotheses before running the mutation experiments, to make the falsifiability of the case study explicit:

**H1 (coverage):** $\mathrm{coverage}_{\mathrm{NOETHER}}(N) = 1.0$ by construction; $\mathrm{coverage}_{\mathrm{NOETHER}}(L)$ and $\mathrm{coverage}_{\mathrm{NOETHER}}(B)$ are strictly less than 1.0.

**H2 (unique detection):** Set $N$ has at least one mutation it uniquely detects, namely a mutation in category (iv) (gradient-reversal sign error), via $\rho_{\text{train-rev}}$.

H1 is a structural prediction of Theorem 1 and the case study can falsify it only if NOETHER's derivation is incorrect. H2 is a non-trivial empirical prediction: it would be falsified if $L$ or $B$ generated an equivalent training-time-reversal MR, or if the mutation was visible to an invariance MR for some unrelated reason.

*Results.* Table 8 reports the case-study numbers. The full row-level outcome matrix (one row per (mr, mutation) pair, 300 rows in total) is provided as supplementary material S3, together with the runner script (`runner.py`) and the analysis script (`analysis.py`) that produce these numbers deterministically from the trained checkpoint. The model under test is an E(3)-equivariant graph neural network [43] (5-class procedural point-cloud classifier; 5,189 parameters; trained on Apple M1 Pro / MPS; validation accuracy 0.93 after 25 epochs).

Table 8. Results of the small-scale comparative case study (Section 5.2). Trained E(3)-equivariant point-cloud classifier; eight 128-point random test clouds per (MR, mutation) pair. The cat-(iv) row is **construct-validity-controlled**: the mutation category was constructed so that $\rho_{\text{train-rev}}$ alone covers it (one defect category per non-empty block of $\mathcal{A}_{\mathrm{equi}}$), and the 5/5 unique-detection figure therefore exhibits construct validity of $\rho_{\text{train-rev}}$ rather than averaged superiority. **Detection numbers for $\rho_{\mathrm{adj}}$ in Set N use the CI-time forward-pass-only formulation of §3.5.4; the alternative debug-time harness-time formulation is available in supplementary S1 but is not used here.**

| | Set N (NOETHER) | Set L (LLM) | Set B (Lit.) |
|---|---|---|---|
| Detection rate | 7/20 | 2/20 | 0/20 |
| Structural coverage ($\mathrm{coverage}_{\mathrm{NOETHER}}$) | 1.00 | 0.40 | 0.20 |
| Unique detections | 5 | 0 | 0 |
| Detected cat. (i) wrong-sign loss | 0/5 | 0/5 | 0/5 |
| Detected cat. (ii) equivariance break | 2/5 | 2/5 | 0/5 |
| Detected cat. (iii) precision degradation | 0/5 | 0/5 | 0/5 |
| Detected cat. (iv) gradient-reversal sign | 5/5 | 0/5 | 0/5 |

*Hypothesis verdicts.* H1 is **retained as a structural sanity check rather than as a falsifiable hypothesis test**. By the framework's construction, $\text{coverage}_{\text{NOETHER}}(N) = 1.00$ holds before any experiment is run; H1's failure can occur only if one of the derivations in Sections 3.5.3–3.5.5 is itself in error. We accordingly use the $\text{coverage}_{\text{NOETHER}}$ values, 1.00 for Set N, 0.40 for Set L (the $G$ and $\mathcal{L}^*$ blocks are reached by an LLM-prompted MR), and 0.20 for Set B (only $\mathcal{L}^*$ via Shin et al.'s idempotency MR; the other four literature MRs are out-of-scope under $\mathcal{A}_{\text{equi}}$), as a *structural-prior diagnostic*, not as a fault-detection metric: the gap quantifies what the algebraic prior contributes that prompt-based and literature-derived MR sets lack on this particular algebra. The load-bearing comparative result of the case study is H2.

H2 is **consistent with the data, but its verdict is construct-validity-controlled**: Set N uniquely detects all five category-(iv) mutations, and in every case the detector is $\rho_{\text{train-rev}}$. Sets L and B detect zero cat-(iv) mutations: neither corpus contains an MR exercising the SGD-trajectory time-reversal property, which is exactly what NOETHER's $\mathcal{T}^*$ block predicts they would miss without an algebraic warrant. *This contrast exhibits construct validity of* $\rho_{\text{train-rev}}$ *as a gradient-reversal probe, not NOETHER's superiority on a defect distribution sampled neutrally from real-world bug reports* (the mutation set was constructed to cover one defect category per non-empty block of $\mathcal{A}_{\text{equi}}$, so cat-(iv)'s category was selected because $\rho_{\text{train-rev}}$ alone covers it). The unique-detection asymmetry between Set N and Set B is statistically significant at $\alpha = 0.05$ on the case study's mutation set (paired McNemar exact two-sided $p = 0.016$; unpaired Fisher exact $p = 0.008$); between Set N and Set L it is borderline ($p_{\text{McNemar}} = 0.063$, $p_{\text{Fisher}} = 0.13$) given Set L's overlap with Set N on two cat-(ii) mutations. Wilson 95% confidence intervals on detection rates are $[0.18, 0.57]$ for Set N, $[0.03, 0.30]$ for Set L, and $[0.00, 0.16]$ for Set B; the intervals N vs B are non-overlapping. The full pairwise comparison matrix (McNemar $b/c$ counts, Fisher 2×2 tables, Wilson CIs) is in supplementary material S3 (`table4.json::pairwise_stats`).

*Construct-validity caveat for H2.* The mutation set was constructed to cover one defect category per non-empty block of $\mathcal{A}_{\text{equi}}$; in particular, cat-(iv) was selected because it targets the $\mathcal{T}^*$ block that $\rho_{\text{train-rev}}$ alone covers. The 5/5 unique-detection result therefore exhibits *construct validity* of $\rho_{\text{train-rev}}$ as a gradient-reversal probe rather than NOETHER's superiority on a defect distribution sampled neutrally from real-world bug reports. That weaker reading is still informative: it shows that the framework's $\mathcal{T}^*$-derived MR is non-redundant relative to LLM-prompted and literature-derived MRs.

*What is and is not detected: a framework boundary, not an edge case.* The result that no MR set detects category-(i) wrong-sign mutations is the most informative cell of Table 8 and we choose to foreground it as a *framework boundary* rather than an edge case. A sign-flipped classification head still satisfies SO(3)-rotation invariance and permutation invariance (the equivariance contract is intact), still passes inference idempotency, satisfies the symmetrised adjoint identity, and is monotone under point-density sub-sampling. The eight blocks of Hypothesis 1 contain no *label-consistency* block, and the wrong-sign-loss fault class is precisely the kind a label-consistency MR would catch. This is the label-consistency out-of-scope class enumerated in Remark 3: ML program families that ship with labelled training data are a candidate ninth block. We do not absorb this case into "out-of-scope for MR testing" in some over-broad sense; the original motivation for MR testing is precisely the absence of an oracle, so framing label-consistency MRs as fundamentally out-of-scope would over-claim the framework's boundary.

Similarly, cat-(iii) precision-degradation mutations on a 16-dim hidden state are not detectable at our chosen tolerances; tighter $\tau$ would surface them at the cost of more baseline false positives. A tolerance-sensitivity sweep over $\tau \in \{10^{-3}, 10^{-4}, 10^{-5}\}$ is reported in supplementary S3 (`tau_sweep.json`) and shows monotone increase in detection at the cost of monotone increase in baseline false-positive rate. Cat-(ii) equivariance-break mutations are detected by both Set N and Set L through $\rho_{\text{rot}}$ / $L_{\text{rot}}$ (parity is expected: rotation-invariance is the structural property the LLM is most likely to rediscover from the system description; the case study's discriminating cell is cat-(iv)).

*Interpretation conditions, stated in advance.* We commit in advance to the following readings of the eventual numerical outcomes:

- If H1 fails (i.e. $\text{coverage}_{\text{NOETHER}}(N) < 1.0$): one of the derivations in Sections 3.5.3–3.5.5 is incorrect and must be revised; the framework's transfer claim does not stand on this benchmark.
- If H1 holds and H2 fails (i.e. no unique detection in cat. iv): the framework's coverage prediction is correct but the unique-MR claim is not differentiated by this mutation set; the LLM or literature baseline produced an equivalent gradient-reversal probe.
- If H1 and H2 both hold: the case study is consistent with NOETHER's transfer claim. We do *not* claim this would establish superior fault-detection on average; the comparison's denominator (20 mutations, one model) is too small.
- If $\text{detection}(N) < \text{detection}(L)$: this is consistent with the framework's design (NOETHER prioritises structural coverage, not raw detection on a particular mutation set) and is reported transparently rather than concealed.

*5.2.1 A small DeepCrime-style real-fault pilot.* To begin closing the gap between protocol and result, we executed a $n = 5$ pilot extension of the case study with five mutation operators systematically derived from the DeepCrime taxonomy [25] (which extracts 35 mutation operators from real DL fault studies). The pilot operators are post-training mutations on the trained EGNN checkpoint and address category labels analogous to DeepCrime's: cat-v-01 *loss-reduction-like* (head-weight scaled by $1/N_{\text{classes}}$), cat-v-02 *activation change* (tanh saturation), cat-v-03 *layer removal* (head zeroed), cat-v-04 *bias removal*, cat-v-05 *weight re-init* (Glorot). The pilot was run with `runner_pilot.py` against the same N/L/B MR sets, the same test set of eight clouds, and the same tolerance as the main case study. Results are deterministic given the seeds in supplementary S3.

Table 9. DeepCrime-style real-fault pilot, $n = 5$ mutations on the trained EGNN checkpoint. Wilson 95% CIs are given for context; pairwise Fisher-exact $p$-values are reported in supplementary S3 (`deepcrime_pilot_stats.json`).

| Set | Detected / 5 | Wilson 95% CI |
|---|---|---|
| N (NOETHER) | 2/5 (cat-v-01, cat-v-03 via $\rho_{\text{train}}$) | $[0.12, 0.77]$ |
| L (LLM) | 0/5 | $[0.00, 0.43]$ |
| B (Lit.) | 0/5 | $[0.00, 0.43]$ |

*Reading the pilot (inferential verdict).* At $n = 5$ the appropriate paired-binary test is McNemar's exact on the discordant counts $(b, c) = (2, 0)$ for both Set N vs Set L and Set N vs Set B: two-sided exact $p = 0.500$

Table 10. Paired contingency for Set N vs Set L on the $n = 5$ DeepCrime pilot. The Set N vs Set B contingency is identical because Set B also detected $0/5$. The appropriate paired-binary test is McNemar's exact test on the $(b, c) = (2, 0)$ discordant counts: two-sided exact $p = 0.500$ (binomial $X = 0$ in $n = 2$ trials with $p_0 = 0.5$ under $H_0$ of equal Set N / Set L performance); equivalently, an unpaired Fisher exact $2 \times 2$ on rows = Sets, columns = (detected, missed) yields two-sided $p = 0.444$. Both numbers fail to reject $H_0$ at $\alpha = 0.05$ because $n = 5$ is below the threshold at which a $2/5$ vs $0/5$ contrast is inferentially decisive; per-pair statistics in supplementary S3 `deepcrime_pilot_stats.json`.

| | Set L detected | Set L missed | Total |
|---|---|---|---|
| Set N detected | 0 | 2 | 2 |
| Set N missed | 0 | 3 | 3 |
| Total | 0 | 5 | 5 |

(and the unpaired-Fisher analogue yields two-sided $p = 0.444$ on the same contingency); the pilot is therefore underpowered for an inferential conclusion at $\alpha = 0.05$. The 2/5 vs 0/5 vs 0/5 detection contrast is reported as descriptive evidence consistent with the direction of the framework's $\mathcal{L}^*$-block prediction, not as a hypothesis confirmation. The pilot's load-bearing claim is that the comparative-evaluation infrastructure runs end-to-end against real-fault-style mutations on the trained checkpoint; non-vacuousness of the $\mathcal{L}^*$-block prediction and the parameter-distribution candidate-ninth-block reading are candidate evidence requiring a larger $n$.

*Interpretation of the two detection events (mechanism, not inference).* For descriptive context only, the detection mechanism on the two events: $\rho_{\text{train}}$ tests training-size limit invariance, that is, for inputs the model classifies confidently at full training, the prediction should remain stable when the head is exposed to a fresh small-batch fine-tuning step. cat-v-01 (head weight scaled by $1/N_{\text{classes}} = 1/5$) collapses head-output magnitudes uniformly toward zero, softens the softmax, and changes the argmax on inputs near classification boundaries; post-fine-tune predictions drift away from pre-fine-tune predictions and the inference-stability invariant fails. cat-v-03 (head weight zeroed) makes every output identically zero pre-softmax, again breaking inference stability. cat-v-02 (tanh saturation), cat-v-04 (bias removal), and cat-v-05 (Glorot re-init) preserve enough head signal for the inference-stability check to pass on the eight test clouds within the chosen tolerance, so $\rho_{\text{train}}$ does not fire on them. The mechanism is therefore "$\rho_{\text{train}}$'s inference-stability check fails when head magnitude is perturbed past a margin-of-error threshold for boundary inputs", a mechanism-level statement independent of the underpowered sample size.

The pilot is the first empirical handhold beyond the constructed mutation set; the larger comparative-evaluation protocol below remains as committed work for follow-up. The three undetected mutations (cat-v-02 activation change, cat-v-04 bias removal, cat-v-05 weight re-init) are *consistent with, but do not establish*, the parameter-distribution candidate ninth block of Remark 3 item (vi); a panel of $n \geq 20$ mutations on more than one architecture is required to establish the block-extension claim. The pilot's role here is to motivate the candidate, not to confirm it; we report it alongside the metric-stability candidate of Remark 5 as the two most actionable extension targets.

*Threats specific to this case study.* The most material threats are: (a) **MR-budget asymmetry**: GPT-4 might generate more than five MR candidates given a different prompt and a larger budget; (b) **baseline selection**: the literature MR set is restricted by the rule "applicable to point-cloud classifiers", which may exclude relevant MRs; (c) the mutation set is **hand-constructed** rather than mined from real defect logs, and importantly, *constructed to cover one defect category per non-empty block of* $\mathcal{A}_{\mathrm{equi}}$, with cat-(iv) selected because it targets the $\mathcal{T}^*$-block MR $\rho_{\text{train-rev}}$. The 5/5 unique-detection result for cat-(iv) therefore exhibits *construct validity* of $\rho_{\text{train-rev}}$ as a gradient-reversal probe, not NOETHER's superiority on a defect distribution sampled neutrally from real-world bug reports; (d) the case study uses a compact EGNN stand-in for a full SE(3)-Transformer, so architecture-specific empirical numbers do not generalise; and (e) Set L is a single GPT-4 sample at temperature 0 with a fixed seed (the prompt and raw output are recorded in supplementary S3 `prompt_log.md`); a different LLM, prompt, or temperature would yield different MRs and the case study does not characterise that variability. Section 5.2 does not generalise beyond this case-study scope; the comparative evaluation and real-bug protocols below describe the empirical extensions that address threats (a)–(c) directly.

*Comparative evaluation against published baselines (protocol).* The single hand-constructed mutation set above is supplemented by a comparative protocol that runs Set N alongside two independent automated pipelines on a shared subject set. The protocol fixes the following measurable quantities, leaving only the result population to subsequent revisions of this paper.

**Subjects.** (i) GenMorph's published 23 Java-method benchmark [3] restricted to subjects whose induced operator algebra is non-trivial under the structural decomposition (we expect $\approx 14$ subjects after filtering on whether $\mathcal{A}_P$ has at least one non-empty block beyond $G$); (ii) two ML benchmarks built on DeepCrime real-fault mutation operators [25]: an MNIST classifier and a CIFAR-10 classifier under DeepCrime's 24 mutation operators systematically extracted from real DL fault taxonomies. Replacing the hand-constructed 20 mutations with DeepCrime's published real-fault operators directly addresses the construction-bias threat (c) above.

**Compared methods.** Set N (NOETHER, derived from $\mathcal{A}_P$); Set M (MR-Scout-mined [60] from existing test suites); Set G (GenMorph-evolved via genetic programming [3]); Set L (LLM-prompted with the same prompt template as in the §5.2 case study); Set B (literature MRs).

**Metrics.** Per-subject mutation-detection rate (Set $\cap$ killed mutants / total killed); real-bug-detection rate (where bug logs are available); false-positive rate on baseline (un-mutated) inputs; coverage with respect to the algebraic block decomposition $\text{coverage}_{\text{NOETHER}}$.

**Statistical tests.** Wilson 95% CI on per-set detection rates; pairwise McNemar exact tests for paired Set-vs-Set comparisons; pairwise Fisher exact tests for unpaired comparisons; Bonferroni correction for the 10 pairwise tests across 5 sets.

**Pre-registered hypothesis (H3a, vs GP-evolved baseline).** Three sub-claims:

**H3a.1 (Detection on D1, per-block).** On the algebra-disrupting stratum (D1, Definition in §5.3.2), Set N's per-block kill rate is competitive with or superior to Set G's on at least one operative block within scope (per-block comparison; aggregate D1 dominance is *not* pre-registered as the load-bearing reading, see §5.3.7).

**H3a.2 (Complementarity).** Set N and Set G exhibit non-trivial complementarity in mutant coverage on the D1 stratum (each set kills mutants the other misses); a non-zero $N$-only kill count on at least one operative block is the operational test.

**H3a.3 (Cost-axis).** Set N's MR-generation cost is asymptotically lower than Set G's (Theorem 2 polynomial-time decidability versus Set G's $\approx$ 30-min stochastic GP search per SUT, detailed in Table 17). The cost-axis claim is independent of the detection-axis claims and is read on the cost-axis only.

**Pre-registered hypothesis (H3b, vs LLM-assisted baseline).** On the structural-coverage diagnostic, $\text{coverage}_{\text{NOETHER}}(N) > \text{coverage}_{\text{NOETHER}}(L_{\text{ensemble}})$, where Set $L_{\text{ensemble}}$ is the LLM-assisted representative drawn from a multi-vendor $\times$ multi-temperature LLM-MR-generation protocol that subsumes the single-sample probes of Shin [47] and Zhang et al. [66]. The diagnostic is interpreted as a structural-prior signal (NOETHER's algebraic prior contributes block coverage that prompt-based LLMs lack on $\mathcal{A}_P$), *not* as a fault-detection-superiority metric. **Verdict (tested, borderline-PASS at the structural reading)**: a 2-vendor (DeepSeek, ChatGPT) $\times$ 5-temperature ensemble harvested 487 LLM-proposed MRs across the 10 §5.3.3 SUTs (100 samples); 43.5% (212/487) of the proposals match a Set N block-template under deterministic translation, and the matchable subset reproduces Set N's kill rate on the aligned-mutant substrate (34/70 = 0.486 for the ensemble union, identical to Set N on the same 70 mutants). The remaining 56.5% explore invariants outside the 8-block frame (associativity, identity / inverse element, monotonicity on SUTs lacking an $\text{O}_{\text{le}}$-block representative); these have no in-protocol kill measurement here and constitute the structural-coverage gap. NOETHER's $\text{coverage}_{\text{NOETHER}}$ remains 1.00 by construction; the LLM ensemble's block-template coverage of Set N's catalogue is 43.5% on the matchable subset. A third-vendor (Anthropic Claude) replication is committed in follow-up (d.set-l-claude) of supplementary S4 (`future_work.md`).

**Pre-registered hypothesis (H3c, vs mining-based baseline).** At cold start (no seed test corpus), Set N is operationally derivable from $\mathcal{A}_P$ alone, while MR-Scout [60]'s reach is upper-bounded by the seed-suite's induced relations; an adapted-from-published-artifact estimate of MR-Scout's reach on the §5.3.3 substrate is reported in Table 17 and contextualised in §5.3.7.

**Pre-registered hypothesis (continuous).** H4 (*detection-rate non-inferiority on real faults*): on DeepCrime real-fault mutants, Set N's detection rate is within $\Delta = 0.10$ of the best non-NOETHER set's detection rate, with $\Delta$ fixed in advance.

The harness for this protocol is provided as supplementary S3 `comparative_baseline/` with adapter scripts for each baseline; replication of the protocol on a different subject set requires only swapping the subject directory.

*Real-bug evaluation (protocol).* To address threat (c) directly, we mine cat-(i)–(iv) faults from public bug reports of e3nn [22] and PyTorch Geometric [18], the two reference SE(3)-equivariant libraries. Target: 10 confirmed real-bug commits with associated test cases, one per cat-(i)–(iv) where available. For each bug:
(1) the pre-fix source code is checked out into a frozen reference checkpoint;
(2) all five MR sets (N, M, G, L, B) are run against the buggy code on the test inputs from the bug report;
(3) detection is recorded as "MR fired = True" iff at least one MR in the set surfaces the buggy behaviour at the published tolerance.

The construct-validity caveat at the start of this subsection then naturally resolves: defects are no longer "selected to cover one block per defect category"; they are mined from a fixed, pre-existing defect distribution that the framework was *not* designed against. The protocol is provided as supplementary S5 `real_bugs/` with one folder per bug containing the issue link, fix commit, test fixture, and per-MR detection outcome.

### 5.3 An empirical test of the structural decomposition: $\mathcal{L}^*$-block blindness on homogeneity-preserving mutators

A generative framework that cannot make falsifiable quantitative predictions is rhetoric, not science. The previous two sections establish that NOETHER *generates* MRs in two domains with structurally distinct operator algebras (§3.4, §3.5); generation alone does not test the framework's central methodological claim, that the structural decomposition (§3.1.9) is empirically operative as a mechanism rather than a derivation-bookkeeping device. This section reports an empirical test of one quantitative consequence of that claim. NOETHER predicts that an MR derived from the linearity-and-scaling block ($\mathcal{L}^*$) is necessarily silent on any mutator that preserves homogeneity of degree 1. PIT's default mutator set has this property by direct calculation. The prediction is therefore that the $\mathcal{L}^*$-block MRs in our SUT set will kill near-zero PIT mutants. We refer to this prediction as *$\mathcal{L}^*$-block blindness*. The prediction is sharp, quantitative, and derivable from public information without consulting any data. The test of the prediction is the central content of this section. Pooled head-to-head numbers against an automated SOTA baseline (GenMorph [3]), structural coverage extension, and cross-pipeline MR rediscovery are reported as corroborating evidence; the section's central claim does not depend on them.

#### 5.3.1 *The prediction: $\mathcal{L}^*$-block blindness.*

*The MR shape.* The $\mathcal{L}^*$ block of $\mathcal{A}_P$ under Hypothesis 1 contains scaling and linearity operators. CONSTRUCT-MP's `Translate` step (Definition 13) carries scaling-block invariants to MRs of the form

$$L_{\text{scale}}: \quad f(\lambda \mathbf{x}) \;=\; \lambda \cdot f(\mathbf{x}), \qquad \lambda > 0, \tag{7}$$

on positively-homogeneous-of-degree-1 programs. The shape is fixed by $\mathcal{A}_P$ structure; it is not chosen with mutation testing in mind.

*The mutator semantics.* PIT 1.7.4 [16] ships a default mutator set comprising seven canonical operator classes: arithmetic-operator swap (AOR; $+ \leftrightarrow -$, $\times \leftrightarrow \div$), conditional boundary mutation, increment mutation, return-value swap (`return zero`, `return one`, `return null`), conditional negation (NCM), constant replacement, and member-call removal. Each AOR mutation that preserves the program's domain of definition transforms a homogeneous-of-degree-1 function $f$ into another homogeneous-of-degree-1 function $f'$, by the elementary fact that the four arithmetic operators in question commute with positive scalar multiplication of all inputs in the relevant homogeneous case (addition and subtraction are degree-1 homogeneous; multiplication and division shift degrees by $\pm 1$, but the swap $\times \leftrightarrow \div$ on a properly typed expression returns a degree-preserving result on the homogeneous-of-degree-1 sub-grammar).

*The prediction.* A composite of these two facts gives the central prediction. If $f$ satisfies $L_{\text{scale}}$ and the mutator $\mu$ preserves homogeneity of degree 1, then $\mu(f)$ also satisfies $L_{\text{scale}}$. Therefore the paired-MR test

$$\left|f(\lambda \mathbf{x}) - \lambda f(\mathbf{x})\right| \;\le\; \tau \quad \text{vs.} \quad \left|\mu(f)(\lambda \mathbf{x}) - \lambda \mu(f)(\mathbf{x})\right| \;\le\; \tau$$

returns the same verdict on $f$ and on $\mu(f)$ for any $\mathbf{x}$, $\lambda$. The kill rate of $L_{\text{scale}}$ against the homogeneity-preserving subset of PIT's mutators is *identically zero* by direct calculation, and against the full default mutator set is approximately zero up to a small fraction contributed by mutators that break homogeneity (return-value swap to `zero` or `one`; conditional-negation when the conditional carries a constant offset; constant replacement when the constant participates additively).

*Falsifiability.* The prediction is falsified if the observed $L_{\text{scale}}$ kill rate is materially above zero on a substrate where the prediction's homogeneity precondition is met. We define "materially above zero" in advance as a single $L_{\text{scale}}$ MR killing a third or more of its PIT mutants on more than one of the SUTs in the substrate. The prediction passes if the observed kill rate is $\leq 1/3$ on at least five of the six SUTs admitting an $L_{\text{scale}}$ MR. The 1/3 threshold and the "more than one SUT" quantifier were committed to git (`configs/d4j_algebra_rich_criterion.json`) before the per-MR kill-count files; under threshold sensitivity in the grid $\{1/4, 1/3, 1/2\} \times$ {more than zero, more than one, more than two}, the verdict remains *Confirmed* on 5/6 SUTs at all 9 grid cells, with `hypotSig` as the single SUT crossing every threshold; the result is therefore robust to plausible threshold variation. *Outlier-handling rule.* An outlier SUT (kill rate $> 1/3$) is rescued from falsification only if all of its killed mutants are independently classified as homogeneity-breaking under the framework's mutator-semantics taxonomy (MATH-swap of $\times \leftrightarrow \div$ on bivariate degree-1 inputs $\Rightarrow$ degree-changing; RC-replacement of $\sqrt{x^2+y^2}$ with a constant $\Rightarrow$ fixed-output breaking scale-invariance; VR-unconditional-zero return $\Rightarrow$ fixed-output sub-rule). Per-mutant classification must be logged alongside the kill data, and the classification must precede inspection of which mutants the outlier's rescue depends on. The rule was codified in the pre-registration config on 2026-05-15 to make explicit a rule on which the §5.3.4 `hypotSig` analysis had relied in implicit rather than written form. Under the codified rule, `hypotSig`'s two killed mutants (`return_zero_doubles_VR` and `Math.sqrt_replaced_with_one_RC`) both classify as homogeneity-breaking, so the original 5/6 verdict stands; future cross-codebase substrates inherit the rule as a written test.

*Public-information argument.* Equation (7) is a consequence of $\mathcal{A}_P$ structure and CONSTRUCT-MP's `Translate` step, both fully specified in §3.2. PIT's default mutator set is public [16]. The composition of the two is a mathematical consequence requiring no experimental input. The prediction is therefore ex-ante in the strong sense that it is *derivable from public information without consulting any data this paper produces.* We track the prediction's commitment to git in supplementary S7 (`d4j/`) alongside the SUT-selection criterion of §5.3.3.

*5.3.2 PIT mutator and structural invariant compatibility.* The $\mathcal{L}^*$-blindness prediction generalises: each PIT default mutator either preserves or breaks the invariant of each NOETHER block, and an MR derived from a block fires on a mutant only if that mutant breaks the block's invariant. Table 11 maps the seven PIT default mutator categories [16] against the eight NOETHER blocks and gives the typical-case verdict ("$\circ$" = preserves the block invariant, the block's MR is typically blind to the mutant; "$\times$" = breaks the block invariant, the block's MR can typically detect the mutant; "$\sim$" = case-dependent on the SUT-specific block instantiation).

The matrix stratifies mutants binary: *algebra-disrupting* (D1, at least one $\times$ in a populated column) vs *algebra-preserving* (D2, all $\circ$ in populated columns; $\sim$ resolved by SUT-specific overrides). The framework

Table 11. Typical-case compatibility of PIT default mutator categories with the eight NOETHER blocks. Cells reflect the dominant case for mutators in the category as applied to a generic algebra-rich SUT; SUT-specific exceptions are catalogued in supplementary S7 (d4j/). The cells ⟨MATH, $\mathcal{L}^*$⟩ and ⟨RETURN_VALS, $\mathcal{L}^*$⟩ are both "∘" (homogeneity-preserving), and the $\mathcal{L}^*$-blindness result of §5.3.4 is the empirical specialisation of these two cells.

| PIT mutator category | $G$ | $O_{\leq}$ | $T^*$ | $\mathcal{T}^*_{\text{rev}}$ | $\mathcal{L}^*$ | $\mathcal{D}^*$ | $\mathcal{E}^*$ | $\mathcal{B}^*_{\text{rel}}$ |
|---|---|---|---|---|---|---|---|---|
| CONDITIONALS_BOUNDARY | ∘ | × | ∘ | ∘ | ∘ | ∼ | ∘ | ∘ |
| INCREMENTS | × | × | × | × | × | ∼ | ∘ | ∘ |
| INVERT_NEGS | ∼ | × | ∼ | × | ∼ | ∼ | ∘ | ∘ |
| MATH (op swap) | × | ∼ | × | × | ∘ | × | ∼ | ∘ |
| NEGATE_CONDITIONALS | × | × | ∼ | × | ∼ | × | ∼ | × |
| RETURN_VALS (zero/one) | ∘ | ∘ | × | × | ∘ | ∘ | ∼ | ∘ |
| VOID_METHOD_CALLS | ∘ | ∘ | ∘ | ∘ | ∘ | × | ∼ | ∘ |

predicts Set N's D2 kill rate is near zero by construction: a mutant that preserves all invariants is structurally invisible to algebraic MRs. D1 is therefore Set N's appropriate fault-detection denominator; D2 is the territory of complementary techniques (random / GP / LLM-prompted MRs). The $\mathcal{L}^*$-blindness result of §5.3.4 is the corresponding D2-cell specialisation on the wider SUT substrate.

*Kill-set overlap on the head-to-head substrate.* On the $n = 62$ PIT mutants pooled across the eight head-to-head SUTs, 22 are killed by both Set N and Set G, 4 are killed by Set N only, 18 are killed by Set G only, and 18 are killed by neither. The 18 G-only kills constitute a candidate scope-mismatch pool (mutants reachable by Set G's GP-evolved oracle but missed by the algebraic Set N construction), and the 18 jointly-missed mutants further partition into D2 mutants outside Set N's algebraic reach and D1 mutants both sets miss for input-coverage reasons. The $\mathcal{L}^*$-only specialisation (Table 12) is the cleanest D2-cell estimate on the wider SUT substrate, since the ⟨MATH, $\mathcal{L}^*$⟩ and ⟨RETURN_VALS, $\mathcal{L}^*$⟩ cells of Table 11 are both "∘".

*Per-mutant D1 / D2 classification with SUT-specific overrides.* Per-mutant classification applies Table 11 with SUT-specific overrides (in `configs/sut_block_decomposition.json` + `configs/sut_block_overrides.json`) resolving the "∼" cells. After the multi-LLM equivalent-mutant exclusion of §5.3.7 (p. 57), the $n = 62$ head-to-head pool drops to $n = 57$ (52 D1 + 5 D2; all 5 excluded mutants were originally in the 10-mutant D2 stratum, so the D1 stratum is unchanged). Stratified kill rates: Set N D1 $26/52 = 0.500$ Wilson 95% $[0.369, 0.631]$; Set G D1 $37/52 = 0.712$ $[0.577, 0.817]$; Set N D2 $0/5 = 0.000$ $[0.000, 0.434]$; Set G D2 $3/5 = 0.600$ $[0.231, 0.882]$. The D2 point estimate (0.000) is consistent with the prediction $\leq 10\%$, but the Wilson upper bound does not exclude the 10% ceiling at $\alpha = 0.05$; the cross-codebase commons-math pilot (§5.3.8 (b.cm)) corroborates the direction at $2/29 = 6.9\%$. Set G's 0.600 on D2 confirms that D2 is a Set N-specific scope boundary, not a substrate-wide ceiling, since a GP-evolved generic baseline is unconstrained by the algebra. On D1, the cross-block aggregate gap (McNemar exact two-sided $p = 0.019$) decomposes per-block in §5.3.7 (gap concentrated on $G$ and $\mathcal{L}^*$; partially offset by Set N's edge on the translation sub-class $G_{\text{tr}}$ of the symmetry block $G$ (10/17 vs 8/17); the per-block decomposition is the appropriate substrate-level reading.

### 5.3.3 Test design.

*Substrate.* PIT 1.7.4 with the default mutator configuration serves as the shared substrate. Each (subject, MR) pair is evaluated through (i) JUnit test-class codegen, (ii) a 2-pass surefire green-suite filter that drops MR instances flaky on the unmutated SUT, and (iii) PIT mutation testing parsed to a per-mutant kill vector.

*Pre-registered SUT criterion.* The selection rule is committed in `configs/d4j_algebra_rich_criterion.json` on branch `feat/d4j-algebra-rich` of the experiment repository *before* any new evaluation data existed. The criterion references package roots (under `org.apache.commons.math3.*`, covering subpackages `ode`, `linear`, `transform`, `analysis.solvers`, `distribution`, `optim`, `fitting`, `stat.regression`, `complex`, `geometry`, `fraction`, plus their `math.*` legacy counterparts), per-package block-coverage hypotheses, and method-signature constraints; it contains no bug-id and no kill-rate references and cannot be tuned by evaluation outcomes. The git timestamp chain (criterion → inscope filter → Set N derivation → Set G GP rerun → pooled M1) is the auditable proof of pre-registration. To strengthen the third-party verifiability of the pre-registration, the SHA-256 hash of `configs/d4j_algebra_rich_criterion.json` together with the matching commit hash is deposited in supplementary S7 (`d4j/`) alongside the experiment artifact; reviewers may verify the deposit against the criterion file directly.

*Subjects and MR yield.* Ten SUTs admit at least one non-trivial NOETHER block beyond $G$: `midpoint`, `exactLog2`, `isSequence` (boolean predicate), `clamp`, `signum`, `ComplexSignal.add` (instance method), `gcdSig`, `lcmSig`, `hypotSig`, `powerSig`. Six of these admit an $L_{\text{scale}}$ MR (those whose induced algebra populates the $\mathcal{L}^*$ block under positive-scalar homogeneity): `midpoint`, `clamp`, `signum`, `gcdSig`, `lcmSig`, `hypotSig`. The remaining four do not admit a degree-1 scaling MR by their algebraic structure (`exactLog2` carries no positive-degree scaling; `isSequence` is a boolean predicate; `ComplexSignal.add` is a complex-arithmetic instance method whose scaling block is distinct in formulation; `powerSig` populates $T_2^*$ rather than $\mathcal{L}^*$). The prediction is tested on the six SUTs admitting $L_{\text{scale}}$.

Set N contains 30 hand-derived NOETHER MRs, two to four per SUT, one or more per non-empty block of each SUT's induced algebra. Set G is harvested by rerunning GenMorph's `genmorph.py gen` mode at seed = 11 on the 10 SUTs; Set G is used in §5.3.7 as the head-to-head comparator and plays no role in the central prediction test of §5.3.4.

*Metrics.* Per-MR kill counts on PIT mutants are the primary measurement for the central prediction test. Pooled M1 and McNemar exact paired tests are reported in §5.3.7 as corroborating context.

*5.3.4 Central result: $\mathcal{L}^*$-block blindness, confirmed.* The $L_{\text{scale}}$ MR's per-SUT kill rate on PIT mutants of the six $\mathcal{L}^*$-admitting SUTs is reported in Table 12. The total denominator is 44 mutants. The total kill count is 2.

*Reading.* On 5 of 6 SUTs the prediction is confirmed at the strongest reading (zero $L_{\text{scale}}$ kills). On the sixth SUT (`hypotSig`) the kill rate is 2/4, above the per-SUT falsification threshold. The pooled rate $2/44 \approx 4.5\%$ remains an order of magnitude below the average Set N M1 of 0.486 (the average kill rate across all Set N MRs on the same substrate; see §5.3.7), and the per-SUT failure pattern is isolated to one SUT.

Table 12. Per-SUT kill rate of the $L_{\text{scale}}$ MR on PIT mutants of the six SUTs admitting an $\mathcal{L}^*$-block scaling MR. The prediction of §5.3.1 is that the kill rate is near-zero by composition of the homogeneity-preserving property of PIT's default mutators with the scaling structure of $L_{\text{scale}}$. Pooled across the six SUTs: $2/44 \approx 4.5\%$, well below the $1/3$ falsification threshold on each SUT and below it on $5/6$ SUTs (`hypotSig` alone reaches $2/4 = 50\%$, explained in the text).

| SUT | $L_{\text{scale}}$ kill / mutants | *Verdict* |
|---|---|---|
| `midpoint` | 0 / 3 | Confirmed |
| `clamp` | 0 / 7 | Confirmed |
| `signum` | 0 / 6 † | Confirmed (no $L_{\text{scale}}$ kill) |
| `gcdSig` | 0 / 9 | Confirmed |
| `lcmSig` | 0 / 11 | Confirmed |
| `hypotSig` | 2 / 4 | Outlier; explained below |
| pooled (all six) | 2 / 44 | 4.5% |

† For `signum` the $\mathcal{L}^*$-block MR is positive-scalar invariance ($\text{signum}(\lambda x) = \text{signum}(x)$ for $\lambda > 0$) rather than degree-1 homogeneity; the prediction applies to this MR shape *a fortiori* since the right-hand side does not depend on $\lambda$.

*The `hypotSig` outlier.* The kill on two `hypotSig` mutants is consistent with the prediction's caveat in §5.3.1, that the small fraction of homogeneity-breaking mutators (return-value swap to a constant; constant-replacement that participates additively in the expression) is the residual non-zero contribution. Inspection of `mutants_killed_set_n.csv` for `hypotSig` (path in supplementary S7 (`d4j/`)) identifies the two killed mutants as `KILLED return_zero_doubles_VR` and `KILLED Math.sqrt_replaced_with_one_RC`: the first replaces the entire return path with the constant zero, the second replaces the inner `Math.sqrt` call with the constant 1.0. Both mutators violate degree-1 homogeneity directly. The two detection events are therefore consistent with the prediction's quantitative tail rather than evidence against it.

*Falsification verdict.* By the criterion stated in §5.3.1, the prediction is falsified if a single $L_{\text{scale}}$ MR kills $\geq 1/3$ of its PIT mutants on more than one SUT. Observed: one SUT (`hypotSig`, 2/4) above the per-SUT threshold; five SUTs (`midpoint`, `clamp`, `signum`, `gcdSig`, `lcmSig`) at zero. The prediction passes. We read this as direct empirical witness for the operative-mechanism reading of the structural decomposition: a block whose generator is a continuous symmetry of the mutator set contributes MRs that are quantitatively silent on mutator-induced defects, and the silence is observed in the data.

*5.3.5 Corroborating per-block patterns.* The prediction-test of §5.3.4 establishes the $\mathcal{L}^*$-block result as the section's central claim. The remaining populated sub-blocks — the finite-group symmetry $G$ and its translation sub-class $G_{\text{tr}}$, together with an out-of-frame idempotence probe ($\mathcal{I}^*$, not among the eight blocks of Definition 13; a candidate ninth block per Remark 3) — carry weaker, qualitative predictions that the data also confirms; they corroborate the operative-mechanism reading without providing a falsifiable quantitative test.

*$G_{\text{tr}}$ translation sub-class (grid shift, period).* Translation MRs carry the highest single-MR kill rates on the numeric SUTs: `midpoint T_shift` 3/3, `powerSig T_exp_step` 8/12, `exactLog2 T_double` 4/10, `clamp T_shift` 3/7. Translation invariance is the property most directly perturbed by PIT's arithmetic-operator swaps; $G_{\text{tr}}$-derived MRs detect this perturbation reliably. The qualitative prediction (high single-MR rate on numeric SUTs) holds.

*$G$ block (group, symmetry).* Symmetry MRs are moderately to highly effective when the SUT exposes the appropriate symmetry: `signum G_negate` 4/6, `midpoint G_swap` 1/3, `ComplexSignal.add G_swap` 2/3. The qualitative prediction (kill rate correlates with SUT-side symmetry exposure) holds.

*$G$-block framework boundary on recursive-normalising SUTs.* The two Euclidean-style SUTs `gcdSig` and `lcmSig` exhibit Set N 0/7 on the $G$ block (Table 15); Set G achieves 7/7 on the same mutants, confirming the mutants are detectable by some MR set and that the gap is therefore Set N-specific. Reading the SUT source clarifies why this is not an MR-design defect but a framework-boundary case: `gcdSig` and `lcmSig` both open with a sign-correcting prologue (`a = a < 0 ? -a : a;` on lines 67–68 and 81–82 respectively) that absorbs the $G$-block sign-flip identity before the recursive Euclidean reduction fires. The grid-synthesised $G$-block MR (`gcdSig(a, b) = gcdSig(-a, -b)`, encoding the canonical $\sigma =$ sign-flip involution) is therefore vacuously satisfied on the SUT's normalised input domain, independent of any downstream mutation. The framework's scope precondition (Definition 13) requires that the SUT's input domain expose the algebraic action under test; for these two SUTs the $G$-action is structurally absorbed by the SUT's own normalisation prologue, so $G$-block reasoning is orthogonal to the recursive Euclidean algorithm's algebraic core (which is closer to a divisibility lattice $\mathcal{D}$-block / reversibility $\mathcal{R}$-block than to a pre-normalisation symmetry $G$). The 0/7 Set N kill rate is the correct $G$-block reading on these SUTs under the framework's scope precondition, not a missed opportunity for MR refinement.

*$\mathcal{I}^*$ idempotence probe (out-of-frame; candidate ninth block).* Idempotence MRs kill few mutants under the paired-MR DSL: `I_idem` $\approx 0$ across most SUTs; `powerSig I_zero_exp` 2/12 is a degenerate single-input case. This reflects an expressivity limit of the JIR/JOR shape, not an absence of idempotence structure in the SUTs (this probe lies outside the eight-block frame; see Remark 3). The qualitative prediction (low kill rate under paired-MR DSL) holds.

The four block predictions taken together, including the central $\mathcal{L}^*$ falsification test of §5.3.4, are consistent with the operative-mechanism reading of the structural decomposition.

*5.3.6 Two convergent witnesses.* Two further observations from the same SUT set are independently informative. Neither is a statistical hypothesis test; both are non-rate evidence that the algebraic-decomposition thesis predicts and that the head-to-head Set G run incidentally reveals.

*Witness 1: cross-pipeline MR rediscovery.* On `midpoint`, GenMorph's GP independently evolves three MRs that map onto Set N's blocks (Table 13). Set N is derived a-priori from the operator algebra of `midpoint`; Set G is searched by mutation-killing fitness with no algebraic structure as input. The two pipelines converge on the same algebraic primitives. Convergence under independent epistemic processes is direct corroboration of the operative-generator reading.

Table 13. Cross-pipeline rediscovery on `midpoint`: GP-evolved MRs (Set G) and their algebra-derived counterparts (Set N).

| GP-evolved MR (Set G) | Set N counterpart | Block |
|---|---|---|
| `SwitchParams??1@2` | `G_swap` (commutativity) | $G$ |
| `NumericAddition?1.000000?1` | `T_shift` (translation by 1) | $G_{\mathrm{tr}}$ |
| `NumericMultiplication?0.500000?2` | `L_scale` (scaling, approx.) | $\mathcal{L}^*$ |

*Witness 2: structural coverage extension.* GenMorph's GP pipeline fails on two of the ten SUTs for plumbing reasons: `ComplexSignal.add` (instance method) hits an XStream `NullPointerException` on receiver deserialisation; `isSequence` (boolean predicate) yields ungrammatical JOR strings. Both SUTs admit Set N MRs derived directly from the SUT signature (commutativity for `ComplexSignal.add`; monotonic-window translation for `isSequence`) and run to completion, covering 8 of 70 PIT mutants on which only Set N is defined. Set N's algebraic-derivation route operates on the SUT signature, independent of the SUT-deserialisation and JOR-grammar infrastructure GenMorph depends on; the structural-coverage extension is an architectural property of the derivation, not an empirical accident.

*5.3.7 Head-to-head at GenMorph's published budget.* This subsection decomposes an aggregate Set G dominance per algebraic block, with cost-axis and D2-stratum framework prediction layered alongside.

**Boundary of contribution (head-to-head restatement)**

*What this subsection claims, and what it does not.* (1) **Aggregate verdict**: Set N is dominated by Set G on the D1 stratum (McNemar $p = 0.0043$ pooled, $p = 0.019$ on D1 only). (2) **Per-block reading**: the gap concentrates on the $G$ block (Set G 7/7 vs Set N 0/7 on commutativity-targeting mutants); on the $G_{\mathrm{tr}}$ translation sub-class Set N dominates (10/17 vs Set G's $G_{\mathrm{tr}}$ kills). (3) **Cost-axis**: independently of detection-rate, Set N is derivable in polynomial time from $\mathcal{A}_P$ (Theorem 2) while Set G requires a $\approx$ 30-min stochastic GP search per SUT. (4) **D2-stratum framework prediction**: Set N's $\leq 10\%$ D2 kill-rate prediction (operative invariant preserved under structurally-non-disrupting mutants) is an ex-ante framework signal that no inductive baseline can derive; pre-registered cap holds at $2/29 = 6.9\%$ on the cross-codebase Commons-Math substrate. The head-to-head is not the framework's load-bearing claim: per-block algebraic derivability, structural coverage extension (two SUTs only Set N reaches), and the D2 prediction are.

**On the algebra-disrupting D1 stratum at GenMorph's published 30-min GAssert budget, Set N is dominated by Set G in the aggregate (McNemar exact two-sided $p = 0.0043$ pooled and $p = 0.019$ on D1 only, $n = 62$ post-equivalent-mutant exclusion).** The paper does not assert head-to-head superiority on D1. The framework's contribution on the head-to-head substrate is read as (i) algebraic derivability, (ii) per-block complementarity (Set G alone kills 15 D1 mutants Set N misses, Set N alone kills 4 D1 mutants Set G misses), and (iii) an out-of-scope D2-stratum framework prediction ($\leq 10\%$ kill rate) that no inductive baseline can derive ex-ante. The D1 pooled comparison is reported below for protocol-completeness rather than as the framework's verdict. The central empirical claim of the section is established at §5.3.4 (the $\mathcal{L}^*$-blindness falsifiable prediction confirmed on 5/6 SUTs) and does not depend on the head-to-head outcome.

The head-to-head is run at GenMorph's published 30-min GAssert budget; a prior 1-min run is retained as a sensitivity reference. Per-SUT entries are reported as directional only; no per-SUT statistical-significance claim is asserted, since $8 \times 2 = 16$ paired comparisons across the two budgets exceed the family-wise control of an uncorrected $\alpha = 0.05$ (Holm–Bonferroni-adjusted threshold $\alpha/16 \approx 0.003$; no per-SUT contrast meets this threshold).

Table 14. Per-SUT PIT mutation kill counts, Set N versus GenMorph-evolved Set G at GenMorph's published 30-min GAssert budget (primary) and at a 1-min GAssert sensitivity rerun, on the 10 algebra-rich Java SUTs of §5.3.3. Set G is structurally absent on `ComplexSignal.add` (instance-method deserialisation) and `isSequence` (boolean-predicate JOR grammar) under both budgets (§5.3.6). Set N's kill vector is held fixed across budgets. Pooled across the 8 head-to-head SUTs ($n = 62$ mutants), 30-min budget primary: Set N = 26, Set G = 40, McNemar exact two-sided $p = 0.0043$. The $\Delta$ (30-min) column reports Set N − Set G at the 30-min budget; note labels are directional descriptors only and do not assert per-SUT statistical significance (see prose). $n = 62$ **is underpowered for a paired hypothesis test at** $\alpha = 0.05$ **in two-sided form.**

| SUT | mutants | Set N | Set G (30-min) | Set G (1-min) | $\Delta$ (30-min) | note |
|---|---|---|---|---|---|---|
| `ComplexSignal.add` | 3 | 2 | N/A | N/A | | instance method (Set G N/A) |
| `midpoint` | 3 | 3 | 3 | 3 | 0 | tie |
| `exactLog2` | 10 | 4 | 0 | 0 | +4 | directional only |
| `isSequence` | 5 | 0 | N/A | N/A | | boolean predicate (Set G N/A) |
| `clamp` | 7 | 5 | 5 | 5 | 0 | tie |
| `signum` | 6 | 4 | 4 | 4 | 0 | tie |
| `gcdSig` | 9 | 1 | 6 | 5 | −5 | directional only |
| `hypotSig` | 4 | 2 | 4 | 4 | −2 | directional only |
| `lcmSig` | 11 | 0 | 8 | 8 | −8 | directional only |
| `powerSig` | 12 | 7 | 10 | 10 | −3 | directional only (2 Set N MRs ineligible, see footnote) |
| **pooled (head-to-head, n=62)** | | **26** | **40** | **39** | −14 | McNemar two-sided $p = 0.0043$ (30-min) |

*Per-block head-to-head and complementarity (Primary, PIT arm).* Each Set N MR derives from exactly one NOETHER block, so the natural head-to-head denominator is per-block rather than the operative-block union. Across the eight head-to-head SUTs, PIT's stock mutator catalogue exercises three operative (sub-)blocks: $G$ (finite-group symmetry: swap / negate), $\mathcal{L}^*$ (scale / homogeneity), and $G_{\mathrm{tr}}$, the translation sub-class of the symmetry block $G$ (grid shift / period). For each block $b$, $n_b$ counts mutants whose (SUT, mutator) entry in `configs/sut_block_overrides.json` marks block $b$ as broken ("×"). Per-block kill counts and the complementarity partition (both / $N$-only / $G$-only / neither) appear in Table 15.

Table 15. Per-block head-to-head on the three operative blocks exercised by PIT 1.7.4 across the eight in-scope SUTs of §5.3.3. Each row's denominator $n_b$ counts mutants violating block $b$'s invariant; complementarity cells "both" / "$N$-only" / "$G$-only" / "neither" partition $n_b$. Wilson 95% intervals are reported alongside point rates. The five operative blocks PIT 1.7.4 does not exercise ($O_{\leq}, \mathcal{T}^*_{\mathrm{rev}}, \mathcal{D}^*, \mathcal{E}^*, \mathcal{B}^*_{\mathrm{rel}}$) are addressed by the construct-trace consistency check on hand-crafted block-targeted mutants (now in supplementary S9, item "D_E_implementation_consistency"); those results are design-implied and not used as independent fault-detection evidence. Auxiliary rows report the D1 aggregate and the unmapped bucket (PIT mutants from cells without an override entry, principally `RETURN_VALS` and `MATH` on SUTs where the Table 5 reconstruction fallback assigns D1 but does not specify broken blocks; lower-bound caveat, not a participating row).

| Block | $n_b$ | Set N kills | rate (95% CI) | Set G kills | rate (95% CI) | both | $N$-only | $G$-only | neither |
|---|---|---|---|---|---|---|---|---|---|
| $G$ (sym.) | 11 | 2 | 0.182 [0.051, 0.477] | 9 | 0.818 [0.523, 0.949] | 2 | 0 | 7 | 2 |
| $\mathcal{L}^*$ | 24 | 10 | 0.417 [0.245, 0.612] | 16 | 0.667 [0.467, 0.820] | 8 | 2 | 8 | 6 |
| $G_{\mathrm{tr}}$ | 17 | 10 | 0.588 [0.360, 0.784] | 8 | 0.471 [0.262, 0.690] | 7 | 3 | 1 | 6 |
| *D1 aggregate, PIT-covered blocks (secondary)* | 52 | 26 | 0.500 [0.369, 0.631] | 37 | 0.712 [0.577, 0.817] | 22 | 4 | 15 | 11 |
| *unmapped (lower-bound caveat)* | 25 | — | — | — | — | — | — | — | — |

The unmapped auxiliary row counts 25 PIT mutants from cells whose D1/D2 override table does not specify broken blocks; it is a lower-bound caveat on per-block denominators (each unmapped mutant is D1-classified but its block membership is unknown), not an additional denominator added to the $G + \mathcal{L}^* + G_{\mathrm{tr}}$ union. The D1 pool is $n = 52$ pre and post equivalent-mutant exclusion (Table 16 row 1).

The per-block reading distinguishes three regimes:

- $G$ **block: Set G dominates.** Set N kills 0/7 of the mutants Set G exclusively reaches; the Set N hits concentrate on `signum` (2/4), while `gcdSig` and `lcmSig` contribute 0/7 for Set N. The Euclidean-style SUTs are structurally unreachable by NOETHER's $G$-block construction on this grid-synthesised catalogue: `gcdSig` normalises its inputs via `a < 0 ? -a : a` before the recursive `gcd(a,b)=gcd(b, a mod b)` fires, absorbing the sign-flip invariant; `lcmSig` inherits the same prologue. Whether this calls for an $\mathcal{R}$-block re-synthesis or for documenting recursive-normalising SUTs as $G$-orthogonal is committed as follow-up (e.4).
- $\mathcal{L}^*$ **block: complementary coverage.** Both Sets kill 8 mutants in common, 2 are $N$-only (`exactLog2`), 8 are $G$-only, 6 are jointly missed; the union covers $18/24 = 75\%$, above either Set's individual rate. Set N's exclusive reach on `exactLog2` quantifies the coverage-extension witness of §5.3.6 per-block; Set G's exclusive reach on `gcdSig`+`lcmSig` compensates for the $G$-block gap.
- $G_{\mathrm{tr}}$ **translation sub-class: Set N edge (underpowered).** Of 17 translation-violating mutants, Set N kills 10 and Set G kills 8 ($N$-only 3, $G$-only 1, both 7, neither 6); union $11/17 = 64.7\%$. Set N is +11.7 pp in rate and +2 in exclusive count on its own sub-block, but the Wilson intervals [0.360, 0.784] and [0.262, 0.690] overlap substantially. We report the per-block edge as directional, consistent with the $G_{\mathrm{tr}}$ design prediction rather than inferential at $\alpha = 0.05$.

*Coverage of the remaining five operative blocks.* PIT 1.7.4's stock mutator catalogue does not exercise the remaining five operative blocks ($O_{\leq}$, $\mathcal{T}^*_{\mathrm{rev}}$, $\mathcal{D}^*$, $\mathcal{E}^*$, $\mathcal{B}^*_{\mathrm{rel}}$) on the §5.3.3 substrate. To exercise the pipeline end-to-end on these blocks, we hand-crafted 5 mutants per uncovered block (25 total) that violate the targeted invariant of a specific known Set N MR for each block. These mutants are construct-trace probes (each one is by design within the reach of the Set N MR it targets); the resulting kill counts are reported in supplementary S9 (Appendix E) for pipeline-correctness verification and Set G incidental-reach quantification, but they are *not* used as evidence for H3a.1 in this section, since construct-trace circularity precludes treating them as independent fault-detection measurements.

*What the per-block reading supports, and what it does not.* The framework's empirical contribution on the present substrate is *block-dependent*. On the three PIT-covered operative blocks the reading is targeted per-block: $G_{\mathrm{tr}}$ shows the design's directional advantage and complementarity; $\mathcal{L}^*$ shows complementarity in both directions (Set N reaches what Set G misses on `exactLog2`; Set G reaches what Set N misses on `gcdSig` / `lcmSig`); $G$ shows the design's current limit on the substrate's two Euclidean-style SUTs. The appropriate reading is therefore *block-targeted precision plus complementarity, with documented per-block design gaps on the PIT-covered substrate*, rather than a single "head-to-head winner" metric. The aggregate D1 head-to-head is dominated by Set G (McNemar $p = 0.019$, see paragraph below), but the dominance collapses into the per-block profile: it is driven primarily by the $G$-block reading on `gcdSig` + `lcmSig` and by Set G's denser $\mathcal{L}^*$ coverage on those same SUTs.

*Aggregate D1 head-to-head (Secondary, cross-block, $n = 52$).* For continuity with prior reporting conventions, the cross-block aggregate D1 head-to-head is: Set N $\mathrm{M1}_{\mathrm{D1}} = 26/52 = 0.500$ (Wilson 95% CI [0.369, 0.631]), Set G $\mathrm{M1}_{\mathrm{D1}} = 37/52 = 0.712$ ([0.577, 0.817]); exact McNemar two-sided $p = 0.019$ on the D1 stratum with discordant pairs $(b, c) = (15, 4)$, paired risk difference $\mathrm{RD}_{\mathrm{paired}} = (b-c)/n = (15-4)/52 = +0.212$ favouring Set G, and odds ratio $\mathrm{OR} = b/c = 15/4 = 3.75$. The aggregate gap (which favours Set G) collapses into the per-block profile above: it is driven primarily by Set G's near-complete reach on $G$ (`gcdSig`

+ `lcmSig`) and by its denser $\mathcal{L}^*$ coverage; it is partially offset by Set N's $G_{\mathrm{tr}}$ edge but not enough to close the cross-block aggregate. The aggregate is presented as secondary because it averages over per-block strengths and weaknesses that the per-block table makes visible.

*Framework prediction on the D2 stratum (algebra-preserving, $n = 5$).* After equivalent-mutant exclusion (see footnote at §5.3.7 below), the 10 originally-classified D2 mutants reduce to $n = 5$: five of the ten were LLM-ensemble-judged equivalent on the SUT's declared input domain and are excluded from both Sets' denominators by symmetric construction. On the remaining $n = 5$ D2 mutants, Set N kill rate $= 0/5 = 0.000$ (Wilson 95% CI [0.000, 0.434]); the point estimate is consistent with the framework prediction kill rate$_{\mathrm{D2}} \leq 10\%$, but the Wilson upper bound at 0.434 does not exclude the 10% ceiling at $\alpha = 0.05$; the cross-codebase commons-math pilot (§5.3.8 (b.cm)) corroborates the direction at $2/29 = 6.9\%$ (Wilson 95% CI $[0.012, 0.221]$), and inferential confirmation of the $\leq 10\%$ ceiling requires a pooled sample of $n \geq 30$. Set G kills $3/5 = 0.600$ (Wilson 95% CI [0.231, 0.882]) on the surviving D2 stratum: the GP-evolved generic baseline is unconstrained by the algebra and incidentally detects D2 mutants through input-domain edge cases, confirming that D2 is a Set N-specific scope boundary rather than a substrate-wide structural ceiling. The framework's prediction therefore separates Set N (algebra-bound) from Set G (generic) on the D2 stratum; the D2 reading is the framework's own falsifiability commitment (§5.3.4), not a head-to-head metric.

*Pooled M1 rates (auxiliary, scope-mismatched, $n = 57$).* For completeness against prior reporting conventions, the post-eq- exclusion pooled M1 rates over D1 $\cup$ D2 are: Set N $= 26/57 = 0.456$ (Wilson 95% CI [0.334, 0.584]), Set G $= 40/57 = 0.702$ ([0.573, 0.805]), McNemar two-sided $p = 0.0043$[2]. We report this pooled comparison as scope-mismatched: the 5 D2 mutants in the denominator are not in Set N's theoretical reach by construction, so the pooled gap overstates Set N's effective miss rate within its declared scope. At the 1-min sensitivity budget, Set N is unchanged from the kill vector at the 30-min budget when Set G is the only variable; Set G's pooled M1 is 0.557 ([0.441, 0.668]). The 30-min rerun changes Set G's per-SUT counts on one of the eight head-to-head SUTs (`gcdSig`, $5 \to 6$); seven SUTs are unchanged, none regresses. A 30$\times$ budget increase therefore buys +1 Set G mutant on a single SUT under PIT mutation.

*LLM-assisted baseline (Set L ensemble, $n = 70$ aligned).* For the LLM-assisted comparator of §5.2.1's H3b, we ran a 2-vendor (DeepSeek, ChatGPT) $\times$ 5-temperature ($t \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$) ensemble on the same 10-SUT substrate, at a multi-LLM-sample scale that subsumes the single-sample probes of Shin et al. [47] and Zhang et al. [66]. The harvest is 100 LLM samples yielding 487 proposed MRs; deterministic template-matching against Set N's per-SUT block catalogue translates 212 (43.5%) into executable JIR/JOR pairs (the remaining 56.5% are either compound multi-input properties or out-of-block algebra, e.g. associativity, identity / inverse elements, monotonicity on SUTs whose Set N catalogue lacks an $\mathrm{O}_{\mathrm{le}}$-block representative). On the aligned $n = 70$ pooled mutant substrate (the full 10-SUT substrate including the two SUTs structurally N/A for Set G), per-LLM pooled kill rates are ChatGPT $34/70 = 0.486$ Wilson 95% CI [0.372, 0.600], DeepSeek $33/70 = 0.471$ [0.359, 0.587], ensemble union $34/70 = 0.486$ [0.372, 0.600].

[2] Pooled discordant pairs $(b, c) = (18, 4)$ on the $n = 57$ paired denominator. The D1-only McNemar separates at $(b, c) = (15, 4)$, $p = 0.019$ (§5.3.7 Aggregate D1 paragraph); the additional 3 discordant pairs that the pooled comparison picks up come from the D2 stratum (Set G kills 3 of the 5 surviving D2 mutants, Set N kills 0 of 5; all 3 are $b$-cell discordances). The pooled McNemar is thus *strengthened* by including the D2 stratum, but at the cost of mixing the within-scope D1 comparison with the out-of-scope D2 stratum that Set N is not designed to detect.

Set L's union kill vector is a strict subset of Set N's on this substrate (34/34 overlap; 0 Set-L-exclusive kills), as a structural consequence of the template-matching translator: every Set L MR is by construction a byte-identical copy of a Set N pair, so Set L can match but not exceed Set N's per-MR kill power. The substantive finding is that the 2-vendor ensemble identifies enough of Set N's catalogue to reproduce Set N's reach on the matchable subset; on the unmatchable 56.5%, Set L explores invariants outside the 8-block frame that the framework's $\text{coverage}_{\text{NOETHER}}$ diagnostic correctly identifies as structurally orphaned. A third-vendor (Anthropic Claude) extension is committed in follow-up (d.set-l-claude) of supplementary S4 (`future_work.md`); full per-(LLM, SUT, temperature) breakdowns in `docs/set_l_phase2_results.md` of the experiment repository.

*Equivalent-mutant denominator (resolved).* The original PIT-SURVIVED denominator ($n = 62$) does not separate semantically distinct mutants from PIT false-positives. Every head-to-head mutant passed through a two-stage filter: the 44 mutants killed by at least one of Set N or Set G are auto-classified non-equivalent; the 18 jointly-missed mutants go through a progressive multi-LLM vote (DeepSeek + ChatGPT at temperature 0.0, with Anthropic Claude Opus tiebreaker when stage-1 voters disagree). Outcome: 5 voted equivalent (all `ConditionalsBoundaryMutator` on `gcdSig`, `lcmSig`, `powerSig` where `a < 0 ? -a : a` prologues absorb the sign-flip), 13 non-equivalent. The 5 equivalents fall entirely within the algebra-preserving (D2) stratum, leaving the per-block D1 denominators unchanged. The eq-excluded total is $n = 57$ (52 D1 + 5 D2 after the (e.1) v2 override reclassified 3 `PrimitiveReturnsMutator` mutants from D1 to D2). The 2+1 voting scheme was calibrated against the pilot's manual analyst session (§5.3.4); the single pilot-vs-LLM disagreement case resolved in favor of the pilot verdict.

Pooled rates exhibit the scope-discriminance pattern predicted by NOETHER's scope declaration (Definition 13): on the wider utility-method baseline (23 methods) (supplementary S5; mixed in-scope and boundary), Set N's pooled M1 is 0.288 vs Set G's 0.363; on the in-scope algebra-rich substrate (eq-excluded $n = 57$), Set N moves to 0.456 and Set G to 0.702 (D1 subset: 0.500 vs 0.712). Both methods benefit from the algebraically richer substrate; the scope declaration is empirically discriminating.

*Two-stratum head-to-head summary (Primary tabulation for H3a verdict).* Table 16 reports the two strata (D1, D2) each with four metrics: total mutants, equivalents excluded, Set N kill / survive counts and rate (Wilson 95% CI), Set G kill / survive counts and rate, and the complementarity partition ($N$-only, $G$-only, both, neither). The headline reading is per-block (Table 15); the D1 aggregate (Table 16) and pooled (auxiliary, preceding paragraph) are reported for continuity with prior conventions.

Table 16. Two-stratum head-to-head on the eight-SUT head-to-head substrate (post equivalent-mutant exclusion). D1 = algebra-disrupting, D2 = algebra-preserving. Complementarity cells report mutant counts; the operative test for H3a.2 is the $N$-only column on D1.

| Stratum | $n$ (post-exclusion) | equivalents excluded | Set N kill / surv. | Set N rate [Wilson 95%] | Set G kill / surv. | Set G rate [Wilson 95%] | both / $N$-only / $G$-only / neither | McNemar $p$ |
|---|---|---|---|---|---|---|---|---|
| D1 | 52 | 0 | 26 / 26 | 0.500 [0.369, 0.631] | 37 / 15 | 0.712 [0.577, 0.817] | 22 / 4 / 15 / 11 | 0.019 |
| D2 | 5 | 5 | 0 / 5 | 0.000 [0.000, 0.434] | 3 / 2 | 0.600 [0.231, 0.882] | 0 / 0 / 3 / 2 | 0.25 |
| pooled | 57 | 5 | 26 / 31 | 0.456 [0.334, 0.584] | 40 / 17 | 0.702 [0.573, 0.805] | 22 / 4 / 18 / 13 | 0.0043 |

*H3a verdict split into three sub-claims.* **H3a.1 (Detection on D1, per-block).** Per Table 15 on the pre-registered PIT-covered substrate of three populated blocks: *mixed*. On the $G_{\text{tr}}$ translation sub-class

Set N achieves $10/17 = 0.588$ versus Set G $8/17 = 0.471$, with three $N$-only exclusive kills, a directional edge consistent with the algebra-induced prediction; the $n = 17$ block-level sample is underpowered for an inferential test at $\alpha = 0.05$ (Wilson intervals overlap). On the $\mathcal{L}^*$ block Set N achieves $10/24 = 0.417$ versus Set G $16/24 = 0.667$, with the complementarity partition $N$-only $= 2$ and $G$-only $= 8$ identifying a coverage-extension witness on `exactLog2` and a denser-coverage region on `gcdSig` / `lcmSig` respectively. On the $G$ block the substrate's two Euclidean-style SUTs (`gcdSig`, `lcmSig`) absorb the $G$-action through their normalisation prologue before the recursive reduction (a documented framework boundary, see $G$-block framework boundary on recursive-normalising SUTs); the $G$-block 0/7 Set N reading on those two SUTs is the framework-correct boundary verdict rather than a missed-opportunity MR-design defect, but the block-level rate (2/11 Set N vs 9/11 Set G) favours Set G on the substrate as reported. The aggregate D1 head-to-head on the PIT-covered stratum is dominated by Set G ($26/52 = 0.500$ vs $37/52 = 0.712$, McNemar $p = 0.019$, Table 16); the per-block decomposition makes visible that the gap is concentrated on the $G$ block plus $\mathcal{L}^*$, partially offset by Set N's $G_{\mathrm{tr}}$ edge. Coverage of the remaining five operative blocks ($O_{\leq}$, $\mathcal{T}^*_{\mathrm{rev}}$, $\mathcal{D}^*$, $\mathcal{E}^*$, $\mathcal{B}^*_{\mathrm{rel}}$) via hand-crafted block-targeted mutants is reported in supplementary S9 (Appendix E) as a construct-trace consistency check; per supplementary S9 (Appendix E), those results are design-implied and are not used as independent evidence for H3a.1 in this paragraph.

**H3a.2 (Complementarity).** *Supported on the PIT-covered substrate.* The complementarity partition on the 52-mutant PIT-covered D1 stratum (Table 16, row 1) reports 4 $N$-only kills, 15 $G$-only kills, 22 both-killed and 11 neither, distributed across the $G_{\mathrm{tr}}$ and $\mathcal{L}^*$ blocks (per Table 15). Union coverage on the PIT-covered substrate is $22 + 4 + 15 = 41/52 = 78.8\%$ (both-killed plus exclusively-Set-N-killed plus exclusively-Set-G-killed), modestly above Set G alone's $37/52 = 71.2\%$ and materially above Set N alone's $26/52 = 50.0\%$. The complementarity is asymmetric: Set G's exclusive contribution (15 kills) is roughly four times Set N's (4 kills), reflecting the substrate's predominance of generic algebra-disrupting mutations that Set G's GP-evolved generic MRs catch, with Set N's exclusive reach concentrated on algebraically-distinctive structures (`exactLog2`'s $\mathcal{L}^*$ pattern) where the algebra-induced MR has no generic counterpart in Set G's catalogue.

**H3a.3 (Cost-axis).** *Supported.* See Approximate-parity-at-lower-cost reading (H3a.3 verdict) below for the detailed cost-axis analysis.

**D2 stratum (framework prediction; not part of H3a).** Set N D2 kill rate $= 0/5 = 0.000$ (Wilson 95% CI $[0.000, 0.434]$); the framework's $\leq 10\%$ prediction is consistent with the direction observed, but $n = 5$ is insufficient for $\alpha = 0.05$ confirmation (the Wilson upper bound 0.434 does not exclude the 10% ceiling). Set G achieves $3/5 = 0.600$ on the same stratum, an out-of-scope contribution for Set N by construction (Set G is not algebra-bound) and a signal that D2 is a *Set N-specific scope boundary* rather than a substrate-wide ceiling. The $\mathcal{L}^*$-blindness result of §5.3.4 and the structural-extension witness on the two SUTs where Set G is N/A (§5.3.6) are unaffected.

*Comparator scope and three-SOTA-category coverage.* The head-to-head reported here delivers the GP-evolved-baseline arm of a three-SOTA-category protocol against one representative per category. The GP-evolved representative is GenMorph (Ayerdi et al. [3]), the SOTA among genetic-programming MR identification pipelines on Java SUTs at the time of submission, run at its published 30-min GAssert budget. The LLM-assisted representative is a multi-vendor × multi-temperature LLM-MR-generation protocol (the same arm reported inline in §5.3.7 as Set $L_{\text{ensemble}}$); this section's earlier Set L is a single-sample $n = 1$ GPT-4 probe (§5.2) and is now superseded by the 2-vendor (DeepSeek, ChatGPT) $\times$ 5-temperature ensemble reported in §5.3.7. A third-vendor (Anthropic Claude) extension running alongside Shin [47], GPTMR [65], and AutoMT [33] is committed as post-acceptance follow-up in supplementary S4 (item (d.set-l-claude)). The mining-based representative is MR-Scout (Sun et al. [60]); on this section's substrate, MR-Scout's reach is reported as an estimate adapted from MR-Scout's published-artifact reach figures rather than a full re-execution (see Table 17 for the cost-axis interpretation; full re-execution is follow-up (d)). NOETHER's cost-axis profile relative to all three categories is given in Table 17; the matrix-driven D1/D2 mutant stratification (Table 11) operationalises the detection-axis claim that NOETHER is appropriate to algebra-disrupting mutants and out-of-scope by design on algebra-preserving mutants. Per-SUT delta entries in Table 14 are *directional descriptors only*; under Holm-Bonferroni correction for the $8 \times 2 = 16$ paired comparisons across the two budgets, the per-SUT family-wise threshold is $\alpha/16 \approx 0.003$, which no per-SUT contrast meets. The two paired hypothesis tests reported on this substrate are the scope-matched D1 McNemar ($p = 0.019$ at the 30-min budget, exact two-sided, $n = 52$ post eq-exclusion, discordant pairs $(b, c) = (15, 4)$) and its auxiliary pooled counterpart ($p = 0.0043$ at $n = 57$, discordant $(b, c) = (18, 4)$). The pooled gap is strengthened by Set G's 3/5 kills on the D2 stratum, an out-of-scope contribution for Set N by construction, which is why pooled and D1-stratified $p$ values now differ.

#### 5.3.8 *Threats to validity and committed future work.*

*(a) Prediction-commitment timestamp.* The $\mathcal{L}^*$-blindness prediction of §5.3.1 is derivable from public information without consulting data, and is committed to git in supplementary S7 (`d4j/`) alongside the SUT-selection criterion before the per-MR kill-count files; a reader who distrusts the timestamp chain can re-derive the prediction independently from §3.2 and [16].

*(b) Set G budget asymmetry.* GenMorph's published 30-min GAssert configuration is reported alongside a 1-min sensitivity rerun (Table 14) that handicaps Set G conservatively; the 30-min budget leaves the $\mathcal{L}^*$-blindness result and the structural-extension finding unchanged.

*(c) Sample size.* $n = 70$ across 10 SUTs is underpowered for a paired head-to-head verdict at $\alpha = 0.05$; the §5.3.7 reading is the honest disclosure that Set N is dominated by Set G in the aggregate D1 stratum (McNemar $p = 0.019$ on D1; $p = 0.0043$ pooled), with the framework's contribution read at the per-block, cost-axis, and D2-prediction layers rather than aggregate fault-detection superiority. The $\mathcal{L}^*$-blindness test on $n = 44$ across 6 SUTs is per-SUT (one third or more on more than one SUT), not pooled.

*(d) Set G structural absence is upstream-snapshot-relative.* The two N/A SUTs of §5.3.6 reflect the state of GenMorph upstream at our snapshot. An upstream patch would restore Set G coverage and weaken the structural framing of the coverage-extension witness, though the architectural point (algebra-derivation operates on the SUT signature, independent of GP plumbing) survives the patch.

*(e) Substrate selection.* The evaluation restricts itself to programs with explicit mathematical-physics equations, by NOETHER's scope declaration (Definition 13); the kill-rate gain over the utility-method reference is reported as scope-discriminance evidence (claim D), not as a generalisable performance improvement on every Java SUT.

*(f) Three SOTA-category coverage with explicit baseline-strength caveats.* The protocol of §5.2.1 commits Set N against one representative per SOTA category: a GP-evolved baseline (GenMorph [3]), an LLM-assisted baseline (subsuming the single-sample probes of [33, 47, 65, 66]), and a mining-based baseline (MR-Scout [60]). Baseline-strength asymmetry: (i) GP-evolved arm delivered in full at GenMorph's published 30-min budget on a single seed (seed = 11); the single-seed and author-designed-substrate concerns are addressed by an additional paired multi-seed check on GenMorph's *own* published 23-method benchmark (the `gcd`/`sin` subset, all 12 published GP seeds; supplementary `routeB`). Under this neutral substrate Set G's dominance is seed-robust (pooled McNemar exact $p = 2.97 \times 10^{-11}$ on `gcd` and $p = 0.012$ on `sin`; Set N $\geq$ Set G on only 4/12 `gcd` seeds), and seed = 11 is among the most Set N-favourable seeds (Set N's highest count, 13, against one of Set G's lowest, 11), so the single-seed head-to-head does not overstate Set G's advantage. (ii) LLM-assisted arm delivered as a 2-vendor (DeepSeek-V3, ChatGPT-4o-mini) $\times$ 5-temperature ensemble (100 samples); the third commercial closed-vendor frontier (Anthropic Claude) is committed as post-acceptance follow-up (d.set-l-claude), and the current Set L results are reported as a 2-vendor proxy rather than the SOTA-strongest configuration. (iii) Mining-based arm reported through an adapted estimate of MR-Scout's reach derived from its published artifact figures; full re-execution committed as follow-up (d). METRIC+ [52] is not run head-to-head because no automated identification pipeline is publicly available; the qualitative-plus-quantitative coverage contrast appears in §5.2 (sorting worked example) and §5.3.3 (per-block coverage diagnostic). The cost-component breakdown of all four methods is in Table 17.

*5.3.9 MR-generation cost.* NOETHER's cost profile differs qualitatively from each of the three SOTA-category representatives across four independent components (algorithmic time, human effort, LLM-token cost, seed-corpus dependency); Table 17 reports each separately, with full derivation methodology (token-cost protocol scale, per-SUT human-effort breakdown, MR-Scout seed-suite scope) in supplementary S4 (`cost_breakdown.md`).

Table 17. MR-generation cost components for the four methods compared in this section. "Cost dimension" rows are independent (an LLM-arm token cost does not amortise GP wall-time and vice versa); the NOETHER human-effort estimate aggregates Set N's derivation of 30 MRs across 10 SUTs at $\approx 1$ h per SUT under CONSTRUCT-MP's four-step procedure, amortising across SUTs that share an operator algebra. Full methodology in supplementary S4 (`cost_breakdown.md`).

| Cost component | NOETHER | GP (GenMorph) | LLM (Xu 2024) | Mining (MR-Scout) |
|---|---|---|---|---|
| Algorithmic time / SUT | poly(\|gen\|) minutes (Thm. 2) | 30 min wall, stochastic | seconds API latency | minutes after corpus |
| Human effort / family | $\approx 10$ h $\mathcal{A}_P$ distillation | none after harness setup | none after prompt template | seed-suite assembly substantial |
| Token cost (USD) / family | 0 | 0 | \$5–50 (\$50–500 with CoT) | 0 |
| Seed-corpus dependency | none | none | none | required (mining input) |
| Determinism | deterministic | seed-/budget-dependent | prompt-/temperature-dependent | corpus-dependent |
| Cold-start capable | yes | yes (after harness) | yes (after prompt) | no (needs corpus) |

The table operationalises three structural advantages of NOETHER as distinct cost-axis arguments rather than as a single "best" verdict: (i) vs. GP, polynomial-time deterministic construction (Theorem 2) replaces a 30-min stochastic per-SUT search, with one-time human cost amortising across same-algebra SUTs; (ii) vs. LLM-assisted, the coverage$_{\text{NOETHER}}$ diagnostic on §5.2 (1.00 vs. 0.40) operationalises the algebraic-prior gap that prompt-based generation does not close, at zero token cost; (iii) vs. mining-based, NOETHER is operative at cold start (no seed test suite required), the regime in which most algebra-rich scientific-computing programs arrive on a tester's desk. These are cost-profile and coverage-profile claims that hold within the framework's scope precondition, not fault-detection-superiority claims.

*Approximate-parity-at-lower-cost reading (H3a.3 verdict).* H3a.3's cost-axis claim holds when detection is read on the per-block primary (Table 15). On $G_{\mathrm{tr}}$, Set N edges Set G with three $N$-only kills at zero per-SUT generation cost beyond the one-time CONSTRUCT-MP derivation ($\approx$ 1 h human, amortising across SUTs sharing $\mathcal{A}_P$); Set G's matching count costs a 30-min GAssert GP-search per SUT ($\approx$ 4 h wall for the eight SUTs in parallel-4) plus the Major + JaCoCo + Randoop pipeline assembly. On $\mathcal{L}^*$, the sets are complementary ($N$-only 2, $G$-only 8 on 24); Set N's reach is reproducible without re-running GP, whereas Set G requires re-tuning budget and seed when the SUT changes. *Quantitative reading*: a directional $G_{\mathrm{tr}}$ edge (10/17 vs 8/17, $n = 17$ underpowered) plus $\mathcal{L}^*$ complementarity, at $\approx$ 1 h amortised human cost versus $\approx$ 30 minutes per-SUT GP search; NOETHER's cost amortises across new SUTs in the same algebra family while Set G's restarts from scratch. The per-block edge does not overturn Set G's D1 aggregate dominance (acknowledged at the head of §5.3.7); the $G$-block gap on the Euclidean SUTs ($G$-block framework boundary on recursive-normalising SUTs) is a framework boundary, not a cost-axis trade-off. H3a.3 is therefore *supported*: comparable within-scope per-block detection at lower amortised generation cost.

*Committed future work (16 items).* The full committed-future-work table (16 entries) is in supplementary S4 (`future_work.md`); key items completed: (i) METRIC+ head-to-head on Sun 2021's 4 subjects with PIT 1.7.4 + Major cross-tool replication (pooled McNemar $p = 0.625$ / 0.211 both NS, 92.6% both-kill on the PIT substrate; details in supplementary S8); (b.cm) Commons-Math pilot (pooled Set N 10/77 = 13.0%, D2 prediction passes 2/29 = 6.9%); (d.set-l) 2-of-3-vendor LLM-ensemble harvest (487 MRs, 212 executable, ensemble matches Set N on the matchable subset 34/34); (e), (e.2), (e.4), (g) the D1/D2 labelling pipeline, equivalent-mutant exclusion, $G$-block Euclidean-SUT documentation, and construct-trace consistency check. The remaining 8 pending items (multi-seed GP replication; full 38-D4J extension; MR-Scout re-execution; Anthropic third-vendor; generic-mutation independent test on the 5 PIT-unexercised blocks; external-transfer reactor-physics corpus; two narrow grid-MR predicate tightenings) are the principal post-publication work programme.

*5.3.10 Summary of evidence.* The section's central claim is a single quantitative falsifiable prediction confirmed: $\mathcal{L}^*$-block blindness on homogeneity-preserving mutators (§5.3.4). The prediction is derivable ex-ante from §3.2 and from PIT's public mutator specification; it is observed in the data on 5/6 SUTs admitting an $L_{\mathrm{scale}}$ MR; the sixth SUT's exception on two mutants is accounted for by the prediction's quantitative tail. The operative-mechanism reading of the structural decomposition is, in this empirical sense, supported.

The corroborating evidence is fourfold:

- Other per-block patterns (the $G$ symmetry and $G_{\mathrm{tr}}$ translation sub-classes, plus the out-of-frame $\mathcal{I}^*$ probe; §5.3.5) are qualitatively consistent with the algebraic predictions.
- Cross-pipeline MR rediscovery on `midpoint` (§5.3.6, witness 1) shows two independent epistemic processes converging on the same algebraic primitives.
- Structural coverage extension on the two SUTs where Set G is structurally absent (§5.3.6, witness 2) is an architectural consequence of algebra-derivation that the head-to-head substrate incidentally reveals.

- Head-to-head against GenMorph (§5.3.7) reads per-block, not as a single winner. On $G_{\mathrm{tr}}$ Set N shows a directional edge (10/17 vs 8/17, three exclusive kills) consistent with its design prediction; on $\mathcal{L}^*$ Set N and Set G are complementary (union 18/24; Set N exclusively reaches `exactLog2`, Set G exclusively reaches denser regions of `lcmSig` / `gcdSig`); on $G$ Set G dominates (9/11 vs 2/11) because the substrate's two Euclidean- style SUTs absorb the sign-flip invariant in their normalisation prologue (Table 15, Set N 0/7 on `gcdSig` + `lcmSig`, a $G$-block reading on D1 mutants, not the D2 prediction). The framework's falsifiable D2 prediction (kill rate $\leq$ 10%) is consistent with the data (Set N D2 kill rate 0/5 = 0.000, Wilson 95% CI [0.000, 0.434]; $n = 5$ is insufficient for $\alpha = 0.05$ confirmation, but the observed direction is on-prediction); the per-block reading is the appropriate substrate-level contribution metric.

The full per-SUT outcome matrix, the per-block kill table for all 30 Set N MRs, the $L_{\mathrm{scale}}$-mutant attribution for `hypotSig`, the GP raw output and the `.jir`/`.jor` translation, the McNemar $b/c$ counts, the prediction's git timestamp, and the test-gate harness (`tests/run.sh`) are in supplementary S7 (`d4j/`).

### 5.4 Relationship with METRIC and METRIC+

METRIC and METRIC+ pioneered the categorical scaffolding that any structural theory of MR identification must respect; NOETHER's MetaPatterns are descendants of that line of work. The point of departure is grounding. METRIC and METRIC+ derive their categories through expert curation and validate them by empirical coverage. NOETHER derives MetaPatterns through algebraic construction and proves closure over the algebra-induced MR space through Theorem 1.

*Worked example: METRIC+ categories for a sorting library, mapped onto NOETHER blocks.* For a numerical comparison-sort library, METRIC+'s eleven input-domain $\times$ output-relation category pairs (the 9-category METRIC base extended by 2 output-domain pairs in METRIC+, per [52] Table II) collapse onto two NOETHER blocks: $G$ (comparator-permutation symmetry $\mathfrak{S}_n$) and $O_{\leq}$ (partial order). Table 18 records the mapping cell by cell: 9 of 11 category pairs map to $G$ or $O_{\leq}$; the remaining 2 ($D_5$ additive shifts, $R_4$ multiplicative scaling) are out-of-scope by construction for a comparison sort whose semantics ignore order-preserving univariate input transformations. The 11 $\rightarrow$ 2-block compression operationalises NOETHER's deflationary direction: a NOETHER-aware METRIC+ user can prune the 9 category-pair enumerations to two algebraically distinct MetaPatterns ($m_{\mathrm{inv}}$ via $G$, $m_{\mathrm{mono}}$ via $O_{\leq}$) without loss of MR coverage. This is a qualitative-plus-quantitative coverage contrast, not a head-to-head fault-detection comparison: METRIC+ is a category-enumeration scaffold without an automated identification pipeline, so kill-rate head-to-head requires re-implementing METRIC+ from prose (committed as follow-up (i) of supplementary S4).

*Small-scale manual METRIC+ derivation on three §5.3.3 SUTs.* To strengthen the qualitative-plus-quantitative reading above, we apply METRIC+'s 11 D$\times$R category framework manually to three SUTs of the §5.3.3 substrate selected to span bivariate input arity and distinct algebraic structure: `midpoint` (bivariate linear), `hypotSig` (bivariate non-linear, scale-symmetric), and `powerSig` (bivariate non-linear, multiplicative). For each SUT we enumerate which METRIC+ category pairs yield a non-vacuous Set-MP MR; classify the algebra-block of each non-vacuous Set-MP MR under $\mathcal{D}(\mathcal{A}_P)$; and identify which Set-N MR (if any) covers the same algebraic content. Table 19 records the analysis. The full PIT-based head-to-head

Table 18. Mapping of 11 input-domain ($D$) × output-relation ($R$) METRIC+ category pairs [12, 52] onto NOETHER blocks for a deterministic comparison-sort library under $\mathcal{A}_{\text{sort}}$. The sort-specific specialisation is operationalised here; 9 pairs collapse onto two non-empty blocks ($G$, $O_{\le}$); 2 pairs ($D_5$, $R_4$) are out-of-scope.

| METRIC+ category pair | NOETHER block | Comment |
|---|---|---|
| $D_1$ (permutation of input list) | $G$ ($\mathfrak{S}_n$) | $m_{\text{inv}}$: sort output invariant |
| $D_2$ (append element to input) | $O_{\le}$ | $m_{\text{mono}}$: sort respects added element |
| $D_3$ (remove element from input) | $O_{\le}$ | $m_{\text{mono}}$: inverse of $D_2$ |
| $D_4$ (replace element with permuted copy) | $G$ ($\mathfrak{S}_n$) | $m_{\text{inv}}$: variant of $D_1$ |
| $D_5$ (add constant to all inputs) | out-of-scope | no MR (sort semantics ignore additive shifts) |
| $D_6$ (concatenate two sorted lists) | $G$ ($\mathfrak{S}_n$) | $m_{\text{inv}}$: merge property |
| $R_1$ (output element-wise equality) | $G$ | identity oracle (degenerate MR) |
| $R_2$ (output permutation equality) | $G$ ($\mathfrak{S}_n$) | $m_{\text{inv}}$ |
| $R_3$ (output prefix equality) | $O_{\le}$ | $m_{\text{mono}}$ on prefixes |
| $R_4$ (multiplicative scaling of input) | out-of-scope | no MR (sort semantics ignore multiplicative scaling) |
| $R_5$ (subset relation on output) | $O_{\le}$ | $m_{\text{mono}}$ on subsets |
| *Coverage summary* | $9 \to G \cup O_{\le}$, 2 out-of-scope | NOETHER block compression: $11 \to 2$ non-empty MetaPatterns |

kill-rate comparison is committed as supplementary S4 item (i); the present manual analysis is the algebra-block mapping component of that follow-up and is reported here to make the framework-vs-METRIC+ coverage relationship concrete on the head-to-head substrate.

Table 19. Manual METRIC+ D×R framework applied to three SUTs of the §5.3.3 substrate (bivariate input arity). For each (SUT, METRIC+ pair) cell: "–" if vacuous on that SUT's algebra; "→ block" identifies the NOETHER block of the non-vacuous Set-MP MR; the second column lists the Set-N MR (if any) covering the same algebraic content. **Across the three SUTs, METRIC+ yields 6, 5, and 3 non-vacuous Set-MP MRs respectively (the remaining 5, 6, and 8 D×R pairs are vacuous because the SUT's arity is fixed, the output is scalar, or the input-transformation is structurally out-of-scope for the SUT's algebra); each non-vacuous Set-MP MR maps to a NOETHER block also covered by Set-N's 5 algebra-derived MRs. Conversely, three Set-N MRs ($m_{\text{adj}}$, $m_{\text{train-rev}}$, $m_{\text{conv}}$) have no METRIC+ D×R counterpart.** The structural reading: NOETHER's $G/O_{\le}$ block coverage subsumes METRIC+'s $D_1$–$D_6$/$R_1$–$R_5$ output on bivariate-input SUTs in this substrate, while NOETHER's $T^*/\mathcal{T}^*/\mathcal{L}^*$ blocks add MR templates the D×R framework does not enumerate. The kill-rate projection in the rightmost column anticipates that on these three SUTs Set-MP's reach would be bounded above by Set-N's $G/O_{\le}$ contribution; a full PIT execution to confirm this is committed as follow-up (i).

| METRIC+ pair | `midpoint` | `hypotSig` | `powerSig` |
|---|---|---|---|
| $D_1$ (input swap) | $\to G$ ($\rho_{\text{swap}}$ via $m_{\text{inv}}$) | $\to G$ ($\rho_{\text{swap}}$) | – (non-symmetric) |
| $D_2$ (append element) | – (fixed arity) | – (fixed arity) | – (fixed arity) |
| $D_3$ (remove element) | – (fixed arity) | – (fixed arity) | – (fixed arity) |
| $D_4$ (replace with permuted) | $\to G$ (variant of $D_1$) | $\to G$ | – |
| $D_5$ (add constant to inputs) | $\to O_{\le}$ ($\rho_{\text{trans}}$ via $m_{\text{mono}}$) | out-of-scope (non-linear) | out-of-scope |
| $D_6$ (scale inputs) | $\to O_{\le}$ ($\rho_{\text{scale}}$ via $m_{\text{mono}}$) | $\to \mathcal{L}^*$ ($\rho_{\text{scale}}$ via $m_{\text{conv}}/L_{\text{scale}}$) | $\to \mathcal{L}^*$ |
| $R_1$ (output element equality) | $\to G$ (identity oracle) | $\to G$ | $\to G$ |
| $R_2$ (output permutation eq.) | – (scalar output) | – (scalar) | – (scalar) |
| $R_3$ (output prefix equality) | – (scalar) | – (scalar) | – (scalar) |
| $R_4$ (output multiplicative) | $\to O_{\le}$ | $\to \mathcal{L}^*$ | $\to \mathcal{L}^*$ |
| $R_5$ (output subset) | – (scalar) | – (scalar) | – (scalar) |
| *Set-MP non-vacuous* | 6 / 11 | 5 / 11 | 3 / 11 |
| *Set-MP → blocks* | $\{G, O_{\le}\}$ | $\{G, \mathcal{L}^*\}$ | $\{G, \mathcal{L}^*\}$ |
| *Set-N coverage of those blocks* | yes ($m_{\text{inv}}$, $m_{\text{mono}}$) | yes ($m_{\text{inv}}$, $L_{\text{scale}}$) | yes ($m_{\text{inv}}$, $L_{\text{scale}}$) |
| *Set-N MRs without Set-MP counterpart* | $m_{\text{adj}}$, $m_{\text{train-rev}}$, $m_{\text{conv}}$ (at limits) | same three blocks | same three blocks |

The structural finding from Table 19 is that METRIC+'s D×R framework, *within Sun et al. 2021's published input–output category catalogue of 11 pairs [52] as instantiated on these three SUTs*, produces a strict subset of NOETHER's block coverage on the bivariate-input subsample of the §5.3.3 substrate: every non-vacuous Set-MP MR maps to a NOETHER block already covered by Set-N's algebra-derived MRs, and three Set-N blocks ($T^*$, $\mathcal{T}^*_{\text{rev}}$, $\mathcal{L}^*$ at limits) have no METRIC+ counterpart *at this category-pair granularity.*

The implication for fault detection on these three SUTs is that Set-MP's expected kill rate on $\mathcal{T}^*$ or $\mathcal{T}^*_{\text{rev}}$-violating mutants is zero by construction, whereas Set-N's $\rho_{\text{adj}}$ and $\rho_{\text{train-rev}}$ MRs are by-construction sensitive to such mutants (cf. §5.3.7's $G_{\text{tr}}$ translation sub-class where Set-N kills 10/17). The full PIT-based head-to-head against Set-MP (supplementary S4 item (i)) would convert this structural prediction into a measured kill-rate contrast; the present small-scale manual analysis establishes the algebra-block component of that contrast on three representative SUTs.

*Scope-precondition validation on the METRIC+ benchmark corpus.* A complementary, more directly comparator-side analysis applies NOETHER's structural decomposition to Sun et al.'s four benchmark subjects [52]: `SPHONE` (China Unicom phone-bill calculator, 107 LOC), `SBAGGAGE` (Air China baggage-billing service, 101 LOC), `SEXPENSE` (sales-department expense reimbursement, 117 LOC), and `SMEAL` (airline catering meal-ordering service, 150 LOC). All four are business-rule billing programs whose semantics are categorical+numeric rather than mathematical/physical. Applying NOETHER's structural test to each subject's category-choice specifications yields the per-block non-emptiness verdicts in Table 20.

Table 20. NOETHER structural scope analysis on Sun et al. 2021 METRIC+ benchmark subjects [52]. Per-block entries: **Y** non-empty (**Y**), **P** partial (**P**, block applies on a restricted regime such as linear-pricing tier or within-class permutation), – empty. Source-code-level instance MR cardinality (last column) is reported by Sun et al. 2021 Table 17; NOETHER's MetaPattern cardinality (second-to-last column) is the equivalence-class summary at the algebra-block level. The two cardinalities differ by 2–3 orders of magnitude because METRIC+'s output scales combinatorially with category-choice products while NOETHER's output is an equivalence-class summary; the analysis below interprets this contrast as *complementary* rather than *competitive*.

| Subject | $G$ | $O_\leq$ | $T^*$ | $\mathcal{T}^*_{\text{rev}}$ | $\mathcal{L}^*$ | $\mathcal{D}^*$ | $\mathcal{E}^*$ | $\mathcal{B}^*_{\text{rel}}$ | NOETHER MPs | METRIC+ MRs |
|---|---|---|---|---|---|---|---|---|---|---|
| `SPHONE` | – | **Y** | – | – | **P** | – | – | – | 2 | 142 |
| `SBAGGAGE` | **P** | **Y** | – | – | **P** | – | – | – | 3 | 735 |
| `SEXPENSE` | – | **Y** | – | – | **P** | – | – | – | 2 | 1130 |
| `SMEAL` | **P** | **Y** | – | – | **Y** | – | – | – | 3 | 3152 |

*Legend:* **Y** = non-empty block; **P** = partial (restricted regime); – = empty. "NOETHER MPs" = Set N MetaPattern cardinality; "METRIC+ MRs" = instance-level MR cardinality reported by Sun et al. 2021 (Table 17), comparable only after either expanding NOETHER MetaPatterns to instance MRs (via the category-choice enumeration) or contracting METRIC+ MRs to algebra-block equivalence classes.

The analysis establishes three findings. First, NOETHER is *in-scope* on all four METRIC+ subjects, contradicting the worst-case reading that NOETHER applies only to mathematical or physical program families: business-rule billing programs admit at least the $O_\leq$ block (monotonicity in usage / weight / mileage / passenger count) and partial $\mathcal{L}^*$ (within linear pricing tiers), with $G$ also activating on `SBAGGAGE` and `SMEAL` where input-permutation invariance holds. Second, NOETHER's reach is *narrower* on this corpus than on its three primary instantiation domains: 2–3 of 8 blocks per subject yields 2–3 MetaPatterns, contrasted with 5–7 of 8 blocks on Boltzmann reactor physics and equivariant ML. The scope precondition is therefore neither vacuously broad nor practically empty; it is a continuous gradient in the program family's algebraic richness. Third, the cardinality contrast between NOETHER MetaPatterns (2–3) and METRIC+ instance MRs (142–3152) reflects different but complementary output types: METRIC+'s 11 D×R framework generates instance-level MRs by category-choice combinatorics; NOETHER's structural decomposition produces equivalence-class summaries over the same MR space. A fair head-to-head comparison requires expanding NOETHER's MetaPatterns to instance MRs at matching cardinality or contracting METRIC+'s MRs to equivalence classes at the algebra-block level; the present analysis establishes the algebra-block coverage condition that every NOETHER block on these subjects ($G$, $O_{\leq}$, $\mathcal{L}^*$) maps to a METRIC+ D×R category pair, while three NOETHER blocks that are active on its primary instantiation domains ($T^*, \mathcal{T}^*_{\text{rev}}, \mathcal{D}^*$) are structurally absent from Sun et al.'s business-rule benchmark. Detailed per-subject derivation, including the regime restrictions for partial **P** blocks, is in supplementary S8 (`scope_analysis.md`). The pre-registered protocol for an instance-level head-to-head with matched cardinality (supplementary S8 `protocol_path_a_headtohead.md`) is *executed at reduced scale* and reported below; the full-scale execution (Sun's Java + PIT 1.7.4 + multi-LLM equivalent-mutant vote) remains as supplementary S4 (`future_work.md`) item (i).

*Path A head-to-head: Java/PIT execution on Sun 2021's corpus.* The pre-registered Path A protocol (supplementary S8 `protocol_path_a_headtohead.md`) was executed in two substrates: a reduced-scale Python AST mutation engine (`results_path_a.md`; $n = 219$ mutants) and the full Java + PIT 1.7.4 substrate matching this paper's §5.3.3 tooling (`results_path_a_full.md`; $n = 120$ PIT mutants). We report the Java/PIT results as the primary head-to-head; the Python reduced-scale run is the cross-substrate replication. Java re-implementations of `SPHONE`, `SBAGGAGE`, `SEXPENSE`, `SMEAL` from Sun 2021's prose specification (Tables 7–14) were exercised under both Set N (NOETHER, $\{17, 12, 19, 10\}$ JUnit MR tests via CONSTRUCT-MP + Translate enumeration) and Set MP (METRIC+, $\{61, 15, 19, 9\}$ JUnit MR tests via automated D×R category enumeration); all 162 MR tests pass on the original code. PIT 1.7.4 with the stock mutator catalogue (`DEFAULTS` + `fullMutationMatrix`) generated $14+36+17+53 = 120$ mutants. Pooled head-to-head: Set N $51/120 = 42.5\%$ Wilson 95% CI $[34.0\%, 51.4\%]$ vs Set MP $53/120 = 44.2\%$ $[35.6\%, 53.1\%]$; McNemar exact two-sided $p = 0.625$ (**not rejecting** $\text{H}_{\text{MP2}}$ parity at $\alpha = 0.05$). Per-subject McNemar exact $p$ is 0.500 on SPHONE, 1.000 on the other three; with Bonferroni-correction at $\alpha_{\text{Bonf}} = 0.0125$ **no subject rejects parity**. The complementarity 4-tuple pooled is (both, $N$-only, $MP$-only, neither) $= (50, 1, 3, 66)$: of the 54 mutants killed by either set, $50/54 = 92.6\%$ are killed by both, confirming the two frameworks reach near-identical fault-detection power on this corpus. $\text{H}_{\text{MP1}}$ coverage-subsumption is **not falsified** at the Java/PIT scale; the Python reduced- scale SPHONE falsification (9 MP-only kills, $p = 0.0039$) does not replicate under PIT 1.7.4's bytecode mutators, indicating the prior result was a Python AST mutation artefact. The scope-precondition prediction of `scope_analysis.md` holds intact: only $\{G, O_{\leq}, \mathcal{L}^*\}$ are active on this corpus, with five blocks ($T^*, \mathcal{T}^*_{\text{rev}}, \mathcal{D}^*, \mathcal{E}^*, \mathcal{B}^*_{\text{rel}}$) structurally absent. $\text{H}_{\text{MP3}}$ cost-axis asymmetry is directionally supported (Set N 58 JUnit tests vs Set MP 104 at this enumeration scale; Sun's published full cardinality 142–3152 would multiply METRIC+'s side by another 30–50× without altering NOETHER's 8-block MetaPattern bound). Three remaining protocol deviations at Java/PIT scale (Java re-impl vs Sun's original; MR enumeration below Sun's full cardinality; multi-LLM equivalent-mutant vote not run) are recorded in `results_path_a_full.md §6`; all are data-blind and would not flip the verdict if resolved.

The Path A Java/PIT execution unambiguously *strengthens* this section's complementarity reading: 92.6% both-kill rate, pooled McNemar $p = 0.625$, no per-subject Bonferroni rejection, no falsified hypothesis at the Java/PIT scale. The two frameworks empirically achieve comparable reach on Sun 2021's published corpus, with the framework's contribution being the algebraic warrant for each MR plus the cost-axis advantage of polynomial-time derivation from $\mathcal{A}_P$ rather than combinatorial category enumeration.

A companion *cross-tool replication* on the Major mutation framework (supplementary S8 `results_path_a_major_crosstool.md`) extends the substrate from PIT 1.7.4's $n = 120$ mutants to Major's $n = 555$ mutants (Major's $\approx$ 95-operator catalogue, vs PIT `DEFAULTS`' $\approx 25$, yields a $4.6\times$ larger pool on the same Java sources and JUnit suite). Major's verdict at the pooled level is *concordant* with PIT's: Set N $222/555 = 40.0\%$ Wilson 95% $[36.0\%, 44.1\%]$ vs Set MP $231/555 = 41.6\%$ $[37.6\%, 45.8\%]$, McNemar exact $p = 0.211$ (NS at $\alpha = 0.05$). Per-subject, Major's larger pool exposes *bidirectional reach asymmetries* invisible to PIT: Set MP has 20 $MP$-only kills on SPHONE (McNemar $p = 0.0000$, Set MP exclusive reach driven by (D1, R1) within-partition equivalence on the tier-pricing structure), while Set N has 16 $N$-only kills on SBAGGAGE ($p = 0.0044$, Set N exclusive reach driven by $G$-block special-status invariance $+\ \mathcal{L}^*$ overweight scaling). These asymmetries are *directionally opposite* and *roughly cancel* at the pooled level: $\mathrm{H}_{\mathrm{MP1}}$ is *falsified in both directions per-subject*, which is the strongest possible evidence that neither framework subsumes the other. The pooled parity verdict ($\mathrm{H}_{\mathrm{MP2}}$) is therefore not PIT-specific and not a statistical happenstance: it is the aggregation of substantive subject-specific complementary reaches that two independent mutation tools confirm.

### 5.5 Reassessing PMCM coverage claims: a worked example

NOETHER does not replace PMCM; it re-grounds it. Under inductive grounding, PMCM's pattern grid is itself an empirical artefact: its rows are the inductive MetaPatterns of a particular catalogue, and "100% coverage" attests only that every row has at least one instance. Under NOETHER's grounding, the grid becomes $\mathbb{M}(\mathcal{A}_P)$, algebraically constructed and closed under the operator algebra in the sense of Theorem 1, and the coverage figure becomes a structural rather than nominal claim. This subsection works through the implication on two concrete cases. The deflationary direction is, in our view, NOETHER's most concrete contribution to testing practice and the most resistant to the circularity caveat of Section 3.4.3: revealing that an inductive catalogue *over-counts* structurally distinct patterns does not require $T^*$ or $\mathcal{T}^*$ to have been derived independently from physics, since the over-counting is exposed by the canonical-block ordering itself.

*Case A: A comparison-sort library.* Consider a numerical sorting library, and assume an inductive Pattern–Matrix Coverage catalogue with rows {P1 conservation/invariance, P2 monotonicity, P3 convergence, P4 trajectory, P5 partial-order/bounding} inherited from the reactor-physics MetaPattern taxonomy of §3.4.3. A user reports "100% coverage of the grid with five rows" on a particular MR set.

*NOETHER re-decomposition.* The operator algebra $\mathcal{A}_{\mathrm{sort}}$ for a deterministic comparison sort has: a permutation-symmetry block $G \ni \mathfrak{S}_n$ (the input–output relation is permutation-equivariant in the comparator-permutation sense), a partial-order block $O_{\leq}$ (sorting respects the input order on already-sorted sub-arrays), an empty self-adjoint block $T^*$ (no reciprocity structure), an empty time-reversal block $\mathcal{T}^*$ (irreversible), an empty limit block $\mathcal{L}^*$ (sort is exact, not asymptotic), an empty qualitative-dynamics block $\mathcal{D}^*$, and an empty method-comparison block $\mathcal{E}^*$ *within a single algorithm* (and non-empty if the user is comparing multiple sorting algorithms). Therefore $\mathbb{M}(\mathcal{A}_{\mathrm{sort}}) = \{\, m_{\mathrm{inv}},\, m_{\mathrm{mono}} \,\}$, of size 2.

*Coverage correction.* The PMCM grid with five rows carries three structurally empty rows under $\mathcal{A}_{\mathrm{sort}}$: P3 (convergence) is vacuous because there is no $\mathcal{L}^*$ generator; P4 (trajectory) is vacuous because there is no $\mathcal{D}^*$ generator; P5 (partial-order/bounding) is vacuous because there is no $\mathcal{E}^*$ generator within a single-algorithm scope. The denominator should be 2, not 5. "100% coverage of the grid with five rows" is therefore not a claim about the user's MR set; it is a claim about the inductive catalogue's row count. The structurally meaningful coverage is "$k/2$ of the algebraically required MetaPatterns covered", where $k$ depends on the user's actual MR set.

*Case A-bis: An external ML category set (Murphy et al. 2008) decoded against a feedforward classifier.* To rule out the self-selection concern, we apply the same decoding to Murphy et al.'s categorisation of MRs for ML applications into six classes [38, 42, 59], an independently published taxonomy (additive, multiplicative, permutative, invertive, inclusive, exclusive). For a generic feedforward image classifier $f : \mathbb{R}^n \to \{1, \dots, K\}$ with no inherent input symmetry, the induced algebra $\mathcal{A}_{\mathrm{FFN}}$ has only a single non-trivial generator under small additive noise (in $O_{\leq}$ or $\mathcal{L}^*$), so $\mathbb{M}(\mathcal{A}_{\mathrm{FFN}}) \subseteq \{m_{\mathrm{stab}}\}$. Three of Murphy's classes (multiplicative, permutative, invertive) are *structurally vacuous* on a vector image classifier without input symmetries; the remaining three (additive, inclusive, exclusive) collapse to the same $m_{\mathrm{stab}}$ MetaPattern. The structurally meaningful denominator for "coverage of Murphy's six classes" on a generic vector image classifier is therefore 1, not 6; full per-class decoding (including the antipodal $\mathbb{Z}/2$ exception, point-cloud / bag-of-features cases, and the bilinearity caveat) is in supplementary S9 (`pmcm_case_abis_full.md`). Murphy et al. explicitly intended their six classes as a checklist for selecting MRs, not as a coverage denominator; the target of this correction is subsequent papers that report Murphy-grid coverage figures on tasks lacking the relevant input symmetries [42]. The deflationary direction here is independent of $T^*$ and $\mathcal{T}^*$, so does not inherit the prediction-circularity caveat of §3.4.3.

*Case B: A reactor-physics solver.* The reactor-physics MetaPattern catalogue discussed in Section 3.4.3 provides a less obvious case. The catalogue has 5 rows (P1–P5). Under NOETHER's re-grounding into $\mathbb{M}(\mathcal{A}_{\mathrm{Boltz}}) = \{m_{\mathrm{inv}}, m_{\mathrm{mono}}, m_{\mathrm{adj}}, m_{\mathrm{rev}}, m_{\mathrm{conv}}, m_{\mathrm{dyn}}, m_{\mathrm{cmp}}\}$, the catalogue is in fact *under-counted* by two rows ($m_{\mathrm{adj}}$ and $m_{\mathrm{rev}}$ are absent), even though for the Bateman-only sub-family it would be over-counted (P4 trajectory and P5 partial-order/bounding are absorbed differently into $\mathcal{D}^*$ and $\mathcal{E}^*$). The deflationary direction is thus not uniformly toward smaller grids; it is toward *the algebraically determined grid for the algebra in question*, which may be larger or smaller than the inductive grid.

*Quantification.* For a fixed program family $\mathcal{F}$ with operator algebra $\mathcal{A}_{\mathcal{F}}$, define

$$\mathrm{coverage}_{\mathrm{NOETHER}}(\mathcal{R}, \mathcal{F}) = \frac{|\{\, m \in \mathbb{M}(\mathcal{A}_{\mathcal{F}}) \mid m \cap \mathcal{R} \neq \emptyset \,\}|}{|\mathbb{M}(\mathcal{A}_{\mathcal{F}})|}$$

for an MR set $\mathcal{R}$. This is the algebraically grounded analogue of PMCM's empirical coverage. Where the inductive PMCM grid agrees with $\mathbb{M}(\mathcal{A}_{\mathcal{F}})$, the two metrics coincide; where they disagree, the NOETHER metric is the structurally meaningful one. We note that $\mathrm{coverage}_{\mathrm{NOETHER}}$ inherits Theorem 1's scope: it is a coverage measure over algebra-induced MRs, not over arbitrary properties; out-of-scope MRs (Appendix A) are by construction outside the denominator.

*Implication for empirical-adequacy practice.* Reports of high pattern-grid coverage in published MR-evaluation studies should, where possible, be cross-checked against the algebraic decomposition of the program family under test. A report of "coverage $= c$" should be qualified with which row decomposition

$c$ is computed against; reports of $c$ over inductive catalogues without such qualification are not directly comparable across papers.

### 5.6 Empirical evidence for the Invariance-Blindness Theorem

This subsection reports the empirical evidence supporting Theorem 3 and its corollaries. Three threads are tested on independent material: (E1) the linear-algebraic Faithfulness condition (Definition 18) at finite test budget; (E2) detection complementarity with a neutral cross-implementation differential oracle (the differential-oracle corollary of §3.3); (E3) the discretisation-floor regime in which the differential oracle has zero recall under a false-positive budget. The full harness, mutation catalogues, paired analyses, and raw outputs are available in supplementary material S10 (`supplementary/S10_noether_homefield/`) and are not re-derived here.

*E1: Faithfulness rank check (`fa_rank_check.py`, `fa_block_classification.py`).* For the fault class $\Theta = \mathbb{R}^{N\times N}$ ($N = 8$), the defect $E_s(L)$ is linear in $L$ and Faithfulness reduces to $\mathrm{rank}(J_{\mathrm{test}}) = \mathrm{rank}(J_{\mathrm{full}})$. The check is performed on three symmetry instances and on the self-adjoint block, with analytic kernel dimensions for cross-validation:

- Translation $\mathbb{Z}_N$: 1 generator pins the commutant of all 7 group elements (rank $56 = 56$); kernel dimension $8 = N$ (circulant subspace).
- Reflection $\mathbb{Z}_2$: 1 generator (rank $32 = 32$); kernel dimension $32 = 2(N/2)^2$.
- Dihedral $D_N$: 2 generators pin the commutant of all 16 group elements (rank $59 = 59$).
- Self-adjoint $T^*$ ($E(L) = L - L^\top$): the upper-triangle witness set (15 rows) pins the dense witness set (36 rows) at rank $15 = 15$; kernel dimension $21 = N(N+1)/2$ (symmetric matrices).

Faithfulness holds in every case at the analytic kernel dimension, confirming the Reachability Lemma (Lemma 1) numerically for $\{G, T^*\}$. The per-block classification confirms that the remaining blocks ($O_\leq$, $\mathcal{T}^*_{\mathrm{rev}}$, $\mathcal{L}^*$) fall outside the linear-equality regime of Theorem 3 (cone, matrix-inverse, and norm-ratio nonlinearity respectively); only the sufficient direction applies to those blocks.

*E2: Detection complementarity (paired McNemar over three SUTs).* A pooled paired analysis of the algebra-MR battery against a neutral cross-implementation differential oracle is run on three multi-implementation SUTs covering thermal and fluid PDE families (2-D advection–diffusion, multigroup radiation diffusion at $G = 2$, and 2-D Gray–Scott reaction–diffusion; `paired_pooled_analysis.py`). Per-SUT McNemar exact tests are reported in supplement S10; the pooled discordants ($b$ = MR-only, $c$ = differential-only) and the stratified Cochran–Mantel–Haenszel test are:

- Total real mutants $N = 104$ across three strata; pooled MR-killed 79, differential-killed 50, union 89 (85.6%), neither 15;
- Pooled $b = 39$ (MR-only) and $c = 10$ (differential-only); pooled McNemar exact $p = 3.85\times10^{-5}$, stratified CMH $\chi^2(1) = 17.16$ ($p = 3.43 \times 10^{-5}$);
- $b > 0$ *and* $c > 0$ in every stratum (kernels cross, not nest): the two oracles are complementary in the sense of the differential-oracle corollary of §3.3.

The pooled asymmetry $b \gg c$ records that the MR battery has higher raw recall than the differential oracle on the tested mutation catalogue; the substantive claim under the joint-kernel corollary of §3.3 is the strict union improvement (union $= 89 > 79$ = best single oracle), which records the predicted joint-kernel reduction. Honest qualification: high MR recall is a feature of this mutation catalogue, not of the framework in general; we do not read the asymmetry as differential-oracle inferiority.

*E3: Discretisation floor and false-positive regime.* On the 2-D advection–diffusion SUT the differential oracle is run under a tolerance $\tau = \text{SAFETY} \cdot \delta$, where $\delta$ is the legitimate cross-implementation discretisation gap (calibrated on the pristine pair). The per-fault SAFETY sweep $\{1.5, 2.0, 3.0, 5.0, 10.0\}$ records detected/$n_{\text{real}}$ counts of $\{16, 15, 13, 12, 7\}/29$ with zero baseline false positives at every level (`advdiff-xeval-diff/detection_metrics.json`). The monotone fall-off as SAFETY $\uparrow$ records the floor: faults whose field signature lies below $\delta$ are by construction not separable from legitimate discretisation difference, irrespective of tolerance choice, consistent with the $\tau \to 0$ exact-arithmetic regime to which Theorem 3 refers (Remark 12, R2).

*Scope of this empirical evidence.* E1 is a deterministic linear-algebra check on the fault class $\Theta = \mathbb{R}^{N\times N}$, not a statistical test; it certifies that the per-generator witness sets used in our MR designs are unisolvent for the symmetry and self-adjoint blocks at $N = 8$. E2 is paired and pre-registered across the three SUTs in supplementary S10. E3 is a sensitivity sweep rather than a hypothesis test. The framework's structural claim is the kernel characterization of Theorem 3 (§3.3); E1–E3 are consistent with the theorem and with its three corollaries (§3.3), within the linear operator-implementation fault class (Remark 12, R1).

## 6 THREATS TO VALIDITY AND LIMITATIONS

We consolidate the construct, internal, external, and conclusion validity discussions from across the paper. The organizing principle is the MR-identification research question: binary operator-block coverage measures structural design-space breadth, not MR quality or average fault-revealing effectiveness.

### 6.1 Validity framework

We organise threats following Wohlin et al.'s validity framework [58]: internal, external, construct, and conclusion.

*Internal validity.* Theorem 1's uniqueness depends on the canonical-block ordering of Definition 14. The proof in Appendix A catalogues every block-block interaction for the algebras instantiated. The closure result is over algebra-induced MRs in the sense of Definition 13; out-of-scope MRs that fall outside Definition 13 are catalogued in Appendix A, and two concrete counterexamples on the PWR core diffusion algebra are proved out-of-scope in Appendix A (corresponding to the negative instantiation of §3.7). The latter two jointly establish that Theorem 1′ (Conjecture A, absolute completeness) is false on $\mathcal{A}_{\text{PWR}}$, and identify five independent extensions of `Translate`'s signature as the locus of follow-up theoretical work.

*Construct validity.* The reactor-physics mapping (§3.4.3) provides preliminary evidence; non-physics-domain evidence remains future work. The case study (§5.2) reports cat-(iv) detection 5/5 for Set N because the mutation set was constructed to cover one defect category per non-empty block of $\mathcal{A}_{\text{equi}}$; this exhibits construct validity of $\rho_{\text{train-rev}}$ as a gradient-reversal probe rather than averaged superiority, with the DeepCrime [25] real-fault protocol committed as comparative addition. *Set N's 30 MRs were derived by a single author following CONSTRUCT-MP's four-step procedure*; an LRCA multi-LLM second-rater protocol (DeepSeek + ChatGPT + Anthropic Claude Opus) yields Fleiss' $\kappa = 1.000$ across three

LLMs on $n = 33$ items in the almost-perfect Landis–Koch band, with full per-rater Cohen's $\kappa$ in supplementary S3 (`lrca_audit.md`). A human-pair $\kappa$ replication is committed for follow-up work. Two residual construct-validity threats are disclosed here. First, the inter-rater agreement figures ($\kappa = 0.857$ on the supplementary-MR audit and $\kappa = 1.000$ on the LRCA Set N audit) are drawn from rater panels composed entirely of large language models (DeepSeek, ChatGPT, Anthropic Claude Opus); because these models share substantial pre-training corpora, the agreement should be read as consistency among similarly-trained but non-coordinated raters rather than as independent human-level verification; the $\kappa$ values cannot be interpreted as equivalent to inter-rater reliability from independent human domain experts, and a human inter-rater $\kappa$ study is committed as follow-up rather than claimed in this paper. Second, the four Java subjects (`SPHONE`, `SBAGGAGE`, `SEXPENSE`, `SMEAL`) used in the Path A head-to-head (§5.4) are re-implementations written from Sun 2021's prose specification by the same author who designed the NOETHER framework and CONSTRUCT-MP procedure; this confounds framework-design knowledge with subject-implementation fidelity and constitutes a construct-validity threat that an independent re-implementation would resolve.

*External validity.* Two distinct external-validity questions are in scope: (i) whether the framework's *algebraic* reach (Hypothesis 1) covers all program families admitting an operator algebra, and (ii) whether the §5.3.3 *empirical* substrate generalises across codebases within the framework's scope precondition. On (i), Hypothesis 1 covers a wide swathe of mathematical structure but is not exhaustive; Remark 3 catalogues six candidate ninth-block program-family classes (symplectic, sheaf-theoretic, probabilistic / martingale, topological, label-consistency, empirical-parameter-distribution), of which two have explicit `Translate`-template designs already proposed (metric-stability $M_{\text{lip}}$ at Remark 5, and empirical-parameter-distribution divergence motivated by the §5.2.1 pilot's three undetected mutations) as the most actionable extension targets. On (ii), the 10 SUTs of §5.3.3 are concentrated on a single codebase (`MathSignalClass` + `ComplexSignal`) selected by the pre-registered scope criterion; they satisfy the framework's scope precondition (each admits at least one non-empty NOETHER block beyond $G$), so the substrate confirms applicability within scope rather than tests the framework outside its design intent. A cross-codebase pilot on Apache Commons Math 3.6.1 (3 SUTs, 5 Set N MRs, 77 PIT 1.7.4 mutants) corroborates the within-scope generalisation direction: pooled Set N $10/77 = 13.0\%$ Wilson 95% CI $[7.2\%, 22.3\%]$; $G$-block $6/21 = 28.6\%$; the framework's D2 stratum prediction passes at $2/29 = 6.9\%$. The $n$ is underpowered for $\alpha = 0.05$ hypothesis testing; full numbers, per-mutant kill matrix, and the $\mathcal{L}^*$-orthogonality analysis on bilinear SUTs are in supplementary S4 (`future_work.md` item (b.cm)). Full 38-D4J extension, SciPy Python-bridge solver suite, and Maven-resolved GenMorph head-to-head are committed as follow-up (b). The replication tests scope-internal generalisation, not whether the framework should apply to programs lacking explicit operator-algebraic structure (such programs are out of scope by construction).

*Out-of-construction transferability (held-out).* To test transferability beyond the domains used to curate the block decomposition, the frozen blocks were applied without re-fitting to program families outside the construction set. (i) *Industrial reactor codes.* The 110 expert-approved metamorphic relations of three production codes, BAMBOO-C's SPARK (core depletion) and LOCUST (lattice), and the SACOS subchannel thermal-hydraulics code, are subsumed without exception by the order ($O_{\leq}$) block (Wilson 95% interval on subsumption $[0.966, 1.000]$), with no orphan requiring a ninth block and a further 18 valid relations beyond the initial expert sets falling in the same block. This is a code-level held-out confirmation, but it exercises a single block and so corroborates transferability of $O_{\leq}$ rather than of the full decomposition. (ii) *Non-physics libraries.* On dense numerical linear algebra (tested on `numpy.linalg`) and digital signal processing (tested on `numpy.fft`), both outside the curation set, the algebra-derived relations populate six blocks each ($G$, $T^*$, $O_{\leq}$, $\mathcal{L}^*$, $\mathcal{E}^*$, and a conservation invariant), every relation confirmed executably on the independent library, and the two FA-rank-tight blocks behave as Theorem 1's companion Invariance-Blindness result predicts (the permutation/orthogonal $G$ block and the self-adjoint $T^*$ block attain the tight kernel under a finite generating set). The two legs are complementary: the industrial corpus supplies expert-validated breadth on a single block in domain, while the library instantiations exercise the multi-block structure and the tight blocks out of domain. Corpora, derivations, and executable checks are in supplementary material `S11_n5_industrial` and `S12_n5_crossdomain`.

*Expert monotonicity bias and the coverage argument.* A regularity across the three production codes is that all 110 expert-approved relations are order-block (monotone sensitivity) relations: increasing one input moves an output in a fixed direction. This is consistent with how MR identification proceeds in practice, from parameter-sensitivity intuition, which is exactly the order block; the symmetry, self-adjoint, conservation, limit, and method-comparison blocks require explicit operator-algebraic recognition and are systematically under-sampled. Yet the governing equations of these codes [29, 49] contain precisely those structures: SACOS solves two-fluid mass, momentum, and energy conservation; LOCUST defines its homogenisation by reaction-rate conservation and its lattice by a geometric symmetry group; SPARK is a $k$-eigenvalue neutron balance with a critical-boron fixed point. NOETHER derives the corresponding conservation, reciprocity, and symmetry relations directly from those equations, without expert input (supplementary `S11`). This paragraph supports the coverage claim only: it does not claim that the additional blocks are, on average, more fault-revealing than expert MRs.

*Conclusion validity.* The case study's statistical inferences (§5.2) rest on a factorial cross-product of 20 mutations and 3 MR sets on a single architecture. Wilson 95% confidence intervals on detection rates ($[0.18, 0.57]$ for Set N, $[0.03, 0.30]$ for Set L, $[0.00, 0.16]$ for Set B) are reported; pairwise McNemar exact and Fisher exact tests are reported in supplementary S3 (`table4.json::pairwise_stats`). The denominator (20 mutations on one EGNN model) is sufficient to detect the observed Set-N-vs-Set-B contrast at $\alpha = 0.05$ (Fisher $p = 0.008$) but is not sufficient to characterise the framework's performance distribution across architectures or defect distributions. The expanded comparative-evaluation protocol in §5.2 (MR-Scout [60], GenMorph [3], DeepCrime [25] subjects) raises the denominator and broadens the model coverage, addressing this threat directly.

### 6.2 Limitations beyond validity

Three limitations sit partly outside the standard validity categories. First, NOETHER can identify MR classes that are not executable on a given program because the interface lacks the required controllable input or observable output. Second, the block taxonomy is an empirical curation of recurring mathematical structures rather than a universal axiom; new program families may require new blocks. Third, design-for-testability is an implication rather than a present contribution: exposing interface affordances that make algebraically valid MR classes executable remains future work.

## 7 FUTURE WORK

Future work follows directly from the limitations above: human validation of operator-block classifications, direct empirical study of search-generated MR readability and maintainability, executable validation of the SACOS/SPARK/LOCUST MR classes where production interfaces permit, and design-for-testability guidelines for exposing the controls and observations required by identified MR classes.

### 7.1 Practical engineering guidance for users

This subsection collects practical recommendations for engineers applying NOETHER, particularly for the cases where Hypothesis 1's assumptions are bordering or borderline.

*Audit guidance for infinite-group truncation.* When the symmetry block $G$ is a finitely generated infinite discrete group under truncation parameter $K$ (e.g. $\mathbb{Z}^d$ for an unbounded grid solver), Theorem 1's closure result is over the truncated algebra rather than the full one. We recommend a *$K$-sweep audit*: compute $\rho_{\iota,G}$ at three truncation levels $K \in \{K_0, 2K_0, 4K_0\}$ and pass if detection-rate stability is within $\pm 5\%$ across the three values (the mesh-convergence stability margin of [30, §6.2]). The audit need not be re-run on every CI invocation; once per architectural change is sufficient. The reference implementation (supplementary S1) includes a `k_sweep_audit()` helper.

*When to check Hypothesis 1's sufficiency at deployment time.* For program families outside the three instantiated domains (Boltzmann reactor physics, equivariant ML, relational query optimisers), users should explicitly enumerate the candidate operators of $\mathcal{A}_P$ and check whether each falls into one of the eight blocks. Operators that resist classification are signals that an additional block may be required and should be reported as candidates for an extension of the block taxonomy rather than absorbed into the existing taxonomy.

*Tolerance selection.* Tolerances $\tau$ on continuous-valued MRs (e.g. $\rho_{\text{rot}}$, $\rho_{\text{adj}}$) should be set as $\tau \approx 10^2\,\epsilon_{\text{fp}}$ where $\epsilon_{\text{fp}}$ is the floating-point unit roundoff (so $\tau = 10^{-4}$ for fp32, $\tau = 10^{-12}$ for fp64); the $10^2$ factor is the forward-pass roundoff floor following Higham's standard error-analysis bound [24] for $n \approx 10^3$ scalar operations per output coordinate. The supplementary S3 `tau_sweep.json` documents the $\tau$/false-positive trade-off.

### 7.2 The remaining human role and partial automation of $\mathcal{A}_P$ distillation

NOETHER mechanises the construction downstream of $\mathcal{A}_P$ but assumes that $\mathcal{A}_P$ has been distilled by a human. Three directions may partially automate that upstream step: LLM-assisted operator extraction, static-analysis-based extraction, and empirical-symmetry detection. None of them eliminates the human role for arbitrary programs.

*Auditable algebra-distillation protocol.* The next version of the workflow should make the upstream step reproducible rather than merely expert-curated. We propose a six-step protocol: (1) collect the governing equations, interface contract, and implementation documentation; (2) list candidate operators, domains, codomains, controllable inputs, and observable outputs; (3) map each operator to zero or more current blocks using the definitions in Section 3; (4) record a source identity or inequality for every block assignment;

(5) require two independent reviewers to label each assignment as explicit, latent-but-not-executable, or out-of-scope; and (6) publish the resulting block-coverage matrix with the unresolved operators. This protocol would directly address the main upstream threat in the present paper: a block assignment can be audited even when the derived MR class cannot yet be executed because the production program does not expose the required interface.

## 8 CONCLUSION

The three foundational questions raised in Section 1, origin, boundary, and transferability of MetaPattern sets, lacked a structural answer. NOETHER provides one within an explicit algebraic scope. *Origin*: MetaPatterns are equivalence classes of MRs derived from invariants of an operator algebra $\mathcal{A}_P$. *Boundary*: Theorem 1 gives a no-drop closure invariant for the `Translate`-reachable, algebra-induced MR space (Definition 13), while the negative PWR result and the Invariance-Blindness Theorem identify where that space ends and what its derived MRs cannot see. *Transferability*: the framework's mechanism applies unchanged once a new program family's algebra has been specified, tested within the framework's scope precondition on three structurally distinct operator-algebraic skeletons (Boltzmann reactor physics, equivariant ML, relational query optimisers). Cross-domain empirical superiority and team adoption by PWR-simulator V&V groups, equivariant-ML testing teams, or database-optimiser test groups are open follow-up questions; the present paper establishes the structural transferability of the construction mechanism, not adoption outcomes on each domain.

This grounding reframes several long-standing questions. Empirical adequacy frameworks gain an algebraic warrant for their pattern grid. Structured identification approaches gain an answer to why a given category set is closed under the chosen algebra. Automated pipelines can be constrained or initialised by the constructed MetaPattern set. LLM-prompted MR generation can be recast as algebra-conditioned generation rather than open-ended prompting.

We have not solved the upstream problem of distilling $\mathcal{A}_P$ from program semantics, nor have we eliminated induction from MetaPattern discovery. The structural decomposition that drives the construction is itself an empirical curation of mathematical structures recurrent across program families, stated as Hypothesis 1. The most important follow-up work is therefore upstream: combining LLM-assisted symbolic extraction, formal-methods-based static analysis, and empirical-symmetry detection into a partially automated $\mathcal{A}_P$-distillation pipeline, and testing the structural decomposition on algebras outside its present image (Remark 3 catalogues six program-family classes likely to require additional blocks). For the downstream layer, from $\mathcal{A}_P$ to $\mathbb{M}(\mathcal{A}_P)$, the contribution is narrower and firmer: the construction is mechanical, algebraic closure under `Translate` is provable within the algebra-induced MR space (Theorem 1), and the construction transfers across domains once the algebra has been specified.

**Boundary of contribution (Conclusion restatement)**

**Established.** (i) A no-drop closure invariant under `Translate` given a block decomposition (Theorem 1); (ii) polynomial-time decidability under finite generating-set assumption (Theorem 2); (iii) three non-vacuous instantiations across structurally distinct algebraic skeletons; (iv) a negative instantiation on the PWR core diffusion algebra $\mathcal{A}_{\mathrm{PWR}}$ (§3.7, Appendix A), in which two MRs from the standard PWR safety-analysis literature (non-additivity of rod-bank reactivity worth, second-order mixed $T_{\mathrm{mod}}$-vs-$C_B$ dependence of $k_{\mathrm{eff}}$) are proved not in MR($\mathcal{A}_{\mathrm{PWR}}$), falsifying Theorem 1′ (Conjecture A) on a structurally significant operator algebra and identifying five pairwise-independent extensions of `Translate`'s signature as the locus of repair. **Open.** (a) Whether a Composite-`Translate` extension covering the five obstructions of §3.7 and Appendix A preserves Theorem 1's closure and Theorem 2's polynomial-time decidability; (b) sufficiency of Hypothesis 1's structural list (Remark 3's six out-of-scope classes are candidate ninth blocks); (c) superiority over existing automated MR-identification pipelines on average defect distributions (the comparative-evaluation protocol in §5.2 establishes effects, not averages); (d) elimination of induction (relocated, not eliminated); (e) automation of upstream $\mathcal{A}_P$ distillation (the framework treats $\mathcal{A}_P$ as a given input from a domain expert; mechanising the extraction of operator algebras from program semantics remains an upstream task that NOETHER does not address; §7.2 sketches three partial-automation directions but commits to none). Both Hypothesis 1 (block sufficiency) and `Translate`'s signature (`Translate` sufficiency) are the loci where future induction-eliminating and completeness-establishing work should target.

## 9 DATA AND ARTIFACT AVAILABILITY

To support reviewer verification, the supplementary code and data are archived on Zenodo under DOI 10.5281/zenodo.20250634. For double-blind review, the same materials can be supplied through the submission system as an anonymised archive with the same file manifest and content hash.

*Review-stage anonymised archive.* The following supplementary materials are submitted alongside the manuscript and are available to reviewers through the conference submission system (or, equivalently, an anonymised OpenReview / Zenodo deposit) under SHA-256 content hash. The hash is computed over the concatenated tar-archive of the listed items and reported in the final manuscript at acceptance.

- **S1** Python reference implementation of CONSTRUCT-MP described in supplementary S9 (Appendix D), including the Boltzmann and equivariant-ML instantiations.
- **S2** The full PWR-MR corpus underlying Section 3.4.3, with each MR annotated by canonical block, NOETHER MetaPattern, and source equation. Table 3 corresponds to a subset of 12 MRs selected by the protocol stated there.
- **S3** The SE(3)-equivariant point-cloud testing harness from Section 3.5.3 and Section 5.2, including the mutation generators and MR sets used in the case study comparison.
- **S4** A reproducibility manifest documenting the random seeds, model checkpoints, and dataset versions referenced in the case study.

*Public archival release.* The public archival package is deposited on Zenodo under DOI 10.5281/zenodo.20250634. The camera-ready version will additionally record the final SHA-256 hash of the submission archive. We aim for the *Available* and *Functional* artefact-evaluation badges.

## APPENDICES A, B (WORKED-EXAMPLE MATERIAL)

Material running CONSTRUCT-MP on the heat / continuity / momentum / slowing-down reactor equations (Appendix A), per-MR source provenance for the representative MRs of Table 3 (Appendix B), and worked examples for multi-block-derivable MRs (originally introductory to Appendix C) is provided in supplementary `S9_migrated_appendices/A_B_C_worked.tex`; the body text's claims do not depend on this material beyond the cross-references made in §3.4 and §3.4.3.

## A PROOFS

### Lemma C.1 (Well-foundedness of canonical-block ordering)

LEMMA 2. *The canonical-block ordering of Definition 14 is a strict total order on the eight blocks, and the assignment of any multi-block-derivable MR to its highest-priority block is unique.*

PROOF. The ordering is a strict total order on a finite set by inspection. Given an MR $\rho$ with derivations $\{(s_i, \iota_i)\}_{i=1}^{k}$ from blocks $s_1, \dots, s_k$, the assignment selects $s^* = \max_{>}\{s_1, \dots, s_k\}$. Since the strict order is total, $s^*$ is unique. □

### Per-block instantiations of `Translate`

Definition 12 fixes the signature of `Translate` but defers per-block specifics. Table 21 fills in the canonical input-tuple-generation rule and the resulting MR template for each of the eight blocks.

Table 21. Per-block instantiations of `Translate`.

| **Block** $s$ | **Canonical tuple from base** $x_0$ | **MR template** $\rho_{\iota,s}$ |
|---|---|---|
| $G$ | $(x_i)_{i=0}^{\lvert G\rvert-1} = (g_i \cdot x_0)$ over group orbit | $\forall x_0,\ \forall g \in G:\ P(g \cdot x_0) = \rho(g) \cdot P(x_0)$ |
| $O_{\leq}$ | $(x_1, x_2)$ with $x_1 \leq_\theta x_2$ in the partial order | $x_1 \leq_\theta x_2 \Rightarrow P(x_1) \leq_{\mathcal{Y}} P(x_2)$ |
| $T^*$ | paired tuple $((x_1, P(x_1)), (x_2, P(x_2)))$ in the inner product | $\langle L\, P(x_1), P(x_2)\rangle = \langle P(x_1), L\, P(x_2)\rangle$ |
| $\mathcal{T}^*$ | $(x, \mathcal{T}x)$ with $\mathcal{T}$ the time-reversal involution | $P(\mathcal{T}x) = \mathcal{T}\, P(x)$ on the reversibility sub-domain |
| $\mathcal{L}^*$ | sequence $(x_\theta)$ with $\theta \to \theta_*$ | $\lVert P_\theta - P_{\theta_*}\rVert_* = O(f(\theta))$ at the prescribed rate |
| $\mathcal{D}^*$ | solution trajectory $(\xi(t))_{t\geq 0}$ extracted by $\mathcal{D}$ | qualitative-feature relation (extremum, monotonicity, S-curve) preserved on $(\xi(t))$ |
| $\mathcal{E}^*$ | method pair $(M_1, M_2)$ in the partial order $\preceq_{\mathcal{E}}$ | $\mathrm{err}(M_1) \leq \mathrm{err}(M_2)$ on the prescribed benchmark family |
| $\mathcal{B}^*_{\mathrm{rel}}$ | rewrite pair $(E, E')$ generated by an algebraic-rewriting rule $\mathcal{R} \in \mathcal{R}_{\mathrm{rel}}$ on the idempotent semiring | $\forall D.\ \mathrm{eval}(E, D) =_{\mathrm{bag}} \mathrm{eval}(E', D)$ at the rewriting rule's stated scope |

The per-block rule together with the $\sim_s$ equivalence relation of Definition 11 fully determines $\mathrm{Translate}(\iota, s)$ once the operator family $\Phi \subseteq s$ is fixed. Implementations of these seven cases as Python callables ship in supplementary material S1 (`construct_mp.py`, function `translate` together with the seven `default_extractor_*` routines).

### Theorem 1 (Algebraic Closure under `Translate`), full proof

PROOF. *Existence.* Let $\rho \in \mathrm{MR}(\mathcal{A}_P)$ in the sense of Definition 13. There exist a block $s$, an invariant $\iota \in \mathcal{I}_s$, and a derivation $\rho = \mathrm{Translate}(\iota, s)$. Step 1 of CONSTRUCT-MP places $\iota$ in $\mathcal{I}_s$; step 2 places $\rho$ in $\mathcal{R}(\iota)$; step 3 forms $m_s = \mathcal{R}(\iota)/{\sim_s}$; step 4 returns $m_s \in \mathbb{M}(\mathcal{A}_P)$.

*Uniqueness.* Suppose $\rho$ admits derivations through multiple blocks: $\rho = \text{Translate}(\iota_1, s_1) = \text{Translate}(\iota_2, s_2)$ with $s_1 \neq s_2$. Definition 14's canonical-block ordering selects $s^* = \max_{>}\{s_1, s_2\}$. By Lemma C.1, $s^*$ is unique; the canonical assignment $\rho \mapsto m_{s^*}$ is unique. □

**Theorem 2 (Decidability), full proof**

PROOF. *Step 1 cost.* For each generator $g_i$, compute $g_i$'s contribution to each $\mathcal{I}_s$. Cost: $O(t_i)$ per generator, $O(n \cdot \max_i t_i)$ aggregated.

*Step 2 cost.* 'Translate' is $O(1)$ per invariant. Total: $O(n)$.

*Step 3 cost.* Union-find with path compression: amortised $O(\alpha(n))$ per operation; we use $O(\log n)$ for simpler analysis. Total: $O(n \log n)$.

*Step 4 cost.* $O(7) = O(1)$.

Total: $O(n \cdot \max_i t_i \cdot \log n)$ when $\max_i t_i \geq 1$. □

### C.4 An open problem: absolute completeness

Theorem 1's scope is bounded by Definition 13: the closure is over MRs reachable through `Translate` from a single block invariant. We attempted a stronger statement:

CONJECTURE (THEOREM 1′, ABSOLUTE COMPLETENESS). *Every MR $\rho$ formulable as a property over $\mathcal{A}_P$'s operators is contained in some $m \in \mathbb{M}(\mathcal{A}_P)$.*

A proof would require either (a) a normalisation theorem reducing every operator-algebra MR to a `Translate`-reachable form, or (b) an extension of `Translate` to compositional invariants spanning multiple blocks. We were unable to establish either without imposing additional structural assumptions on $\mathcal{A}_P$. A partial empirical step toward Theorem 1′ is available on a single domain through the supplementary-MR engineering audit reported in §3.4.3: 17 of 18 are placed by majority of three LLM labellers into one of the seven MetaPatterns or $m_{\text{rel}}$ (subsumption 94.4%, Fleiss' $\kappa = 0.857$, raw labels in supplementary S2 `18mr_audit/`); the single orphan localises to the metric-stability class discussed in Appendix 5 and Appendix C.5.2, for which a concrete candidate ninth block is sketched. This is a constructive partial support, not a proof: it is empirical, single-domain, bounded by the LLM-shared-training-data caveat noted in §3.4.3, and does not bound the gap in the absence of the candidate ninth block. The conjecture remains open. Section C.5 below documents three concrete classes of MRs that lie outside Theorem 1's scope and would have to be addressed by any positive resolution of Theorem 1′; §3.7 and Appendix A additionally falsify Theorem 1′ on $\mathcal{A}_{\text{PWR}}$.

### C.5 Out-of-scope MRs: three concrete classes

We document three families of MRs that satisfy the informal description "a property of $P$'s executions over the operator algebra $\mathcal{A}_P$" but are not algebra-induced under Definition 13, and therefore lie outside the scope of Theorem 1. These are not threats to the theorem; they delimit it.

*C.5.1 Probabilistic / distributional MRs without operator-algebraic representation.* Consider the MR "the Shannon entropy $H(f(\mathbf{x}))$ of a classifier's output distribution should not decrease when the input is augmented by Gaussian noise $\mathcal{N}(0, \sigma^2 I)$". The augmentation is parametric in $\sigma$ and is not a group action, a partial-order operator, a self-adjoint operator, a time-reversal involution, a method-comparison operator, or (without further structure) a limit operator on $\mathcal{X}$. The MR constrains a functional $H$ of the output distribution, not a point-wise relation between outputs. There is no $\iota \in \mathcal{I}_s$ for any $s$ from which Translate can reach this MR. Consequently the MR is not algebra-induced under Definition 13 and is not covered by Theorem 1. A positive resolution of Theorem 1′ would require either embedding stochastic perturbations into a probabilistic extension of $\mathcal{A}_P$, or extending `Translate` to functionals over output distributions.

*C.5.2 Adversarial / input-set MRs.* The MR "$\|\delta\|_p \leq \epsilon \Rightarrow f(\mathbf{x}+\delta) = f(\mathbf{x})$" (adversarial robustness in $\ell_p$ ball) constrains $f$'s behaviour on a set defined by a norm constraint, not by a group action. The set $\{\delta : \|\delta\|_p \leq \epsilon\}$ does not in general carry a closed group structure on $\mathcal{X}$, and the MR is not derivable as Translate$(\iota, s)$ for any $s$ in $\mathcal{D}(\mathcal{A}_P)$ as currently defined. Any treatment of this MR within NOETHER's scope would require an eighth block (e.g. a metric-ball block with bounded Lipschitz operators), or a relaxation of `Translate` to non-group input neighbourhoods. A concrete candidate for the eighth block alluded to above is a metric-stability block $M_{\text{lip}} = (\mathcal{X}, d_{\mathcal{X}}) \to (\mathcal{Y}, d_{\mathcal{Y}})$ whose operators are $K$-Lipschitz maps and whose induced MetaPattern $m_{\text{lip}}$ is the equivalence class of pointwise stability inequalities $d_{\mathcal{Y}}(P(x'), P(x)) \leq K \cdot d_{\mathcal{X}}(x', x)$. The corresponding `Translate` template constructs a follow-up input by metric perturbation $x' = x + \varepsilon u$ (norm $\|u\| = 1$, $|\varepsilon| < \delta$) and predicates a $K$-Lipschitz output bound; canonical-block ordering would place $M_{\text{lip}}$ after $\mathcal{B}^*_{\text{rel}}$ since metric structure is independent of the seven existing block invariants (no group action on the perturbation set, no partial order on inputs, no self-adjoint operator, and no parametric refinement family). Theorem 1's closure proof transfers without modification because $m_{\text{lip}}$ is single-block algebraically derived. We do not commit to $M_{\text{lip}}$ as part of the canonical decomposition in this paper, but we record it as the most concrete sub-instance of Remark (iv) (topological invariants) that admits an immediate `Translate` template.

*C.5.3 Compositional MRs across multiple blocks under non-canonical derivations.* The MR "rotating a point cloud by $R$ *and* simultaneously taking the training-size limit $n \to \infty$ should give an output that converges to $f(\mathbf{x})$ at rate $O(1/\sqrt{n})$" invokes both $G$ (rotation) and $\mathcal{L}^*$ (training-size limit) within a single MR. Definition 14's canonical-block ordering assigns this to $G$, but the resulting MR "$f(R \cdot \mathbf{x}) = f(\mathbf{x})$" loses the convergence-rate content. The compositional content is in $\mathbb{M}(\mathcal{A}_P)$ if and only if it can be split into separate single-block invariants (one in $G$, one in $\mathcal{L}^*$); MRs whose semantic content is irreducibly compositional (in the sense that splitting destroys the property's truth value) lie outside Theorem 1's scope. This is the case Theorem 1′ option (b) is targeted at: an extended `Translate` that operates on tuples of block invariants. We have not constructed such an extension that preserves Theorem 1's polynomial-time decidability (Theorem 2), and we leave this as the most concrete sub-problem under Theorem 1′.

The three classes of §C.5.1–§C.5.3 are *abstract* characterisations of MRs outside Theorem 1's scope. Appendix A supplements them with two *concrete instances* drawn from the PWR core diffusion algebra $\mathcal{A}_{\text{PWR}}$, each of which is empirically realised on every conforming PWR core simulator and documented in the standard reactor-physics literature. Together with §3.7, Appendix A establishes that Theorem 1′ (Conjecture A) is false on $\mathcal{A}_{\text{PWR}}$, and identifies five structural obstructions in `Translate`'s signature (operator-spectrum output, homomorphism-failure $\pi$-template, configuration-indexed adjoint structure, higher-order mixed-difference templates, bidirectional joint parametric dependence) that any positive resolution would have to repair.

### C.6 Proofs for the negative instantiation on $\mathcal{A}_{\mathrm{PWR}}$ (§3.7)

This appendix supplies the proofs for Propositions 1 and 2 of §3.7. The proof structure for both is the same: enumerate the eight blocks of $\mathcal{D}(\mathcal{A}_{\mathrm{PWR}})$, instantiate the per-block `Translate` template of Table 21, and verify by inspection that no invariant $\iota \in \mathcal{I}_s$ yields the target MR. The block-by-block exclusions in Proposition 1's proof are the most detailed; Proposition 2 reuses the same exclusion pattern with the obstruction localised to the $O_{\leq}$ block.

*C.6.1 Proof of Proposition 1 (non-additivity is not `Translate`-reachable). Statement.* For $\rho_{\mathrm{nonadd}}$ as in Definition 20 and every $s \in \mathcal{D}(\mathcal{A}_{\mathrm{PWR}})$, every $\iota \in \mathcal{I}_s$: $\mathrm{Translate}(\iota, s) \neq \rho_{\mathrm{nonadd}}$.

*Proof.* We use the exact-form definition of $d\rho$ (Definition 19) throughout, which depends only on dominant eigenvalues $k_{\mathrm{eff}}$ of operators in $\mathcal{A}_{\mathrm{PWR}}$. The adjoint-perturbation reading of §3.7 is the physical motivation but is not invoked in any case below.

**Case** $s = G$**.** By Definition 11, an invariant $\iota \in \mathcal{I}_G$ has the form $(\Phi, \pi)$ with $\Phi \subseteq G$ a finite operator family and $\pi$ a relation on tuples $(x_i, P(x_i))_{i=1}^{k}$ obtained by applying $\Phi$ to a base input $x_0$. By Table 21, the canonical `Translate` template for $G$ is the equivariance schema

$$\mathrm{Translate}(\iota, G) \;\equiv\; \forall x_0\ \forall g \in \Phi:\ P(g \cdot x_0) = \rho(g) \cdot P(x_0).$$

Two structural mismatches with $\rho_{\mathrm{nonadd}}$:

(a) *Operator-spectrum output, not $P$ output.* $\rho_{\mathrm{nonadd}}$ is an inequality between sums of $1/k_{\mathrm{eff}}$ values, where each $k_{\mathrm{eff}}$ is the dominant eigenvalue of a configuration-specific diffusion operator $H_X \in \mathcal{A}_{\mathrm{PWR}}$. The dominant eigenvalue is a *spectral* quantity of the operator, obtained as part of the simultaneous solution $(k_{\mathrm{eff}}, \phi)$ of $H\phi = (1/k_{\mathrm{eff}})\, F\phi$; it is not a function of $P(x)$ alone but a property of the operator $H$ itself. `Translate`'s output relation $\pi$ in Definition 11 ranges over $(\mathcal{X} \times \mathcal{Y})^k$, where $\mathcal{Y}$ is the program output space (flux distributions, in this case). Operator-spectrum quantities are not in $\mathcal{Y}$; they are scalar invariants of operators in $\mathcal{O}$, lying outside `Translate`'s signature by construction.

(b) *Non-additivity is not equivariance.* Even granting a charitable extension to admit $k_{\mathrm{eff}}$ as a derived output, the $G$-template asserts equivariance of the output under the action of $\Phi$: $P(g \cdot x_0)$ is determined by $g$ and $P(x_0)$ through the representation $\rho(g)$. $\rho_{\mathrm{nonadd}}$ asserts that the worth functional $d\rho : \mathcal{O}_{\mathrm{rod}} \to \mathbb{R}_{>0}$ is *not a semigroup homomorphism*: $d\rho(A \cup B) \neq d\rho(A) + d\rho(B)$. Failure of homomorphism is not equivariance failure; it is the absence of an additive structure-preserving map, which has no expression in the equivariance template $\pi$ for $G$. No $\iota \in \mathcal{I}_G$ yields $\rho_{\mathrm{nonadd}}$.

**Case** $s = O_{\leq}$**.** By Table 21, the $O_{\leq}$ template is the absolute-monotonicity schema

$$\mathrm{Translate}(\iota, O_{\leq}) \;\equiv\; \forall x_1, x_2:\ x_1 \leq_\theta x_2 \implies P(x_1) \leq_{\mathcal{Y}} P(x_2).$$

$\rho_{\mathrm{nonadd}}$ is a *quaternary relation* on the four configurations $(x_0, \mathcal{O}^A x_0, \mathcal{O}^B x_0, \mathcal{O}^{A\cup B} x_0)$: it asserts a non-vanishing mixed second difference

$$\begin{aligned}\Delta_{AB}(x_0) \;&= \left[k_{\mathrm{eff}}^{-1}(P(x_0)) - k_{\mathrm{eff}}^{-1}(P(\mathcal{O}^{A\cup B}x_0))\right]\\ &\quad - \left[k_{\mathrm{eff}}^{-1}(P(x_0)) - k_{\mathrm{eff}}^{-1}(P(\mathcal{O}^{A}x_0))\right]\\ &\quad - \left[k_{\mathrm{eff}}^{-1}(P(x_0)) - k_{\mathrm{eff}}^{-1}(P(\mathcal{O}^{B}x_0))\right] \;\neq\; 0\end{aligned}$$

(where the subtractions use Definition 19's positive-worth convention). The $O_{\leq}$ template captures binary monotonicity between two points along a single partial order $\leq_\theta$. The mixed-difference structure of $\rho_{\text{nonadd}}$ requires comparison across four configurations forming a "rectangle" $\{x_0, \mathcal{O}^A x_0, \mathcal{O}^B x_0, \mathcal{O}^{A\cup B} x_0\}$ with two independent perturbation directions (insertion of $A$ and insertion of $B$); no $\leq_\theta$ relates all four pairwise into a single chain. Furthermore, the assertion is *non-vanishing of a difference*, not a directional inequality; it is direction-agnostic, capturing both shadowing ($\Delta > 0$) and anti-shadowing ($\Delta < 0$). The $O_{\leq}$ template's $\leq_{\mathcal{Y}}$ is a directional partial order, not a non-vanishing constraint. No $\iota \in \mathcal{I}_{O_{\leq}}$ yields $\rho_{\text{nonadd}}$.

**Case** $s = T^*$**.** By Table 21, the $T^*$ template asserts $\langle L\, P(x_1), P(x_2)\rangle = \langle P(x_1), L\, P(x_2)\rangle$ for a single self-adjoint operator $L$ in a fixed inner product. The structural mismatch with $\rho_{\text{nonadd}}$ has two components:

(i) *Single operator $L$ vs. configuration-indexed family.* The exact $d\rho$ values entering $\rho_{\text{nonadd}}$ are eigenvalues of *four distinct diffusion operators* $H_\emptyset, H_A, H_B, H_{A\cup B}$, each self-adjoint within its own configuration but pairwise distinct as operators. The $T^*$ template's $L$ is fixed; it does not range over a configuration-indexed family $\{H_X\}_{X\in\mathcal{O}_{\text{rod}}}$. The self-adjointness of any single $H_X$ does not imply or constrain a relation between eigenvalues of *different* $H_X$'s.

(ii) *Adjoint weighting function depends on configuration.* Under the standard adjoint-perturbation reading, each $d\rho(B; X)$ for $X \in \{\emptyset, A\}$ admits the first-order representation

$$d\rho(B; X) \;\approx\; -\frac{\langle \phi_X^\dagger,\, \delta H_B\, \phi_X\rangle}{\langle \phi_X^\dagger,\, F_X\, \phi_X\rangle},$$

with $(\phi_X, \phi_X^\dagger)$ the principal eigenfunctions of $(H_X, H_X^\dagger)$. The Hilbert-space inner product $\langle\cdot,\cdot\rangle = \int(\cdot)(\cdot)\, d\Omega\, dE\, d\mathbf{r}$ is unchanged across $X$. What changes across $X$ is the *adjoint weighting function* $\phi_X^\dagger$, which is the principal eigenfunction of a structurally different adjoint operator $H_X^\dagger$ from $H_\emptyset^\dagger$ (because $H_X$ contains $A$'s absorbing material whereas $H_\emptyset$ does not). $\phi_A^\dagger$ is locally depressed in $A$'s geometric support and globally redistributed elsewhere; this is the adjoint-perturbation root cause of the non-additivity $d\rho(A\cup B) \neq d\rho(A) + d\rho(B)$.

The $T^*$ template's structure (single $L$, fixed inner product, asserting $\langle Lx_1, x_2\rangle = \langle x_1, Lx_2\rangle$) has no place to express "the weighting function $\phi^\dagger$ entering the inner product is itself the principal eigenfunction of a configuration-indexed adjoint operator and changes when configuration changes". This is a configuration-dependent adjoint structure, not a self-adjointness identity on a single operator. No $\iota \in \mathcal{I}_{T^*}$ yields $\rho_{\text{nonadd}}$.

**Case** $s = \mathcal{T}^*_{\text{rev}}$**.** The PWR diffusion operator $-\nabla\cdot D\nabla + \Sigma_a$ is irreversible: it is parabolic (transient form) or elliptic (steady-state form), neither of which admits a time-reversal involution on the relevant solution sub-family. Hence $\mathcal{T}^*_{\text{rev}}$ is empty for $\mathcal{A}_{\text{PWR}}$, and vacuously no $\iota \in \mathcal{I}_{\mathcal{T}^*_{\text{rev}}}$.

**Case** $s = \mathcal{L}^*$**.** By Table 21, the $\mathcal{L}^*$ template is a convergence statement $\|P_\theta - P_{\theta_*}\|_* = O(f(\theta))$ at a parametric limit. $\rho_{\text{nonadd}}$ contains no limit operation: it is a strict inequality at finite, fixed configurations $(\emptyset, A, B, A\cup B, x_0)$. No $\iota \in \mathcal{I}_{\mathcal{L}^*}$.

**Case** $s = \mathcal{D}^*$**.** By Table 21, the $\mathcal{D}^*$ template asserts a qualitative-feature relation (extremum, monotonicity, S-curve) on a solution trajectory $\xi(t)$. $\rho_{\text{nonadd}}$ is a steady-state inequality between four reactivity values; it does not concern trajectory shapes. No $\iota \in \mathcal{I}_{\mathcal{D}^*}$.

**Case** $s = \mathcal{E}^*$**.** By Table 21, the $\mathcal{E}^*$ template compares two *methods* $M_1, M_2$ on a benchmark family. $\rho_{\text{nonadd}}$ compares four *operator configurations* on a single fixed method (the same diffusion solver $P$ in all four worth values). No $\iota \in \mathcal{I}_{\mathcal{E}^*}$.

**Case** $s = \mathcal{B}^*_{\text{rel}}$**.** Per §3.6, $\mathcal{B}^*_{\text{rel}}$ is non-empty only on program families with idempotent-semiring rewriting structure. The PWR diffusion solution operator algebra does not carry such structure (no rewriting rules between core states preserve evaluation under all valid inputs). $\mathcal{B}^*_{\text{rel}}$ is empty for $\mathcal{A}_{\text{PWR}}$. Vacuously no $\iota \in \mathcal{I}_{\mathcal{B}^*_{\text{rel}}}$.

This exhausts $\mathcal{D}(\mathcal{A}_{\text{PWR}})$. For every block $s$ and every $\iota \in \mathcal{I}_s$, $\text{Translate}(\iota, s) \neq \rho_{\text{nonadd}}$. Hence $\rho_{\text{nonadd}} \notin \text{MR}(\mathcal{A}_{\text{PWR}})$. □

*C.6.2 Three obstructions identified by Proposition 1's proof.* The proof identifies three independent structural obstructions in `Translate`'s present definition:

(O1) *Operator-spectrum output is not in $\mathcal{Y}$.* $\rho_{\text{nonadd}}$ asserts a relation between $1/k_{\text{eff}}$ values, where each $k_{\text{eff}}$ is a dominant eigenvalue of an operator $H_X$. `Translate`'s $\pi$ in Definition 11 ranges over $(\mathcal{X} \times \mathcal{Y})^k$; eigenvalues of operators in $\mathcal{O}$ are scalar invariants of those operators, not elements of $\mathcal{Y}$.

(O2) *Output relation is non-additivity (failure of homomorphism), not equivariance, partial order, or self-adjointness.* The worth functional $d\rho : \mathcal{O}_{\text{rod}} \to \mathbb{R}$ is not a semigroup homomorphism; this is a third type of algebraic relation distinct from the equivariance, monotonicity, and self-adjointness expressed by `Translate`'s per-block $\pi$ templates.

(O3) *Adjoint weighting function $\phi^\dagger_X$ entering the $T^*$ block's inner product is configuration-dependent.* While the Hilbert-space measure $d\Omega\, dE\, d\mathbf{r}$ is fixed, the eigenfunction $\phi^\dagger_X$ varies with $X$ because $H^\dagger_X$ varies with $X$. Definition 11 fixes the self-adjoint operator $L$ once; it does not admit a configuration-indexed family $\{L_X\}$ with $L_X$'s spectrum varying with $X$.

A constructive resolution of Proposition 1's obstructions would require `Translate` to admit (i) operator-spectrum output relations (eigenvalues, integrals, ratios as the targets of $\pi$), (ii) homomorphism-failure relations as a $\pi$-template type alongside equivariance / monotonicity / self-adjointness, and (iii) configuration-dependent adjoint structure on $T^*$.

*C.6.3 Proof of Proposition 2 (MTC-vs-boron mixed dependence is not `Translate`-reachable). Statement.* For $\rho_{\text{MTC-bor}}$ as in Definition 21 and every $s \in \mathcal{D}(\mathcal{A}_{\text{PWR}})$, every $\iota \in \mathcal{I}_s$: $\text{Translate}(\iota, s) \neq \rho_{\text{MTC-bor}}$.

*Proof.* The principal obstruction is in the $O_\leq$ block; we treat that case in detail and abbreviate the others.

**Case** $s = O_\leq$**.** By Table 21, the $O_\leq$ template is the absolute-monotonicity schema

$$\text{Translate}(\iota, O_\leq) \;\equiv\; \forall x_1, x_2 :\; x_1 \leq_\theta x_2 \implies P(x_1) \leq_{\mathcal{Y}} P(x_2),$$

a *first-order* statement asserting a binary relation between $P(x_1)$ and $P(x_2)$ at $\theta$-comparable inputs along a single partial-order direction $\leq_\theta$. $\rho_{\text{MTC-bor}}$ asserts a *non-zero second-order mixed partial derivative* of $k_{\text{eff}}$ with respect to two *independent* parameter directions $T_{\text{mod}}$ and $C_B$:

$$\left| \frac{\partial^2 k_{\text{eff}}}{\partial T_{\text{mod}}\, \partial C_B} \right| \;>\; \tau_{\text{MTC-bor}}.$$

Two independent obstructions in the $O_\leq$ template:

(a) *Order vs. mixed-derivative structure.* The mixed second derivative is the limit of a four-point finite-difference quotient over a "rectangle" $\{(T_0, C_0), (T_0 + \Delta T, C_0), (T_0, C_0 + \Delta C), (T_0 + \Delta T, C_0 + \Delta C)\}$:

$$\frac{\partial^2 k_{\text{eff}}}{\partial T_{\text{mod}}\, \partial C_B} \;=\; \lim_{\Delta T, \Delta C \to 0} \frac{k(T_0 + \Delta T, C_0 + \Delta C) - k(T_0, C_0 + \Delta C) - k(T_0 + \Delta T, C_0) + k(T_0, C_0)}{\Delta T \cdot \Delta C}$$

(writing $k$ for $k_{\text{eff}}$). This is structurally a *four-point relation*, not a two-point relation. The $O_{\leq}$ template's $\pi$ relates $(P(x_1), P(x_2))$ pairwise; it has no expression for a four-point combination weighted by $1/(\Delta T \cdot \Delta C)$. Equivalently, the canonical input-tuple-generation rule for $O_{\leq}$ in Table 21 produces tuples $(x_1, x_2)$ with $x_1 \leq_\theta x_2$, not 2-by-2 perturbation rectangles.

(b) *Two independent parameter directions.* The mixed derivative requires *jointly varying* $T_{\text{mod}}$ and $C_B$ along two independent directions. The $O_{\leq}$ template's $\leq_\theta$ is a *single* partial order on $\mathcal{X}$ (or on a single coordinate of $\mathcal{X}$). Even granting the construction of a product order $\leq_T \times \leq_C$ on a 2-D parameter slice of $\mathcal{X}$, the $\pi$ relation still applies along the order chain as a directional inequality, not as a non-vanishing-second-difference statement. The two independent parameter directions are perpendicular, not chained; the $\leq_\theta$ formalism collapses them into one chain only by losing the non-vanishing-mixed-difference content.

Furthermore, $\rho_{\text{MTC-bor}}$'s output is again $k_{\text{eff}}$, an operator-spectrum quantity (cf. Proposition 1 Case $G$ obstruction (a)). This gives a *third* obstruction in $O_{\leq}$: even if mixed-second-derivative structure could be embedded into $\pi$, the output value would not lie in $\mathcal{Y}$. No $\iota \in \mathcal{I}_{O_{\leq}}$ yields $\rho_{\text{MTC-bor}}$.

**Other blocks.**

- *Case $s = G$.* No group action in $\mathcal{A}_{\text{PWR}}$ relates $(T_{\text{mod}}, C_B)$ pairs at different parameter values to each other through equivariance: $T_{\text{mod}}$ and $C_B$ are continuous parameters acting on the cross-section library, not group-action coordinates. Even if a charitable embedding in $G$ were attempted, the same operator-spectrum-output obstruction (Proposition 1 Case $G$ (a)) applies: $k_{\text{eff}}$ is not in $\mathcal{Y}$.
- *Case $s = T^*$.* The MTC operator $\partial/\partial T_{\text{mod}}$ is not self-adjoint in any natural inner product on the diffusion-solution space (the parameter $T_{\text{mod}}$ enters the cross-section coefficients of $H$ rather than $H$'s acting space, so $\partial H/\partial T_{\text{mod}}$ has no self-adjointness structure analogous to $H$ itself). No $\iota \in \mathcal{I}_{T^*}$.
- *Case $s = \mathcal{T}^*_{\text{rev}}$.* Empty for PWR diffusion. No $\iota \in \mathcal{I}_{\mathcal{T}^*_{\text{rev}}}$.
- *Case $s = \mathcal{L}^*$.* $\rho_{\text{MTC-bor}}$ does not assert convergence at a parametric limit; it asserts a non-zero finite value of a mixed second derivative at a *finite* parameter point in the operating envelope. The mixed-derivative limit $\Delta T, \Delta C \to 0$ is the *definition* of the derivative, not the MR's claim; the MR's claim is that the resulting derivative exceeds $\tau_{\text{MTC-bor}}$, which is a non-vanishing condition at a finite parameter value, not a convergence rate. No $\iota \in \mathcal{I}_{\mathcal{L}^*}$.
- *Case $s = \mathcal{D}^*$.* $\rho_{\text{MTC-bor}}$ concerns $k_{\text{eff}}$ as a function of two parameters at steady state; it does not involve a solution trajectory of an underlying ODE/PDE. No $\iota \in \mathcal{I}_{\mathcal{D}^*}$.
- *Case $s = \mathcal{E}^*$.* The comparison is between two parameter regimes $(T_0, C_0)$ and $(T_0 + \Delta T, C_0 + \Delta C)$ of a single fixed method (the same PWR core simulator $P$ in all four eigenvalues), not between two methods. No $\iota \in \mathcal{I}_{\mathcal{E}^*}$.
- *Case $s = \mathcal{B}^*_{\text{rel}}$.* Empty for $\mathcal{A}_{\text{PWR}}$. No $\iota \in \mathcal{I}_{\mathcal{B}^*_{\text{rel}}}$.

This exhausts $\mathcal{D}(\mathcal{A}_{\text{PWR}})$. Hence $\rho_{\text{MTC-bor}} \notin \text{MR}(\mathcal{A}_{\text{PWR}})$. ☐

*C.6.4 Two further obstructions identified by Proposition 2's proof.* The proof identifies two further independent structural obstructions in `Translate`'s present definition, in addition to the three identified by Proposition 1:

(O4) *MR is a non-zero second-order mixed partial derivative, not a first-order relation.* All per-block $\pi$ templates in Table 21 are first-order: equivariance is a first-order identity $P(gx_0) = \rho(g)P(x_0)$; monotonicity is a first-order inequality $P(x_1) \leq P(x_2)$; self-adjointness is a first-order pairing $\langle Lx_1, x_2\rangle = \langle x_1, Lx_2\rangle$. Mixed second differences (and a fortiori higher-order mixed differences) have no expression in any of these.

(O5) *MR involves two independent parameter directions, joined as a 2-by-2 perturbation rectangle, not as a chain.* The $O_{\leq}$ block's $\leq_\theta$ is a single partial-order direction; even with multiple independent partial orders in $\mathcal{I}_{O_{\leq}}$, the $\pi$ template relates pairwise inputs along a *single chosen direction*, not jointly across two perpendicular directions in a finite-difference rectangle.

A constructive resolution of Proposition 2's obstructions would require `Translate` to admit (iv) higher-order mixed-difference $\pi$-templates and (v) bidirectional joint parametric dependence beyond the single-$\theta$ partial order of Definition 11.

*C.6.5 Combined corollary.*

Corollary (Theorem $1'$ is false on $\mathcal{A}_{\text{PWR}}$, two-fold). *The MRs $\rho_{\text{nonadd}}, \rho_{\text{MTC-bor}}$ are each formulable over operators of $\mathcal{A}_{\text{PWR}}$ and each empirically realised on every conforming PWR core simulator. By Propositions 1 and 2, neither is in* $\mathrm{MR}(\mathcal{A}_{\text{PWR}})$. *The structural obstructions O1–O5 identified by the two proofs are pairwise distinct: no single extension of `Translate`'s signature absorbs any two simultaneously, so the joint obstruction is irreducibly five-fold.*

Remark 13 (Open: Composite `Translate`). *A natural follow-up is to define a* Composite `Translate` $\widetilde{\text{Translate}} : \mathcal{I}_{s_1} \times \cdots \times \mathcal{I}_{s_k} \to \mathrm{MR}(P)$ *that combines invariants from multiple blocks under a generalised $\pi$ template admitting (i) operator-spectrum output, (ii) homomorphism-failure relations, (iii) configuration-indexed adjoint structure, (iv) higher-order mixed differences, and (v) bidirectional joint parametric dependence. Whether such an extension preserves Theorem 1's closure (now over $\widetilde{\mathrm{MR}}(\mathcal{A}_P)$) and Theorem 2's polynomial-time decidability is the principal open problem this subsection leaves to future work. Five independent extensions are needed; a single uniform Composite `Translate` covering all of them would be a substantive theoretical contribution.*

### C.7 Worked enumeration of CONSTRUCT-MP on the Boltzmann algebra (migrated)

The line-by-line enumeration of CONSTRUCT-MP Steps 1–4 on $\mathcal{A}_{\text{Boltz}}$, originally Appendix C.7, is provided in supplementary `S9_migrated_appendices/C7_boltzmann_worked.tex`.

## APPENDICES D, E (IMPLEMENTATION + CONSISTENCY CHECK)

A reference implementation of CONSTRUCT-MP (Appendix D) and the construct-trace consistency check on hand-crafted block-targeted mutants (Appendix E, formerly demoted from the H3a.1 evidence base) are provided in supplementary `S9_migrated_appendices/D_E_implementation_consistency.tex`; both are confidence-strengthening illustrative material that does not carry inferential weight for the paper's H3 verdicts.

## DATA AVAILABILITY STATEMENT

The companion artefact for this paper, including the `CONSTRUCT-MP` implementation, the structural decompositions for the three case-study algebras, the Set N MR catalogue (§5.3.3: 36 MRs on

`MathSignalClass` + `ComplexSignal`; ISSUE-009: 5 MRs on Apache Commons Math), the Set G GenMorph rerun output, the LLM-ensemble Set L harvest with token-cost logs, the PIT 1.7.4 mutation traces, and the per-block + D1/D2 aggregation pipeline, is archived on Zenodo under DOI 10.5281/zenodo.20250634. Reproduction scripts (`setup.sh`, `run_all.sh`, `tests/run.sh`) and the experiment-side `CLAUDE.md` collaboration record are bundled. The camera-ready version will record any newer Zenodo version DOI if the replication package is updated during review.


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